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# The Living World

| Q.No. | Question | Answer |
|---|---|---|
| 1. | Increase in mass and increase in number of individuals are twin characteristics of which character of living organisms? | Growth |
| 2. | Growth, reproduction, ability to sense environment are the features of living organisms. (True/False) | True |
| 3. | Who is the Darwin of 20th century? | E. Mayr |
| 4. | The biggest spin off of taxonomic studies was the recognition of sharing of similarities among living organisms both 'A' and 'B' . | A-Horizontally<br>B-Vertically |
| 5. | Early man could easily perceive the difference between inanimate matter and living organisms. (True/False) | True |
| 6. | In majority of higher animals and plants, growth and reproduction are mutually exclusive events. (True/False) | True |
| 7. | What is the common method for multiplication among the fungi, filamentous algae and protonema of mosses ? | Fragementation |
| 8. | Reproduction can not be taken as defining characteristics of living organisms. Why ? | There are many organisms those cannot reproduce |
| 9. | For unicellular organisms like bacteria, unicellular algae and *Amoeba*, 'A' is synonymous with 'B' . | A-Reproduction<br>B-Growth |
| 10. | Properties of tissues are not present in the constituent cells but arise as a result of interactions among the constituent cells. (True/False) | True |
| 11. | The sum total of all chemical reactions occurring in a living body is called ? | Metabolism |
| 12. | What is the most obvious and technically complicated feature of all living organisms ? | Consciousness |
| 13. | The patient is brain-dead and has no self-consciousness. Such patients who never come back to normal life. Such patient is considered living or non-living. | Living |
| 14. | Early man could easily perceive the difference between _____ matter and living organisms. | Inanimate |
| 15. | In Binomial nomenclature, each name has two components the A and the B. | A-Generic name<br>B-Specific epithet |
| 16. | What is the full form of ICBN ? | International Code of Botanical Nomenclature |
| 17. | How many species are known and described till now ? | 1.7-1.8 million |

| | | |
|---|---|---|
| **18.** | The number and types of organisms present on earth, this refers to as ? | Biodiversity |
| **19.** | For plants, scientific names are based on agreed principles and criteria, which are provided by :- | ICBN |
| **20.** | Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate _____. | Their Latin origin |
| **21.** | Biological names are generally in __A__ language and written in __B__. | A-Latin, B-Italics |
| **22.** | All living organisms can be classified into different taxa on the basis of characteristics, this process is called ? | Classification |
| **23.** | According to binomial system, how the first letter of first word and first letter of second word in name of an organism are written respectively ? | Capital and Small |
| **24.** | A need to standardize the naming of living organisms such that a particular organism is known by the same name all over the world. This process is called _____. | Nomenclature |
| **25.** | The word systematics is derived from the Latin word 'Systema' which means ? | Systematic arrangement of organisms |
| **26.** | Taxonomic categories / taxonomic groups are distinct biological entities and not merely morphological aggregates. (True/False) | True |
| **27.** | What are basics of taxonomy ? | Identification<br>Nomenclature<br>Classification |
| **28.** | The earliest classifications were based on the _____ of various organisms. | Uses or economic importance |
| **29.** | Genus comprises a group of related species which has more characters in common in comparison to species of other genera. (True/False) | True |
| **30.** | Felidae and *Felis* represent which taxonomic categories respectively? | Family, Genus |
| **31.** | Write the scientific names of :-<br>(A) Human<br>(B) Potato<br>(C) Leopard | <br>A-*Homo sapiens*<br>B-*Solanum tuberosum*<br>C-*Panthera pardus* |
| **32.** | *Solanum, Petunia* and *Datura* are the genera, belong to which family ? | Solanaceae |
| **33.** | A group of individual organisms with fundamental morphological similarities is called ? | Species |

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| **34.** | Order and other higher taxonomic categories are identified based on the aggregation of characters. (True/False) | True |
| **35.** | Fishes, amphibians, reptiles, birds along with mammals constitute the next higher category called ? | Phylum (Chordata) |
| **36.** | As we go higher from species to kingdom then what will happen to the number of common characteristics ? | They will decrease |
| **37.** | The following sequence shows taxonomic categories in ascending order : Kingdom ← Phylum ← B ← Order ← A ← Genus ← Species<br>What are A and B in this sequence? | A-Family<br>B-Class |
| **38.** | Find the odd one from followings :-<br>Chordata, Carnivora, *Solanum*, Mammalia | *Solanum* |
| **39.** | Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level. Why ? | Because common characters are less in higher taxa. |
| **40.** | Match the column (I & II)<br>**Column-I** **Column-II**<br>(A) Man (i) *Mangifera indica*<br>(B) Housefly (ii) *Triticum aestivum*<br>(C) Mango (iii) *Homo sapiens*<br>(D) Wheat (iv) *Musca domestica* | <br><br>(A) - iii<br>(B) - iv<br>(C) - i<br>(D) - ii |
| **41.** | Find out the odd one with respect to rank or taxonomic categories :- Sapindales, Anacardiaceae, Poales | Anacardiaceae |
| **42.** | Identify A & B :-<br>*Triticum aestivum* → *Triticum* → A → Poales → B → Angiospermae | A - Poaceae<br>B - Monocotyledonae |
| **43.** | The taxonomic studies of various species of plants and animals are useful in agriculture, forestry, industry and in general for knowing our bio-resources and their diversity. (True/False) | True |
| **44.** | The Binomial naming system given by _____is being practiced by biologists all over the world. | Carolus Linnaeus |
| **45.** | In *Mangifera indica* , *Mangifera* represents the _____while *indica*, is a _____. | Genus, specific epithet |
| **46.** | Both the words in a biological name, when handwritten, are_____, or printed in _____to indicate their _____origin. | Latin, italics, Latin |
| **47.** | 'animals', 'mammals' and 'dogs' represent taxa at _____. | different levels |
| **48.** | The word systematics is derived from the_____ word 'systema' which means systematic arrangement of _____. | Latin, organisms |

| | | |
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| **49.** | _____used Systema Naturae as the title of his publication. | Linnaeus |
| **50.** | Classification is not a single step process but involves hierarchy of steps in which each step represents a_____ or _____. | rank, category |
| **51.** | The category is a part of overall taxonomic arrangement, it is called the taxonomic category and all categories together constitute the _____. | taxonomic hierarchy |
| **52.** | Taxonomic studies consider a group of individual organisms with fundamental similarities as a _____. | species |
| **53.** | Each genus may have one or more than one specific epithet representing different organisms, but having _____. | morphological similarities |
| **54.** | _____Comprises a group of related species which has more characters in common in comparison to species of other genera. | Genus |
| **55.** | Felis (cats) in the family _____. | Felidae |
| **56.** | _____, has a group of related genera with still less number of similarities as compared to genus and species. | Family |
| **57.** | All animals belonging to various phyla are assigned to the highest category called _____. | Kingdom |
| **58.** | The taxonomic categories from species to kingdom have been shown in_____ order starting with _____. | Ascending, species |
| **59.** | When the species is determined on the basis of morphological characters then it is called as_____. | taxonomic species |
| **60.** | Ernst Mayr Darwin of_____, & proposed the _____of species. | 20$^{th}$ century , biological concept |
| **61.** | _____classification is easily carried out by using computers and it is based on all observable characteristics. | Phenetic classification or Numerical classification |
| **62.** | Taxonomic hierarchy ,can be descending or ascending arrangement of taxonomic categories | True |
| **63.** | _____in his book "Theories elementaire de la botanique" (Theory of elementary botany) | A.P. de. Candolle |
| **64.** | Occurrence of the number and types of organisms present on earth, is called_____ | Biodiversity |
| **65.** | Anabolism + Catabolism = _____ | Metabolism |
| **66.** | Artificial classification many superficial morphological characters like habit, colour, number and shape of leaves | False |
| **67.** | Natural phylogenetic in this classification, only gross morphology is considered. | False |
| **68.** | The taxonomic studies of various species of plants and animals are useful in agriculture, forestry, industry and in general for knowing our bio-resources and their diversity. | True |

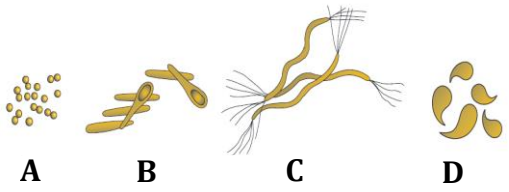
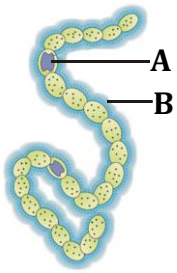
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# Biological Classification

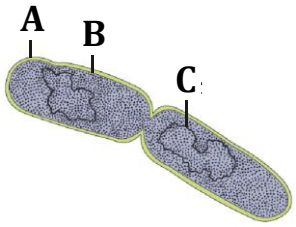
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<table>
<tr><th>Q.No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1.</td><td>Who gave two kingdom system ?</td><td>Carolus Linnaeus</td></tr>
<tr><td>2.</td><td>Two kingdom system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (Autotrophs) and non-photosynthetic (Heterotrophs) organisms. These are demerits of two kingdom system. (True/False)</td><td>True</td></tr>
<tr><td>3.</td><td>Aristotle used simple morphological characters to classify plants into __A__, __B__ and __C__.</td><td>A-trees, B-shrubs, C-herbs</td></tr>
<tr><td>4.</td><td>Though __A__ and __B__ kingdoms have been a constant under all different systems, the und erstanding of what group/organisms be included under these kingdoms have been changing. The __C__ and __D__ of other kingdom have also been understood differently by different scientists over time.
<table>
<tr><th rowspan="2">Characters</th><th colspan="5">Five Kingdom</th></tr>
<tr><td>Monera</td><td>Protista</td><td>A</td><td>Plantae</td><td>Animalia</td></tr>
<tr><td>Cell type</td><td>Prokaryotic</td><td>Eukaryotic</td><td>Eukaryotic</td><td>Eukaryotic</td><td>Eukaryotic</td></tr>
<tr><td>Cell wall</td><td>Noncellulosic (Polysaccharide + amino acid)</td><td>Present in some (cellulose)</td><td>Present (without cellulose)</td><td>Present (cellulose)</td><td>Absent</td></tr>
<tr><td>Nuclear membrane</td><td>Absent</td><td>Present</td><td>Present</td><td>Present</td><td>B</td></tr>
<tr><td>Body Organisation</td><td>Cellular</td><td>C</td><td>Multiceullar /loose tissue</td><td>Tissue/organ</td><td>Tissue/ organ/ organ system</td></tr>
<tr><td>D</td><td>Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophyte-/parasite)</td><td>Autotrophic (Photosynthetic) and Heterotrophic</td><td>Heterotrophic (Saprophytic/ Parasitic)</td><td>Autotrophic (Photosynthetic)</td><td>Heterotrophic (Holozoic-/Saprophytic etc.)</td></tr>
</table>
Observe the above table and answer the following questions (5 to 9).</td><td>A-Plantae<br>B-Animalia<br>C-Number<br>D-Nature</td></tr>
<tr><td>5.</td><td>Identify the A, B, C and D.</td><td>A-Fungi, B-Present, C-Cellular, D-Mode of nutrition</td></tr>
<tr><td>6.</td><td>How many kingdoms include producers and name these kingdoms ?</td><td>Three; Monera, Protista, Plantae</td></tr>
<tr><td>7.</td><td>In which kingdom unicellular, eukaryotic autotrophic organisms are placed ?</td><td>Protista</td></tr>
<tr><td>8.</td><td>In how many kingdom cellulosic cell walled organisms are present ?</td><td>Two– Protista, Plantae</td></tr>
<tr><td>9.</td><td>From above table, how many characters are uncommon between fungi and plantae ?</td><td>Nature of cell wall, Mode of Nutrition, Body organisation</td></tr>
<tr><td>10.</td><td>On the basis of shape, how many groups of bacteria are formed and what are their names ?</td><td>Four<br>Coccus, Bacillus, Vibrium, Spirillum</td></tr>
</table>

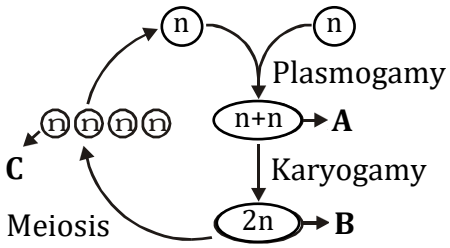
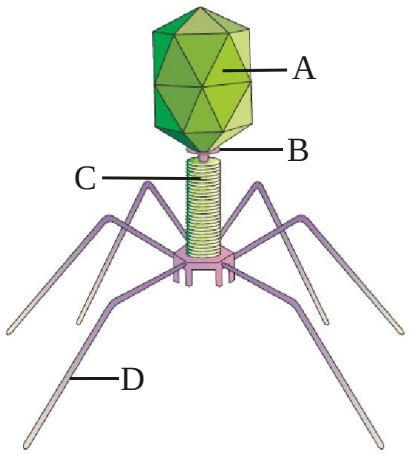
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| **11.** | The criteria for classification are changing over time. Why ? | Because the improvement in our understanding of characters |
| **12.** | All __A__ organisms are grouped under kingdom monera and unicellular eukaryotes are placed in kingdom __B__. | A-Prokaryotic<br>B-Protista |
| **13.** | *Chlamydomonas* and *Amoeba* are classified in kingdom protista according to Whittaker's classification system but by earlier classification system both these organisms are classified in different kingdoms. (True/False) | True |
| **14.** | Identify the following figures (A, B, C and D).<br>  <br>**A** **B** **C** **D** | A-Cocci<br>B-Bacilli<br>C-Spirillia<br>D-Vibrio |
| **15.** | Identify the following figure with structure __A__ and __B__ and write their functions :-<br>  <br>**A**<br>**B** | A-*Nostoc*, Heterocyst<br>Function - Nitrogen fixation<br><br>B-Mucilagenous sheath<br>Function - Protection from drying |
| **16.** | The bacterial structure is very__A__, they are very complex in __B__. | A-Simple, B-Behaviour |
| **17.** | The vast majority of bacteria are heterotrophs. (True/False) | True |
| **18.** | Read the following statements :-<br>(A) Found in the gut of ruminants<br>(B) Complex cell wall structure<br>(C) Survive in marshy habitats<br>Above statements are related to which type of Archaebacteria. | Methanogens |
| **19.** | Compared to many other organisms, which group shows the most extensive metabolic diversity ? | Bacteria |
| **20.** | Which organism are the smallest living cells known and can survive without oxygen ? | Mycoplasma |
| **21.** | Cholera, Typhoid, Tetanus, Citrus canker are well known diseases caused by :- | Bacteria |
| **22.** | In which kingdom, Chrysophytes, Dinoflagellates, Euglenoids, Slime moulds and Protozoans are included ? | Protista |
| **23.** | Which Chrysophytes are the chief producers in the ocean? | Diatoms |

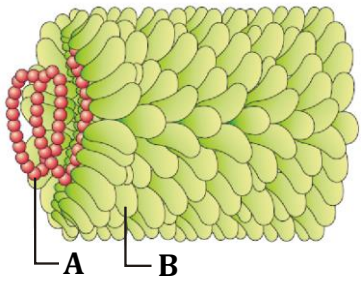
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| **24.** |  <br>Identify the above process in bacteria and identify part A, B & C : | Process : Binary Fission<br><br>A-Cell wall<br>B-Cell membrane<br>C-DNA |
| **25.** |  <br>Identify the above organism :- | *Euglena* |
| **26.** | Which protist possesses two flagella one lies longitudinal and other transversely ? | Dinoflagellates |
| **27.** | Read the following points regarding Protista :-<br>(A) Saprophytic nature<br>(B) Spores with true wall<br>(C) Wall less vegetative phase<br>Above organism should be :- | Slime moulds |
| **28.** | Euglenoids have a protein rich layer called _____ which makes their body flexible. | Pellicle |
| **29.** | _____ are primitive relatives of animals. | Protozoans |
| **30.** | Which organisms are used to make bread and beer ? | Yeasts |
| **31.** | Some hyphae are continuous tubes filled with multinucleated cytoplasm, these are called:- | Coenocytic hyphae |
| **32.** | The cell walls of fungi are composed of :- | Chitin and Polysaccharides |
| **33.** | Fungi prefer to grow :- | Warm & Humid places |
| **34.** | Write the name of three steps of sexual reproduction in fungi :- | 1-Plasmogamy<br>2-Karyogamy<br>3-Meiosis |
| **35.** | Identify the following figure :-<br>  | *Agaricus* |
| **36.** | By which basis kingdom-Fungi is divided into various classes ? | Morphology of mycelium<br>Mode of spore formation and fruiting bodies |

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| **37.** | Identify the A, B and C in the given line diagram :-<br>  <br> | A-Dikaryon<br>B-Synkaryon<br>C-Spores |
| **38.** | Members of which fungal class are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites. | Phycomycetes |
| **39.** | Which fungus is used extensively in bio-chemical & genetic works? | *Neurospora* |
| **40.** | The basidiospores are __A__ produced on the __B__. | A-exogenously<br>B-Basidium |
| **41.** | Identify the A, B and C in the following table :-<br>**Class** / **Member** / **Disease**<br>Ascomycetes / A / Ergot<br>Basidiomycetes / *Puccinia* / B<br>C / *Alternaria* / Blight | A-*Claviceps*<br>B-Rust<br>C-Deuteromycetes |
| **42.** | Mushroom, Bracket fungi & smut fungi belong to which fungal class? | Basidiomycetes |
| **43.** | Most of the members of fungal class ___ are decomposers of litter and help in mineral cycling. | Deuteromycetes |
| **44.** | Life cycle of plants has two distinct phases - the diploid __A__ and the __B__ gametophyte. | A-Sporophyte,<br>B-Haploid |
| **45.** | All of us have suffered the ill effects of common cold or 'flu'.<br>It is caused by :- | Viruses |
| **46.** | Which kingdom is characterized by heterotrophic, eukaryotic organisms that are multicellular and their cells lack cell wall? | Kingdom - Animalia |
| **47.** | Bladderwort and venus fly trap are examples of __A__ plants and *Cuscuta* is a __B__ plant. | A-Insectivorous<br>B-Parasitic |
| **48.** | Identify the following figure with A, B, C and D :-<br>  <br> | Bacteriophage<br>A-Head<br>B-Collar<br>C-Sheath<br>D-Tail fibres |
| **49.** | Who called *contagium vivum fluidum* and what does it stand for ? | M.W. Beijerinck<br>Infectious living fluid |

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| **50.** | Read the following symptoms :-<br>• Mosaic formation<br>• Leaf rolling and curling<br>• Yellowing and vein clearing<br>• Dwarfing<br>Above symptoms are due to infection of. | Viruses |
| **51.** | Identify the following figure and A & B :-<br>  <br> | Tobacco mosaic virus (TMV)<br><br>A-RNA<br>B-Capsid |
| **52.** | Who said that viruses are smaller than bacteria ? | D.J. Ivanowsky |
| **53.** | In lichens, the algal component is known as __A__ and the fungal component as __B__. | A-Phycobiont<br>B-Mycobiont |
| **54.** | Identify the infectious agent on the basis of given information's :-<br>• Discoverer T.O. Diener<br>• Smaller than viruses<br>• Low molecular weight RNA<br>• Absence of protein coat | Viroids |
| **55.** | The causative agent of potato spindle tuber disease is composed of | Only RNA |
| **56.** | Identify the group of organisms, on the basis of given information's :-<br>• Show symbiotic association<br>• Very good pollution indicator<br>• Pioneer to barren rocks | Lichen |
| **57.** | In lichens, __A__ prepare food for __B__ and fungi provide shelter and absorb __C__ for its partner. | A-Algae<br>B-Fungi<br>C-Water and mineral |
| **58.** | _____ (1969) proposed a Five Kingdom Classification. | R.H. Whittaker |
| **59.** | Kingdom Protista has brought together_____, _____ (earlier placed in Algae within Plants and both having cell walls) with *Paramecium* and _____ (which were earlier placed in the animal kingdom which lack cell wall). | *Chlamydomonas, Chlorella*, Amoeba |
| **60.** | _____ are the sole members of the Kingdom Monera. | Bacteria |
| **61.** | Though the bacterial structure is very simple, they are very complex in _____. | Behaviour |
| **62.** | Bacteria are grouped under four categories based on their_____, the spherical Coccus, the rod-shaped _____ the comma-shaped Vibrium and the spiral Spirillum. | Shape, *Bacillus* |

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| **63.** | In Linnaeus' time a _____ system of classification with Plantae and Animalia kingdoms was developed that included all plants and animals respectively. | Two Kingdom |
| **64.** | The _____ system has also been proposed that divides the Kingdom Monera into two domains, leaving the remaining eukaryotic kingdoms in the third domain and there by a six kingdom classification. | Three-domain |
| **65.** | The vast majority of bacteria are_____, i.e., they depend on other organisms or on dead organic matter for food. | Heterotrophs |
| **66.** | _____ differ from other bacteria in having a different cell wall structure and this feature is responsible for their survival in extreme conditions. | Archaebacteria |
| **67.** | _____ are present in the gut of several ruminant animals such as cows and buffaloes and they are responsible for the production of methane (biogas) from the dung of these animals. | Methanogens |
| **68.** | These bacteria are special since they live in some of the most harsh habitats such as extreme _____ (halophiles), _____ (thermoacidophiles) and marshy areas (methanogens). | Salty areas, hot springs |
| **69.** | The_____ (also referred to as blue-green algae) have chlorophyll a similar to green plants and are photosynthetic autotrophs. | cyanobacteria |
| **70.** | They often form _____ in polluted water bodies. | Blooms |
| **71.** | Some of these organisms can fix atmospheric nitrogen in specialised cells called _____ e.g., *Nostoc* and *Anabaena*. | Heterocysts, |
| **72.** | _____ autotrophic bacteria oxidise various inorganic substances such as nitrates, nitrites and ammonia and use the released energy for their ATP production. | Chemosynthetic |
| **73.** | Chemoautotrophic bacteria play a great role in recycling nutrients like _____. | Nitrogen, phosphorous, iron and sulphur |
| **74.** | _____bacteria are most abundant in nature. The majority are important decomposers. | Heterotrophic |
| **75.** | They are helpful in making curd from milk, production of _____, _____ in legume roots, etc. | Antibiotics, fixing nitrogen |
| **76.** | Cholera, typhoid, tetanus, citrus canker are well known diseases caused by different _____ | Bacteria. |
| **77.** | Bacteria reproduce mainly by fission. Sometimes, under _____ conditions, they produce spores. | Unfavourable |
| **78.** | They also reproduce by a sort of sexual reproduction by adopting a primitive type of DNA transfer from one _____ to the other. | Bacterium |
| **79.** | The _____ are organisms that completely lack a cell wall. | *Mycoplasma* |

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| **80.** | They are the smallest living cells known and can survive _____. | Without oxygen |
| **81.** | Many mycoplasma are _____ in animals and plants. | Pathogenic |
| **82.** | All single-celled eukaryotes are placed under _____ but the boundaries of this kingdom are not well defined. | Protista, |
| **83.** | What may be 'a _____' to one biologist may be 'a plant' to another. | Photosynthetic protistan |
| **84.** | In this book we include _____, Dinoflagellates, Euglenoids, Slime moulds and Protozoans under Protista. | Chrysophytes |
| **85.** | Members of Protista are _____. | Primarily aquatic. |
| **86.** | This kingdom forms a link with the others dealing with plants, animals and _____. | fungi. |
| **87.** | Being eukaryotes, the _____ cell body contains a well defined nucleus and other membrane-bound organelles. Some have flagella or cilia. | Protistan |
| **88.** | _____ reproduce asexually and sexually by a process involving cell fusion and zygote formation. | Protists |
| **89.** | This group includes diatoms and _____(desmids). | Golden algae |
| **90.** | They are microscopic and float passively in _____ (plankton). Most of them are _____. | Water currents, photosynthetic |
| **91.** | In diatoms the cell walls form two thin _____, which fit together as in a soap box. The walls are embedded with _____ and thus the walls are indestructible. | Overlapping shells, silica |
| **92.** | Thus, diatoms have left behind large amount of cell wall deposits in their habitat; this accumulation over billions of years is referred to as '_____'. Being gritty this soil is used in polishing, filtration of oils and syrups. Diatoms are the chief '_____' in the oceans | diatomaceous earth, producers |
| **93.** | These organisms are mostly _____ and_____. They appear yellow, green, brown, blue or red depending on the main pigments present in their cells. | Marine, photosynthetic |
| **94.** | The cell wall has stiff _____ plates on the outer surface. | cellulose |
| **95.** | Most of them have_____one lies longitudinally and the other transversely in a furrow between the wall plates. | Two flagella |
| **96.** | Very often, red dinoflagellates (Example: *Gonyaulax*) undergo such rapid multiplication that they make the sea appear red (_____). | Red tides |
| **97.** | _____ released by such large numbers may even kill other marine animals such as fishes. | Toxins |
| **98.** | Majority of them are fresh water organisms found in _____. | Stagnant water |
| **99.** | Instead of a cell wall, they have a protein rich layer called _____ which makes their body flexible | Pellicle |

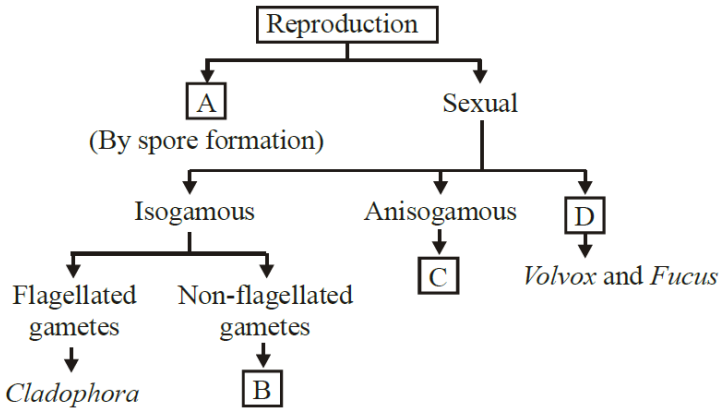
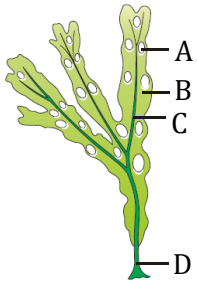
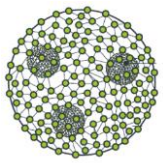
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|---|---|---|
| **100.** | Though they are _____ in the presence of sunlight, when deprived of sunlight they behave like _____ by predating on other smaller organisms. | Photosynthetic, heterotrophs |
| **101.** | Interestingly, the pigments of euglenoids are identical to those present in _____. Example: *Euglena* | Higher plants |
| **102.** | Slime moulds are _____ protists. | Saprophytic |
| **103.** | The body moves along _____ and leaves engulfing organic material. | Decaying twigs |
| **104.** | Under suitable conditions, they form an aggregation called _____ which may grow and spread over several feet. | Plasmodium |
| **105.** | During _____ conditions, the _____ differentiates and forms fruiting bodies bearing spores at their tips. | Unfavourable, plasmodium |
| **106.** | The spores possess _____. They are extremely resistant and survive for many years, even under adverse conditions. The spores are dispersed by _____. | True walls, air currents |
| **107.** | Fusion of protoplasms between two motile or non-motile gametes called _____. | plasmogamy |
| **108.** | Fusion of two nuclei called_____. Meiosis in zygote resulting in haploid spores. | Karyogamy |
| **109.** | In other fungi (ascomycetes and basidiomycetes), an intervening _____ (n + n, i.e., two nuclei per cell) occurs; such a condition is called a dikaryon and the phase is called dikaryophase of fungus. | Dikaryotic stage |
| **110.** | The morphology of the mycelium, _____ and fruiting bodies form the basis for the division of the kingdom into various classes. | Mode of spore formation |
| **111.** | _____ is used extensively in biochemical and genetic work. | *Neurospora* |
| **112.** | Many members like _____ are edible and are considered delicacies. | Morels and truffles |
| **113.** | Karyogamy and meiosis take place in the basidium producing _____. The basidiospores are _____ produced on the basidium (pl.: basidia). | Four basidiospores, exogenously |
| **114.** | Commonly known as _____because only the _____ or vegetative phases of these fungi are known. | Imperfect fungi, asexual |
| **115.** | _____ (1892) recognised certain microbes as causal organism of the mosaic disease of tobacco. | Dmitri Ivanowsky |
| **116.** | _____ (1935) showed that viruses could be crystallised and crystals consist largely of proteins. They are inert outside their specific host cell. | W.M. Stanley |
| **117.** | In plants, the symptoms can be mosaic formation, leaf rolling and curling, yellowing and vein clearing, dwarfing and stunted growth. | True |

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| | | |
|---|---|---|
| **118.** | In 1971, Stanley, discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. | False |
| **119.** | In modern medicine certain infectious neurological diseases were found to be transmitted by an agent consisting of abnormally folded protein. The agent was similar in size to viruses. These agents were called prions. | True |
| **120.** | The most notable diseases caused by Viroids are bovine spongiform encephalopathy (BSE) commonly called mad cow disease in cattle and its analogous variant Cr–Jacob disease (CJD) in humans. | False |
| **121.** | Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi. The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. | True |
| **122.** | The parasitic forms cause diseases such as sleeping sickness. Example: *Trypanosoma*. | True |

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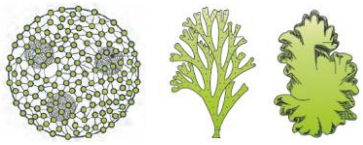
# Plant Kingdom

| Q.No. | Question | Answer |
|---|---|---|
| 1. | Which taxonomy is based on following points :-<br>• chromosome number<br>• chromosome structure<br>• chromosome behaviour | Cytotaxonomy |
| 2. | Match the column I with column II :-<br>**Column-I** **Column-II**<br>(A) Unicellular (i) *Spirogyra*<br>(B) Colonial (ii) Kelps<br>(C) Filamentous (iii) *Volvox*<br>(D) Massive plant bodies (iv) *Chlamydomonas* | A-iv, B-iii, C-i, D-ii |
| 3. | Identify A, B, C and D in given flow chart :-<br>  | A-Asexual<br>B-*Spirogyra*<br>C-*Eudorina*<br>D-Oogamous |
| 4. | Numerical taxonomy which is now easily carried out using computers is based on _____. | All observable characteristics |
| 5. | Which type of sexual reproduction is found in *Fucus* and red algae ? | Oogamous |
| 6. | Identify the given figure with A, B, C & D :-<br>  | *Fucus*<br>A-Air bladder<br>B-Frond<br>C-Mid rib<br>D-Holdfast |
| 7. | Identify the given figures (A and B) :-<br>  <br>(A)<br>  <br>(B) | A-*Volvox*<br>B-*Ulothrix* |

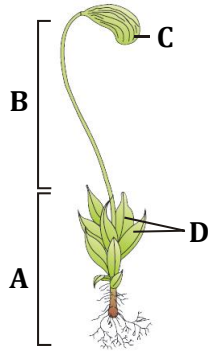
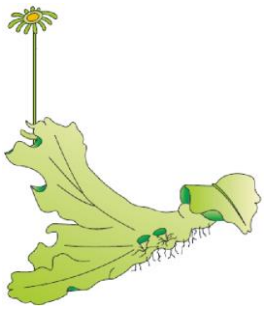
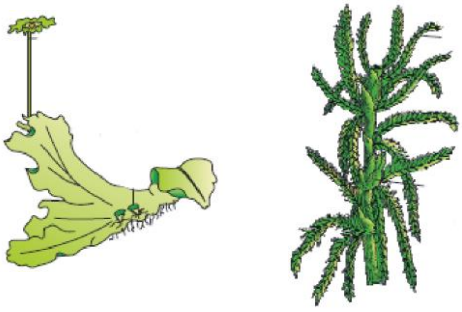


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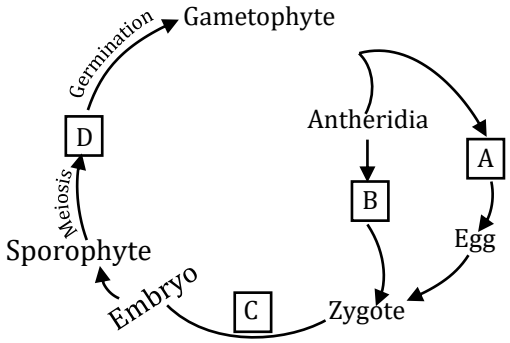
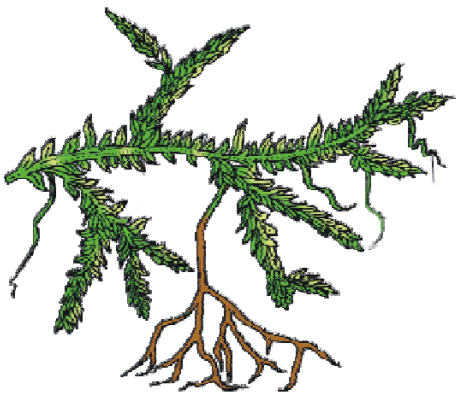
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<table>
<tr><td>8.</td><td>Write down the algal class of following figures :-<br>  <br>(A) (B) (C)</td><td>A-Chlorophyceae<br>B-Phaeophyceae<br>C-Rhodophyceae</td></tr>
<tr><td>9.</td><td>Study the following table and identify (i) (ii) & (iii) :-
<table>
<tr><th>Figure</th><th>Algal Group</th><th>Name</th></tr>
<tr><td>  </td><td>Brown algae</td><td>(i)</td></tr>
<tr><td>  </td><td>(ii)</td><td>Polysiphonia</td></tr>
<tr><td>  </td><td>Green algae</td><td>(iii)</td></tr>
</table></td><td>(i) Laminaria<br>(ii) Red algae<br>(iii) Ulothrix</td></tr>
<tr><td>10.</td><td>Porphyra & Dictyota are belong which algal class respectively</td><td>Rhodophyceae,<br>Phaeophyceae</td></tr>
<tr><td>11.</td><td>Agar, one of the commercial products obtained from __A__ and Gracilaria are used to grow microbes and in preparations of ice-creams and jellies.</td><td>A-Gelidium</td></tr>
<tr><td>12.</td><td>Which are the pigments found in phaeophyceae ?</td><td>Chlorophyll-a, c,<br>Fucoxanthin</td></tr>
<tr><td>13.</td><td>Write the name of eukaryotic unicellular algae which is rich in proteins and used as food supplements :-</td><td>Chlorella</td></tr>
<tr><td>14.</td><td>Most of the green algae have one or more storage bodies, which are located in the chloroplasts are called :-</td><td>Pyrenoids</td></tr>
<tr><td>15.</td><td>Certain marine brown and red algae produce large amounts of hydrocolloids eg. __A__ and __B__ , respectively which are used commercially.</td><td>A-Algin<br>B-Carrageen</td></tr>
<tr><td>16.</td><td>Identify A, B & C in given table :-
<table>
<tr><th>Class</th><th>Major Pigments</th><th>Stored Food</th></tr>
<tr><td>Chlorophyceae</td><td>A</td><td>Starch</td></tr>
<tr><td>B</td><td>Chl. a, c</td><td>Mannitol<br>Laminarian</td></tr>
<tr><td>Rhodophyceae</td><td>Chl. a, d</td><td>C</td></tr>
</table></td><td>A-Chlorophyll a, b<br>B-Phaeophyceae<br>C-Floridean starch</td></tr>
</table>

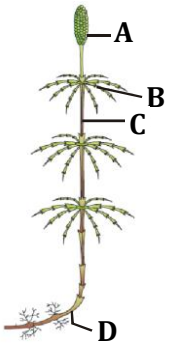
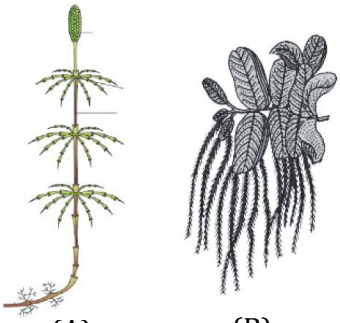
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| | | |
|---|---|---|
| **17.** | In phaeophyceae, the vegetative cells have a cellulosic cell wall usually covered on the outside by a gelatinous coating of _____. | Algin |
| **18.** | Why are the members of Rhodophyceae commonly called red algae? | Due to pre-dominance of r-phycoerythrin |
| **19.** | Floridean starch is similar to _A_ and glycogen in structure. | A-Amylopectin |
| **20.** | In which class of algae, gametes are pyriform (pear-shaped) and bear two unequal laterally attached flagella ? | Phaeophyceae |
| **21.** | In which class of algae sexual reproduction is oogamous and accompanied by complex post fertilization developments ? | Rhodophyceae |
| **22.** | Identify the given figure with A, B, C & D :-   | *Funaria*<br>A-Gametophyte<br>B-Sporophyte<br>C-Capsule<br>D-Leaves |
| **23.** | *Gracilaria* and *Porphyra* are the members of which class of algae ? | Rhodophyceae |
| **24.** | Identify the given figure and also identify the part which produce specialised asexual reproductive structures ?  | *Marchantia*, Gemma cup |
| **25.** | Identify the plants in figures (A & B).  (A) (B) | A-*Marchantia* (Dioecious)<br>B-*Sphagnum* (Monoecious) |
| **26.** | Bryophytes are called amphibians of the plant kingdom. Why ? | Because they can live in the moist soil but depend on water for sexual reproduction |

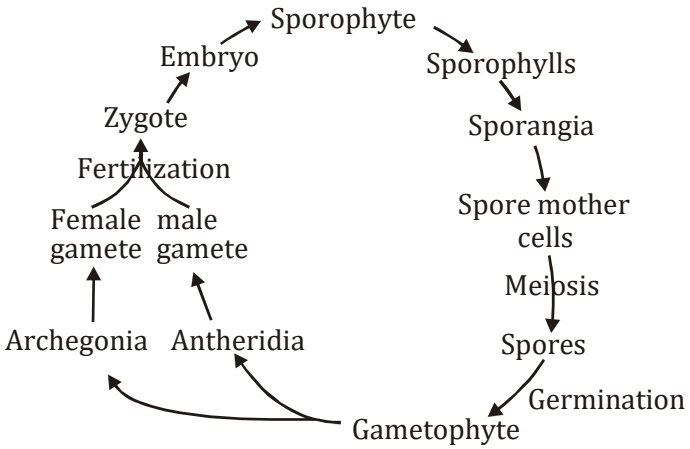
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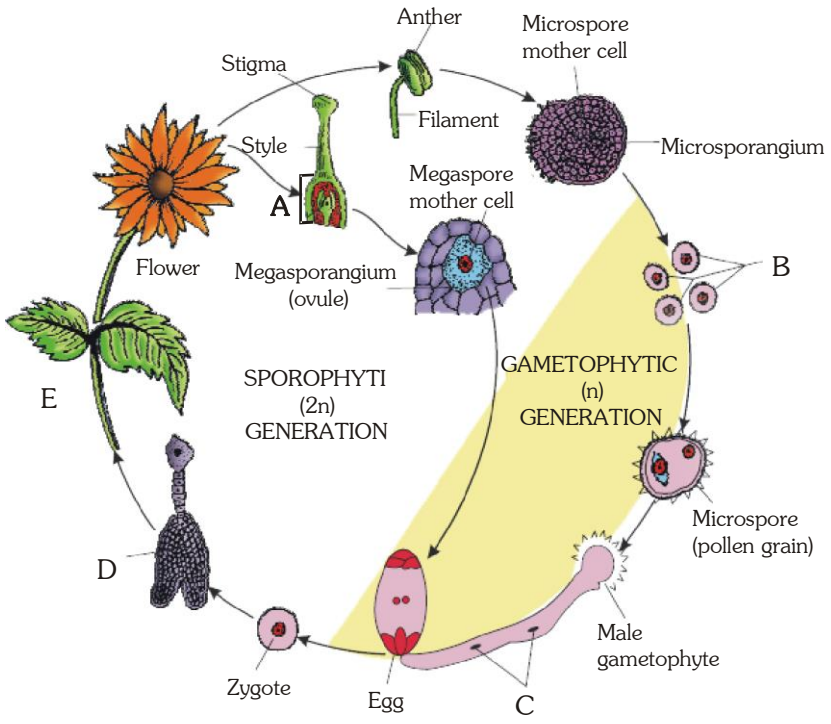
| | | |
|---|---|---|
| **27.** | Study the given flow chart and identify A, B, C & D :-<br>  <br> | A-Archegonia<br>B-Antherozoids<br>C-Mitosis<br>D-Spores |
| **28.** | In liverworts, ___A___ are green, multicellular, asexual buds, which develop in small receptacles called ___B___ located on thallus :- | A-Gemmae<br>B-Gemma cups |
| **29.** | In bryophytes, zygotes do not undergo reduction division immediately. (True/False) | True |
| **30.** | Find the ploidy level of followings with respect to bryophytes :-<br>A-Gametophyte B-Sporophyte<br>C-Spore D-Antherozoids<br>E-Gemmae F-Rhizoids | <br>A-n B-2n<br>C-n D-n<br>E-n F-n |
| **31.** | The moss gametophyte consists of two predominant stages, what are these, and they develop from ? | <table><tr><td>Stages</td><td>Develop from</td></tr><tr><td>Protonema</td><td>Spore</td></tr><tr><td>Leafy gametophyte</td><td>Bud</td></tr></table> |
| **32.** | In bryophytes, the dominant phase in the life cycle is the _____ plant body but in pteridophytes, the main plant body is _____. | Gametophytic<br>Sporophytic |
| **33.** | Which of the following terms are not related with mosses ?<br>Capsule, Buds, Prothallus, Rhizoids, Archegonia, Protonema | Prothallus |
| **34.** | Which part of gametophytic plant body of a moss bears sex organs-antheridia & archegonia? | At the apex of the leafy shoots |
| **35.** | Identify the following figure :-<br>  | *Selaginella* |

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<table>
<tr><td>36.</td><td>Label the A, B, C & D in given figure :-<br> 
</td><td>A-Strobilus<br>B-Node<br>C-Internode<br>D-Rhizome</td></tr>
<tr><td>37.</td><td>Identify the following figures (A & B) :-<br>  <br>(A) (B)</td><td>A-Equisetum<br>B-Salvinia</td></tr>
<tr><td>38.</td><td>Match the column I with column II :-
<table>
<tr><th colspan="2">Column I<br>(Figure)</th><th colspan="2">Column II<br>(Class)</th></tr>
<tr><td>(A)</td><td>  </td><td>(i)</td><td>Sphenopsida</td></tr>
<tr><td>(B)</td><td>  </td><td>(ii)</td><td>Pteropsida</td></tr>
<tr><td>(C)</td><td>  </td><td>(iii)</td><td>Lycopsida</td></tr>
</table></td><td>A-iii<br>B-i<br>C-ii</td></tr>
<tr><td>39.</td><td>How many of the following have heterosporous nature ?<br>Selaginella, Equisetum, Salvinia</td><td>Two<br>(Selaginella, Salvinia)</td></tr>
</table>

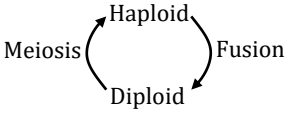
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| | | |
|---|---|---|
| **40.** | Study the following life cycle of *Pteridophytes* :-<br>  <br><br>How many structures represent gametophytic & sporophytic generations, respectively ? | Gametophytic structures = Six [Spore, Gametophyte, antheridia, archegonia, male gamete and female gamete]<br>Sporophytic structures = Six [Zygote, embryo, sporophyte, sporophylls, sporangia and spore mother cells] |
| **41.** | Read the following information's and identify the genus having them :-<br>• Coralloid roots are with BGA<br>• Unbranched stem<br>• Lack ovary wall<br>• Pinnate leaves | *Cycas* |
| **42.** | Match the column I with column II :-<br>**Column-I** / **Column-II**<br>(A) *Pinus* — (i) Precursor of seed habit<br>(B) *Cycas* — (ii) Coralloid roots<br>(C) *Selaginella* — (iii) Mycorrhizal association | A-iii, B-ii, C-i |
| **43.** | The spread of living pteridophytes is limited and restricted to narrow geographical regions. why ? | because they need water for fertilization |
| **44.** | Female cone → Mega sporophylls → Megasporangia → Megaspores<br>For ab?r cell $\xrightarrow{A}$ Microspore $\xrightarrow{B}$ Male gametophyte | Female cone<br>B-Germination |
| **46.** | In gymnosperms, the male and the female gametophytes do not have an independent free-living existence. (True/False) | True |
| **47.** | In gymnosperms, the nucellus is protected by envelops and the composite structure is called ? | Ovule |
| **48.** | How many of the following cells are present in the typical angiospermic embryo sac:<br>A - Egg cell<br>B - Synergids<br>C - Antipodals<br>D - Central cell | <br><br>A-One<br>B-Two<br>C-Three<br>D-One |

| | | |
|---|---|---|
| **49.** | Write the names of male and female sex organs of angiosperms respectively :- | Stamen<br>Carpel or Pistil |
| **50.** | In how many classes angiosperms are divided ? | Two (Dicotyledonae and monocotyledonae) |
| **51.** | Ovules, stamen, ovary, carpel, anther, antipodals, synergids, pollen grains<br>How many structures are NOT produced by female plant of angiosperm ? | Three :- these are<br>Stamen<br>Anther<br>Pollen grains |
| **52.** | Angiosperms range in size from tiny, almost microspic e.g. __A__ to tall trees of e.g. __B__. | A-*Wolfia*<br>B-*Eucalyptus* |
| **53.** | How many nuclei are involved in double fertilization ? | Five [2 male nuclei, 2 polar nuclei, 1 egg nucleus] |
| **54.** | Write down the ploidy level of followings regarding typical angiosperm :-<br>(A) Endosperm (B) Synergids (C) Embryo (D) Pollen grain<br>(E) Embryosac | A-3n, B-n, C-2n, D-n, E-n |
| **55.** | After fertilization, what is the future of the following :-<br>(A) Ovule<br>(B) Synergieds<br>(C) Ovary<br>(D) Antipodals | <br>A-Seed<br>B-Degenerate<br>C-Fruit<br>D-Degenerate |
| **56.** | Identify the A, B, C, D & E in given life cycle of an angiosperm :-   | A-Ovary<br>B-Microspore<br>C-Male gametes<br>D-Embryo<br>E-Sporophyte |

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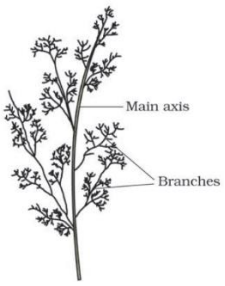
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|---|---|---|
| **57.** | Many algae such as *Volvox, Spirogyra* and some species of *Chlamydomonas* represent _____ pattern of life cycle. | Haplontic |
| **58.** | Which kind of life cycle pattern is shown by all seed-bearing plants? | Diplontic |
| **59.** | The life cycle of any sexually reproducing plant, there is an alternation of generation between __A__ producing by haploid gametophyte and __B__ producing by diploid Sporophyte. | A-Gametes<br>B-Spore |
| **60.** | Which processes are essential for alternation of generation in plant life cycle ? |  <br>Meiosis and fusion of gametes |
| **61.** | In bryophytes, what is the dominant phase of life cycle ? | Gametophytic phase |
| **62.** | "The diploid sporophyte is represented by a dominant, independent, photosynthetic, vascular plant body. It alternates with multicellular saprophytic/autotrophic, independent but short-lived haploid gametophytes".<br>Above pattern of life cycle is represented by which group of plants? | Pteridophyta |
| **63.** | "The short-lived multicellular sporophyte totally or partially dependent on the gametophyte".<br>Above statement is related to which group of kingdom plantae ? | Bryophyta |
| **64.** | Which type of life cycle is found in *Fucus* ? | Diplontic |
| **65.** | Algae usually reproduce vegetatively by __A__, asexually by formation of __B__ and sexually by fusion of __C__. | A-Fragmentation<br>B-Spores<br>C-Gametes |
| **66.** | Match the column-I (Process) with column II (Result)<br>**Column I (Process)** — **Column II (Result)**<br>(A) Fusion of male & female gametes — (i) Spores<br>(B) Mitosis in zygote — (ii) Zygote<br>(C) Meiosis in sporophyte — (iii) Gametophyte<br>(D) Germination of spores — (iv) Embryo | A-ii<br>B-iv<br>C-i<br>D-iii |
| **67.** | The main plant body of a bryophyte is _____ and produce _____. | Gametophyte, Gamete |
| **68.** | The predominant stage in the life cycle of a moss is gametophyte which consists of two stages. The first stage is the_____, which develops directly from a_____.<br>The second stage is the_____ stage, which develops from the _____. | Protonema stage,<br>Spore,<br>Leafy,<br>Secondary protonema |

**69.**

| Classes | Stored Food | Cell Wall | Flagella Number and Position of Insertion |
|---|---|---|---|
| Chlorophyceae | Starch | Cellulose | **C** |
| Phaeophyceae | **A** | **B** | **D** |
| Rhodophyceae | Floridean Starch | Cellulose, pectin and poly sulphate esters | **E** |

A. Mannitol, laminarin
B. Cellulose and algin
C. 2-8, equal, apical
D. 2, unequal lateral
E. Absent

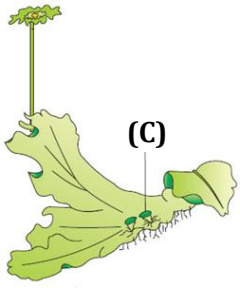
**70.**



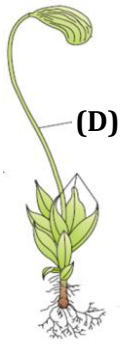

**(A)**



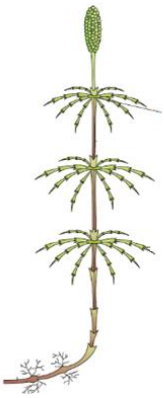
**(B)**



**(C)**



**(D)**



**(E)**

Identify A, B, C, D and E in above given diagrams?

A. *Polysiphonia*
B. *Ulothrix*
C. Gemma cup
D. Seta
E. Node



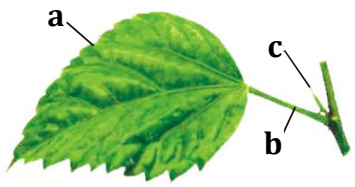
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# Morphology of Flowering Plants

<table>
<tr><th>Q.No.</th><th>Question</th><th>Answer</th></tr>
<tr><td>1.</td><td>Direct elongation of the radicle leads to the formation of _____ root.</td><td>primary.</td></tr>
<tr><td>2.</td><td>Lateral roots includes ___(a)___ and ___(b)___ roots</td><td>(a) secondary<br>(b) tertiary</td></tr>
<tr><td>3.</td><td>The primary root and its branches constitute the _____.</td><td>Tap root system</td></tr>
<tr><td>4.</td><td>When the primary root is short lived and replaced by large number of thin roots which orginates from base of the stem and constitutes the</td><td>Fibrous root system</td></tr>
<tr><td>5.</td><td>The roots which arise from the parts of the plant other than the radicle are called.</td><td>Adventitious roots</td></tr>
<tr><td>6.</td><td>The root apex is covered by thimble like structure called</td><td>Root cap</td></tr>
<tr><td>7.</td><td>The region of meristematic activity is situated few millimeters above root cap. True or false?</td><td>True</td></tr>
<tr><td>8.</td><td>Match the following :<br><table><tr><td>1.</td><td>Tap root</td><td>a.</td><td>Monstera</td></tr><tr><td>2.</td><td>Fibrous root system</td><td>b.</td><td>Wheat</td></tr><tr><td>3.</td><td>Adventitious root</td><td>c.</td><td>Mustard</td></tr></table></td><td><table><tr><td>1 - c</td></tr><tr><td>2 - b</td></tr><tr><td>3 - a</td></tr></table></td></tr>
<tr><td>9.</td><td>Which region of the root-tip have very small cells, thin walled and with dense protoplasm?</td><td>Meristematic zone</td></tr>
<tr><td>10.</td><td>Which part of root-tip is responsible for the growth of the root in length?</td><td>Region of elongation</td></tr>
<tr><td>11.</td><td>Region of elongation is present ___(a)___ to the region of meristem and region of maturation is present ___(b)___ to the region of elongation.</td><td>a - Proximal<br>b - Proximal</td></tr>
<tr><td>12.</td><td>Match the following root modifications :<br><table><tr><td>1.</td><td>For storage</td><td>a.</td><td>Maize</td></tr><tr><td>2.</td><td>Prop root</td><td>b.</td><td>Sugarcane</td></tr><tr><td>3.</td><td>Stilt root</td><td>c.</td><td>Sweet potato</td></tr><tr><td>4.</td><td>For respiration</td><td>d.</td><td>Turnip</td></tr><tr><td></td><td></td><td>e.</td><td>Banyan</td></tr><tr><td></td><td></td><td>f.</td><td>Rhizophora</td></tr></table></td><td><table><tr><td>1 - c, d</td></tr><tr><td>2 - e</td></tr><tr><td>3 - a, b</td></tr><tr><td>4 - f</td></tr></table></td></tr>
<tr><td>13.</td><td>Stilt roots arise from</td><td>Lower node of the stem</td></tr>
<tr><td>14.</td><td>Hanging structures which provide support to branches of banyan trees are called</td><td>Prop roots</td></tr>
<tr><td>15.</td><td>In Rhizophora, pneumatophores help in</td><td>Respiration</td></tr>
</table>

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<table>
<tr><td>16.</td><td>Shoot develops from the ___(a)___ of the embryo</td><td>a - plumule</td></tr>
<tr><td>17.</td><td>Part of the plant which bears nodes and internodes is known as</td><td>Stem</td></tr>
<tr><td>18.</td><td>Stem is generally ___(a)___ when young and later often becomes woody and dark ___(b)___.</td><td>a - Green<br>b - Brown</td></tr>
<tr><td>19.</td><td>Match the following :-
<table>
<tr><td>1.</td><td>Organ of perennation</td><td>a.</td><td>Opuntia</td></tr>
<tr><td>2.</td><td>Stem tendril</td><td>b.</td><td>Citrus</td></tr>
<tr><td>3.</td><td>Stem thorn</td><td>c.</td><td>Euphorbia</td></tr>
<tr><td>4.</td><td>Phylloclade</td><td>d.</td><td>Bougainvillea</td></tr>
<tr><td colspan="2"></td><td>e.</td><td>Zaminkand</td></tr>
<tr><td colspan="2"></td><td>f.</td><td>Cucumber</td></tr>
</table></td><td>1 - e<br>2 - f<br>3 - b, d<br>4 - a, c</td></tr>
<tr><td>20.</td><td>Stem modified into flattened structure in ___(a)___ or fleshy cylindrical green structure in ___(b)___ and perform photosynthesis.</td><td>a - Opuntia<br>b- Euphorbia/Casuarina</td></tr>
<tr><td>21.</td><td>In which plants each node bears a rosette of leaves and a tuft of roots?</td><td>Pistia and Eichhornia</td></tr>
<tr><td>22.</td><td>Which type of sub-aerial modification occur in Jasmine.</td><td>Stolon</td></tr>
<tr><td>23.</td><td>In banana, the growing lateral branch come out ..... giving rise to leafy shoots.</td><td>Obliquely upward</td></tr>
<tr><td>24.</td><td> 

Recognise a, b and c.</td><td>a - Lamina<br>b - Petiole<br>c - Stipule</td></tr>
<tr><td>25.</td><td>Leaves originate from ..... meristem.</td><td>Shoot apical</td></tr>
<tr><td>26.</td><td>Leaf develops at the ___(a)___ and bears a ___(b)___ in its axil.</td><td>a - Node, b - Bud</td></tr>
<tr><td>27.</td><td>The axillary bud later develops into a</td><td>Branch or flower</td></tr>
<tr><td>28.</td><td>The leaves are arranged in _____ manner on stem.</td><td>Acropetal</td></tr>
<tr><td>29.</td><td>Leaf may bear two lateral small leaf like structures which are known as.</td><td>Stipules</td></tr>
<tr><td>30.</td><td>Sheathing leaf base is found in monocots, true or false.</td><td>True</td></tr>
<tr><td>31.</td><td>In some ___(a)___ plants the leaf base may become swollen which is known as ___(b)___.</td><td>a - leguminous<br>b - pulvinus</td></tr>
<tr><td>32.</td><td>Vein provide ___(a)___ to the leaf blade and act as channels of ___(b)___ for water minerals and food materials</td><td>a - Rigidity<br>b - Transport</td></tr>
<tr><td>33.</td><td>The arrangement of veins and veinlets in the lamina of leaf is termed as _____.</td><td>Venation</td></tr>
<tr><td>34.</td><td>When the veinlets form a network, the venation is known as</td><td>Reticulate venation</td></tr>
<tr><td>35.</td><td>When the veins run parallel to each other within a lamina the venation is termed as</td><td>Parallel venation</td></tr>
</table>

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<table>
<tr><td>36.</td><td>In a leaf, when its lamina is entire or when incised, the incisions do not touch the midrib then this type of leaf is known as</td><td>Simple leaf</td></tr>
<tr><td>37.</td><td>When the incisions of leaf lamina reach upto the midrib the leaf is known as</td><td>Compound leaf</td></tr>
<tr><td>38.</td><td>A _____ is present in the axil of petiole in both simple and compound leaves but not in the axil of leaflets.</td><td>Bud</td></tr>
<tr><td>39.</td><td>Rachis is found in _____ leaf.</td><td>Pinnately compound</td></tr>
<tr><td>40.</td><td>In neem plant, which type of leaf is present</td><td>Pinnately compound leaf</td></tr>
<tr><td>41.</td><td>In palmately compound leaf, the leaflets are attached at the ___(a)___ as in silk cotton.</td><td>a - Common point<br>i.e. tip of Petiole</td></tr>
<tr><td>42.</td><td>Match the following :
<table>
<tr><td>1.</td><td>Alternate phyllotaxy</td><td>a.</td><td>Alstonia</td></tr>
<tr><td>2.</td><td>Opposite phyllotaxy</td><td>b.</td><td>Mustard, Chinarose</td></tr>
<tr><td>3.</td><td>Whorled phyllotaxy</td><td>c.</td><td>Guava, Calotropis</td></tr>
</table>
<table>
<tr><td>1 - b</td></tr>
<tr><td>2 - c</td></tr>
<tr><td>3 - a</td></tr>
</table></td></tr>
<tr><td>43.</td><td>Match the following :
<table>
<tr><td>1.</td><td>Leaf tendril</td><td>a.</td><td>Venus fly trap</td></tr>
<tr><td>2.</td><td>Leaf spine</td><td>b.</td><td>Pea</td></tr>
<tr><td>3.</td><td>Insectivorous plant</td><td>c.</td><td>Cactus</td></tr>
</table>
<table>
<tr><td>1 - b</td></tr>
<tr><td>2 - c</td></tr>
<tr><td>3 - a</td></tr>
</table></td></tr>
<tr><td>44.</td><td>Name the plant in which, the leaves are small and short lived and petiole modify in leafy structure and perform photosynthesis.</td><td>Australian acacia</td></tr>
<tr><td>45.</td><td>Flower is a modified</td><td>Shoot</td></tr>
<tr><td>46.</td><td>The arrangement of flower on floral axis is known as</td><td>Inflorescence</td></tr>
<tr><td>47.</td><td>In ___(a)___ inflorescence the main axis continuous to grow.<br>Whereas in __(b)__ inflorescence main axis terminates in a flower.</td><td>a - Racemose<br>b - Cymose</td></tr>
<tr><td>48.</td><td>In racemose type of inflorescence the flowers are arranged in</td><td>Acropetal manner</td></tr>
<tr><td>49.</td><td>In a typical flower, calyx, corolla, androecium and gynoecium are attached on</td><td>Thalamus</td></tr>
<tr><td>50.</td><td>Calyx and corolla are ___(a)___ whorls while androecium, and gynoecium are ___(b)___ whorls of organs.</td><td>a - Accessory<br>b - Reproductive<br>or essential</td></tr>
<tr><td>51.</td><td>When calyx and corolla are not distinct then they are known as</td><td>Perianth</td></tr>
<tr><td>52.</td><td>If a flower has both androecium and gynoecium, it is a _____ flower.</td><td>Bisexual</td></tr>
<tr><td>53.</td><td>If a flower having either only stamens or only carpels then it is a _____ flower.</td><td>Unisexual</td></tr>
</table>

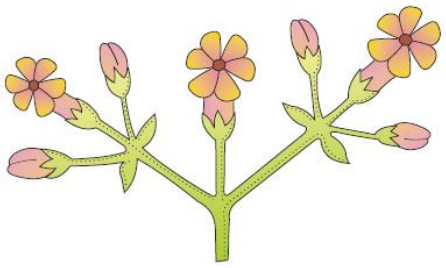
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**54.** Match the following :

| | | | |
|---|---|---|---|
| 1. | Actinomorphic flower | a. | *Mustard, Datura* |
| 2. | Zygomorphic flower | b. | Canna |
| 3. | Asymmetric flower | c. | Chilli |
| | | d. | *Cassia* |
| | | e. | Pea, Bean |

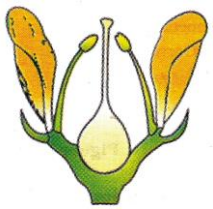
1 - a, c
2 - d, e
3 - b

**55.** Given figure represents which type of inflorescence.



Cymose inflorescence

**56.** Given figure represents which type of flower.



Perigynous flower

**57.** Match the following :-

| | | | |
|---|---|---|---|
| 1. | Hypogynous flower | a. | Cucumber |
| 2. | Perigynous flower | b. | Mustard |
| 3. | Epigynous flower | c. | Chinarose |
| | | d. | Plum |
| | | e. | Guava |

1 - b, c
2 - d
3 - a, e

**58.** In case of perigynous flower, the ovary is — Half inferior

**59.** When the sepals are united then condition is known as — Gamosepalous condition

**60.** When the petals are free then condition is known as — Polypetalous condition

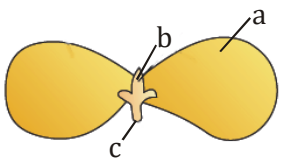
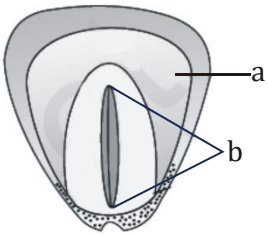
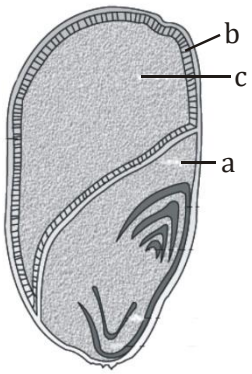
**61.** The mode of arrangement of sepals or petals in floral bud with respect to other members of the same whorl is — Aestivation

**62.** If the margins of sepals or petals overlap one another but not in any particular direction, then this type of aestivation is known as — Imbricate aestivation

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<table>
<tr><td>63.</td><td>Recognise the type of aestivation in a, b, c and d.<br>  <br>a b c d</td><td>a - Valvate<br>b - Twisted<br>c - Imbricate<br>d - Vexillary</td></tr>
<tr><td>64.</td><td>Match the following with respect to aestivation in petals –
<table>
<tr><td>1.</td><td>Valvate</td><td>a.</td><td>Cassia</td></tr>
<tr><td>2.</td><td>Twisted</td><td>b.</td><td>Pea</td></tr>
<tr><td>3.</td><td>Imbricate</td><td>c.</td><td>Calotropis</td></tr>
<tr><td>4.</td><td>Vexillary</td><td>d.</td><td>China rose</td></tr>
<tr><td></td><td></td><td>e.</td><td>Lady finger</td></tr>
<tr><td></td><td></td><td>f.</td><td>Gulmohar</td></tr>
</table></td><td>
<table>
<tr><td>1 - c</td></tr>
<tr><td>2 - d, e</td></tr>
<tr><td>3 - a, f</td></tr>
<tr><td>4 - b</td></tr>
</table></td></tr>
<tr><td>65.</td><td>The pollen grains are produced in (Pollensac / Ovule)</td><td>Pollen sac</td></tr>
<tr><td>66.</td><td>A sterile stamen is known as</td><td>Staminode</td></tr>
<tr><td>67.</td><td>When the stamens are attached to the petals, they are known as</td><td>Epipetalous</td></tr>
<tr><td>68.</td><td>Match the following :-
<table>
<tr><td>1.</td><td>Epitepalous stamen</td><td>a.</td><td>Citrus</td></tr>
<tr><td>2.</td><td>Monoadelphous stamen</td><td>b.</td><td>Pea</td></tr>
<tr><td>3.</td><td>Diadelphous stamen</td><td>c.</td><td>China rose</td></tr>
<tr><td>4.</td><td>Polyadelphous stamen</td><td>d.</td><td>Lily</td></tr>
</table></td><td>
<table>
<tr><td>1 - d</td></tr>
<tr><td>2 - c</td></tr>
<tr><td>3 - b</td></tr>
<tr><td>4 - a</td></tr>
</table></td></tr>
<tr><td>69.</td><td>In rose and lotus, the condition of carpels is (Apocarpous/Syncarpous)</td><td>Apocarpous</td></tr>
<tr><td>70.</td><td>After fertilisation, the ovules develop into ___(a)___ and the ovary matures into a ___(b)___.</td><td>a - Seeds<br>b – Fruit</td></tr>
<tr><td>71.</td><td>The arrangement of ovules within the ovary is known as</td><td>Placentation</td></tr>
<tr><td>72.</td><td>Ripened or mature ovary is known as</td><td>Fruit</td></tr>
<tr><td>73.</td><td>If a fruit is formed without fertilisation of ovary, it is called</td><td>Parthenocarpic fruit</td></tr>
<tr><td>74.</td><td>Which of the following is not correct ?<br>1. Axile placentation - Primrose<br>2. Parietal placentation - Mustard, Argemone<br>3. Free central placentation - Dianthus<br>4. Basal placentation - Sunflower, Marigold</td><td>1<br>(Note : Primrose has free central placentation)</td></tr>
<tr><td>75.</td><td>Ovary is one chambered but it becomes two chambered due to formation of the false septum in ___(a)___ and ___(b)___.</td><td>(a) Mustard<br>(b) Argemone</td></tr>
<tr><td>76.</td><td>Generally, the fruit is consist of ___(a)___ and ___(b)___.</td><td>a - Pericarp, b - Seeds</td></tr>
<tr><td>77.</td><td>In mango and coconut the fruit is known as</td><td>Drupe</td></tr>
<tr><td>78.</td><td>The drupe fruits are developed from ..... and superior ovary.</td><td>Monocarpellary gynoecium</td></tr>
</table>

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| | | |
|---|---|---|
| **79.** | The edible part of mango is | Mesocarp |
| **80.** | The mesocarp of coconut is | Fibrous |
| **81.** | A seed is generally made up of | Seed coat and an embryo |
| **82.** | The seed coat has two layers, the outer ___(a)___ and the inner ___(b)___. | a - Testa<br>b - Tegmen |
| **83.** |  <br>Recognise the a, b and c in the above diagram. | a - Cotyledon<br>b - Plumule<br>c - Radicle |
| **84.** | Orchid seed is (non endospermic or endospermic) | Non-endospermic |
| **85.** | In the seeds of cereal the seed coat and fruit wall are (Fused/Free) | Fused |
| **86.** | In maize, the outer covering of endosperm separates the embryo by a proteinous layer called | Aleurone layer |
| **87.** | In monocots, one large and shield shaped cotyledon is known as | Scutellum |
| **88.** | The plumule and radicle are enclosed in sheaths which are ___(a)___ and ___(b)___, respectively | a - Coleoptile<br>b - Coleorhiza |
| **89.** | Recognise a, b and c in the given diagram :-<br>  | a - Endosperm<br>b - Embryo |
| **90.** | Recognise a, b and c in the given diagram :-<br>  | a - Scutellum<br>b - Aleurone layer<br>c - Endosperm |
| **91.** | Floral formula of mustard is | $\oplus\,♂♀ K_{2+2} C_4\ A_{2+4}\ \underline{G}_{(2)}$ |
| **92.** | Papilionatae is a subfamily of | Leguminosae |
| **93.** | Which structure of petals encloses the stamen and pistil. | Keel |
| **94.** | Floral formula of Pea is | $\%\,♂♀ K_{(5)} C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$ |

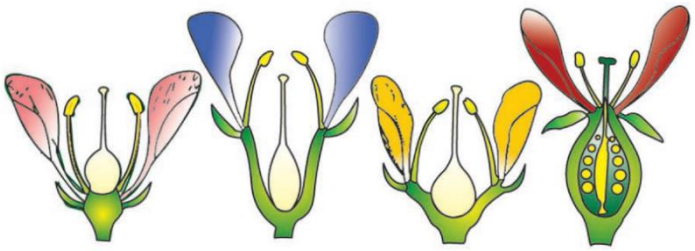
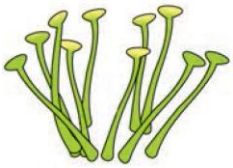
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| | | |
|---|---|---|
| **95.** | In fabaceae family, the gynoecium is monocarpellary, unilocular with many ovules, single style and superior ovary. (True/False) | True |
| **96.** | How many of the plants given below are comes under fabaceae family - Moong, Soyabean, *Indigofera*, Sunhemp, Sesbania, *Lupinus*, Mulaithi | All |
| **97.** | Solanaceae family commonly known as | Potato family |
| **98.** | The floral formula of *Petunia* is | $\oplus \male K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$ |
| **99.** | Persistent calyx is found in the members of (Solanaceae/Liliaceae) | Solanaceae |
| **100.** | How many of the plants given below are comes under solanaceae family :- Tobacco, Tomato, Chilli, Belladona, Ashwagandha, *Petunia*, Makoi | All |
| **101.** | Family Liliaceae is commonly called | Lily family |
| **102.** | Epitepalous condition is found in (Fabaceae/Liliaceae) | Liliaceae |
| **103.** | In the members of family Liliaceae the gynoecium is tricarpellary, syncarpous, trilocular with many ovules and superior ovary. (True/False) | True |
| **104.** | Floral formula of onion is | $Br \oplus \male \overset{\frown}{P_{(3+3)}} A_{3+3} \underline{G}_{(3)}$ |
| **105.** | How many of the plants given below are comes under family Liliaceae :- *Aloe*, *Asparagus*, Tulip, *Gloriosa*, *Colchicum* | All |
| **106.** | In vexillary aestivation, posterior petal is known as | Standard or Vexillum |
| **107.** | Underground modification of stem i.e. bulbs, corms and rhizomes comes under.... family. | Liliaceae |
| **108.** | Fruit of Liliaceae family is | Capsule, rarely berry |
| **109.** |  Rocognise the family which is related to above floral diagram. | Solanaceae family |
| **110.** | Match the following :- <table><tr><td>1.</td><td>Fabaceae</td><td>i.</td><td>Swollen placenta</td></tr><tr><td>2.</td><td>Solanaceae</td><td>ii.</td><td>Diadelphous stamens</td></tr><tr><td>3.</td><td>Liliaceae</td><td>iii.</td><td>Epitepalous condition</td></tr></table> | 1 - ii<br>2 - i<br>3 - iii |
| **111.** | The underground part of the flowering plant is the _____while the portion above the ground forms the _____. | Root system,<br>Shoot system. |
| **112.** | The region of the stem where leaves are born are called _____ while _____are the portions between two nodes. | Nodes and internodes |

| | | |
|---|---|---|
| **113.** | The leaf is attached to the stem by the leaf base and may bear two lateral small leaf like structures called_____. | Stipules |
| **114.** | In _____, the leaf base expands into a sheath covering the stem partially or wholly. | Monocots |
| **115.** | _____ provide rigidity to the leaf blade and act as channels of transport for water, minerals and food materials. | Veins |
| **116.** | A bud is not present in the axil of _____ of the compound leaf. | Leaflet |
| **117.** | Most of dicotyledonous plants leaves possess reticulate venation. | True |
| **118.** | In palmately compound leaves, the leaflets are attached at a common point, at the tip of petiole. | True |
| **119.** | Parallel venation is the characteristic of **some** monocotyledons. | False |
| **120.** | Phyllotaxy is the pattern of arrangement of leaflets on the stem or branch. | False |
| **121.** |  **(a)** **(b)** **(c)** Identify the pattern of arrangement of leaves on the stem in above given diagram and give one example of each phyllotaxy? | **(a)** – Opposite- e.g. Guava **(b)** – Whorled- e.g. *Alstonia* **(c)** – Alternate- e.g. China rose |
| **122.** |  Identify the above given diagram. | Racemose inflorescence |
| **123.** | A flower is a modified shoot wherein the shoot apical meristem changes to floral meristem. | True |
| **124.** | The outer covering of endosperm separates the embryo by a proteinous layer called _____ in monocot seed. | Aleurone layer |
| **125.** | When a flower can be divided into two equal radial halves in any radial plane passing through the centre, it is said to be _____. | Actinomorphic |

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| | | |
|---|---|---|
| **126.** | In *Canna* flower can be divided into two similar halves only in one particular vertical plane. | False |
| **127.** | Flowers with bracts-reduced leaf found at the base of the pedicel are called _____. | Bracteate |
| **128.** |  **(A) (B) (C) (D)** Which of the above given figure are related with rose, plum and peach? | B and C |
| **129.** | Identify given below diagram and what is the name of its unit.  | Androecium Unit - Stamen |
| **130.** | _____ are usually brightly coloured to attract insects for pollination. | Petals |
| **131** |  **(a) (b)** Identify the above given aestivation and given two examples of each? | **(a)** Twisted – e.g. China rose and lady's finger **(b)** Vexillary – e.g. Pea and Bean |
| **132.** | A _____ stamens is called staminode. | Sterile |
| **133.** | When stamens are attached to the petals, they are called as _____ and find in _____. | Epipetalous, Brinjal |
| **134.** |  In the above given placentation, initially ovary is one chambered but due to formation of false septum it becomes two chambered. What is the name of this false septum ? | Replum |
| **135.** | In seeds such as castor, the endosperm is formed as a result of _____. | Double fertilization |

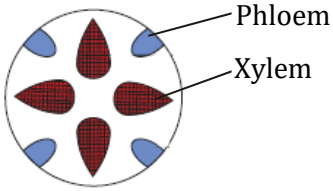
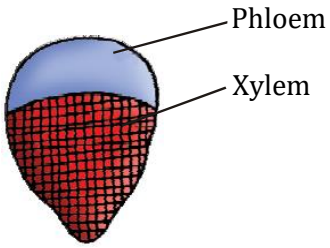
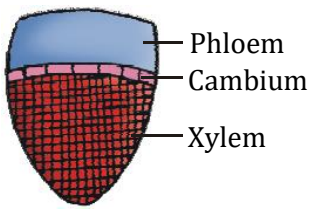
# Anatomy of Flowering Plants

| Q.No. | Question | Answer |
|---|---|---|
| 1. | Study of internal structure of plants is called _____. | Plant Anatomy |
| 2. | A group of cells having a common ___(A)___ and usually performing a common function, called ___(B)___. | (A) Origin<br>(B) Tissue |
| 3. | ___(A)___ in plants is largely restricted to specialised regions of active cell division, called ___(B)___. | (A) Growth<br>(B) Meristem |
| 4. | The term meristem is derived from ___(A)___ word meristos, which means ___(B)___. | (A) Greek<br>(B) Divided |
| 5. | The meristem which occurs at the tips of roots and shoots and produce primary tissues are called | Apical meristems |
| 6. | The meristem that occurs in the mature region of roots and shoots of many plants particularly those that produce woody axis and appear later than primary meristem is called. | Secondary meristem |
| 7. | The structure(s) which is derived from shoot apical meristem and present in axil (s) of leaf (leaves) and are capable of forming a branch or a flower, called .....(s). | Axillary bud |
| 8. | The meristem which occur in grasses between mature tissues and regenerate parts removed by grazing herbivores is known as_____. | Intercalary meristem |
| 9. | Both ___(A)___ and intercalary meristems are ___(B)___ meristems and contribute to the formation of the primary plant body. | (A) Apical meristem<br>(B) Primary |
| 10. | Which simple permanent tissue forms the major component with in plant *organs* ? | Parenchyma |
| 11. | Cells are generally isodiametric.<br>Cells may be spherical, oval, rounded, polygonal or elongated in shape.<br>The cell wall is thin and made up of *cellulose*.<br>The cells may either be closely packed or have intercellular spaces.<br>Above characters of cells related to which type of simple permanent tissue? | Parenchyma |
| 12. | Which simple tissue performs various functions like photosynthesis, secretion and storage ? | Parenchyma |
| 13. | Which tissue occurs as a homogenous layers or in patches below the epidermis in stem of dicotyledonous plants ? | Collenchyma |

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<table>
<tr><td>14.</td><td>__(A)__ consists of cells which are much thick ended at the corners due to the deposition of cellulose, hemicellulose and __(B)__.</td><td>A-Collenchyma<br>B-Pectin</td></tr>
<tr><td>15.</td><td>Which simple permanent tissue provides mechanical support to the growing parts of plant such as young stem and petiole of a leaf.</td><td>Collenchyma</td></tr>
<tr><td>16.</td><td>Which simple permanent tissue consists of long narrow cells with thick and lignified walls having a few or numerous pits.</td><td>Sclerenchyma</td></tr>
<tr><td>17.</td><td>Permanent tissues which are made up of more than one type of cells and these cells work together as a unit.</td><td>Complex tissues</td></tr>
<tr><td>18.</td><td>The presence of vessels in xylem is characteristics feature of __(A)__.</td><td>A-Angiosperms</td></tr>
<tr><td>19.</td><td>Which xylem element has highly thickened walls and obliterated central lumen?</td><td>Xylem fibre</td></tr>
<tr><td>20.</td><td>Which xylem element stores food material in the form of starch or fats and other substances like tannin?</td><td>Xylem Parenchyma</td></tr>
<tr><td>21.</td><td>Plant group in which phloem is composed of sieve tube elements, companion cells, phloem parenchyma & Phloem fibres.</td><td>Angiosperms</td></tr>
<tr><td>22.</td><td>Spermatophytes in which phloem have albuminous cells and sieve cells</td><td>Gymnosperms</td></tr>
<tr><td>23.</td><td>Phloem elements which are long, tube-like structures, arranged longitudinally and are associated with the companion cells.</td><td>Sieve tube elements.</td></tr>
<tr><td>24.</td><td>A mature sieve tube element possesses a peripheral cytoplasm and a large __(A)__ but lacks a __(B)__.</td><td>(A)-Vacuole<br>(B)-Nucleus</td></tr>
<tr><td>25.</td><td>The functions of __(A)__ are controlled by the __(B)__ of companion cells.</td><td>(A)-Sieve tubes<br>(B)-Nucleus.</td></tr>
<tr><td>26.</td><td>Match the column-I with column-II
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>(A)</td><td>Xylem ray parenchyma</td><td>(I)</td><td>Store food material and other substances like resin, latex & mucilage</td></tr>
<tr><td>(B)</td><td>Companion cells</td><td>(II)</td><td>Devoid of protoplasm at maturity</td></tr>
<tr><td>(C)</td><td>Phloem fibres</td><td>(III)</td><td>Radial conduction of water</td></tr>
<tr><td>(D)</td><td>Phloem parenchyma</td><td>(IV)</td><td>Help in maintaining the pressure gradient in the sieve tubes</td></tr>
</table></td><td>A-III, B-IV, C-II, D-I</td></tr>
</table>

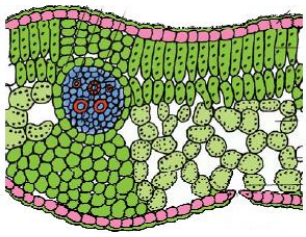
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|---|---|---|
| **27.** | On the basis of their ___(A)___ and ___(B)___, there are three types of tissue systems. | A-Structure/morphology<br>B-Location/Position |
| **28.** | Phloem fibre are made up of ___(A)___ cells. These are generally absent in the ___(B)___. | A-Sclerenchymatous<br>B-Primary Phloem |
| **29.** | Epidermal cells are ___(A)___ with a small amount of ___(B)___ and large vacuole. | A-Parenchymatous<br>B-Cytoplasm |
| **30.** | In dicots, guard cells are ___(A)___ shaped whereas in grasses guard cells are ___(B)___ shaped. | A-bean<br>B-dumb-bell |
| **31.** | The stomatal aperture, guard cells and the surrounding subsidiary cells are together called ___(A)___. | A-Stomatal apparatus |
| **32.** | The root hairs are ___(A)___ elongations of the ___(B)___ cells and help absorb water and minerals from the soil. | A-Unicellular<br>B-Epidermal |
| **33.** | All tissues except ___(A)___ and Vascular bundles constitute the ground tissue system. | A-Epidermis |
| **34.** | In ___(A)___., the ground tissue consists of thin walled chloroplast containing cells and is called ___(B)___. | A-Leaves<br>B-Mesophyll |
| **35.** | Given figure represents which type of vascular bundle ?<br>  | Radial |
| **36.** | Given figure represents which type of vascular bundle ?<br>  | Conjoint, collateral & closed |
| **37.** | Given figure represents which type of vascular bundle?<br>  | Conjoint, collateral & open |
| **38.** | Which type of vascular bundles are found in roots ? | Radial |

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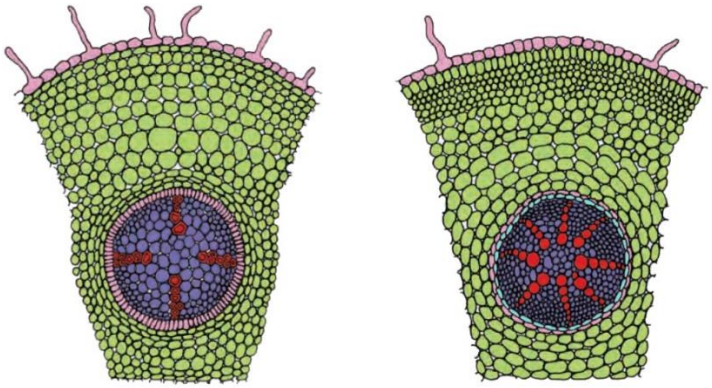
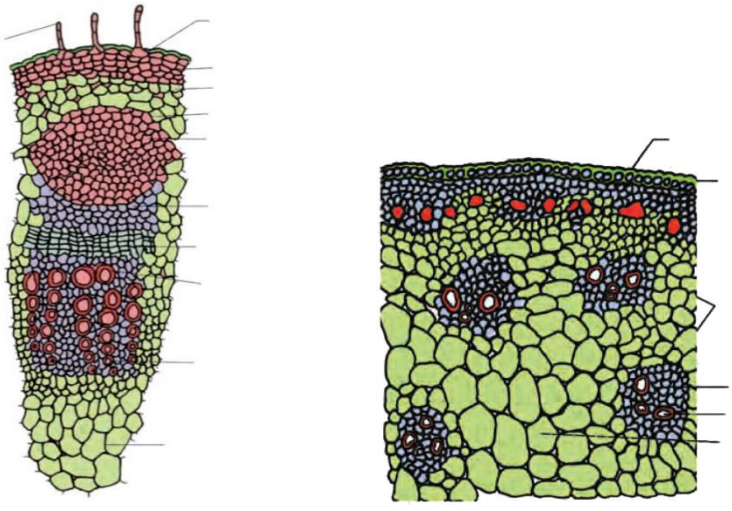
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|---|---|---|
| **39.** | Some of the terms are given below :-<br>**Pericycle, Pith, vascular bundles and medullary rays**<br>Which is not a part of stele in the primary internal structure of Dicot and Monocot roots but part of stele in the primary internal structure of Dicot stem. | Medullary rays |
| **40.** | The cross section of a plant material shows the following anatomical features under microscope:-<br>**(a)** usually more than six xylem bundles with exarch xylem<br>**(b)** pith is large and well developed<br>**(c)** Radial arrangement of xylem and phloem bundles.<br>The plant material should be :- | Monocotyledonous root |
| **41.** | In the internal structure of roots, casparian strips are present in :- | Endodermis |
| **42.** | The cross section of a plant material shows the following anatomical features under microscope.<br>(a) usually two to four xylem and phloem patches with exarch xylem.<br>(b) pith in small or inconspicuous.<br>(c) Radial arrangement of xylem & phloem bundle. The plant material should be. | Dicotyledonous root |
| **43.** | In flowering plants, lateral roots are endogenous in origin and originates from the cells of ...(A).... | A-Pericycle |
| **44.** | In dicotyledonous stem, which layer is also termed as starch sheath? | Endodermis |
| **45.** | List of some of the plant materials is given below :-<br>• Dicotyledonous root<br>• Monocotyledonous root<br>• Dicotyledonae stem<br>• Monocotyledonous stem<br>Which plant material have sclerenchymatous pericycle in patches, located just above phloem bundles in the primary internal structure. | Dicotyledonous stem |
| **46.** | List of some the plant materials is given below :-<br>• Dicotyledonous root<br>• Dicotyledonous stem<br>• Monocotyledonous root<br>Which plant material have bundle cap in the primary internal structure ? | Dicotyledonous stem |

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| | | |
|---|---|---|
| **47.** | In the internal structure of sunflower stem, in between the vascular bundles there are a few layers of radially placed ___(A)___ cells, which constitute ___(B)___. | A-Parenchymatous<br>B-Medullary rays |
| **48.** | Which type of vascular bundles are found in the primary internal structure of dicotyledonous stem (e.g. Sunflower stem) ? | Conjoint, collateral & open |
| **49.** | The cross section of a plant material shows the following anatomical features under microscope:-<br>(a) Sclerenchymatous bundle sheath<br>(b) Many vascular bundles are scattered in parenchymatous ground tissue, each vascular bundle is surrounded by sclerenchymatous bundle sheath<br>(c) Vascular bundles are usually oval shaped.<br>(d) Vascular bundles are conjoint, collateral & closed and xylem is endarch<br>(e) Phloem parenchyma is absent and water containing cavities are present within the vascular bundles<br>The plant material should be :- | Monocotyledonous stem |
| **50.** | Look at the given figure and find the position (Adaxial or abaxial) of xylem in the vascular bundle.<br>  | Adaxial |
| **51.** | In a dorsiventral leaf, palisade parenchyma lies towards ...(A)... epidermis and spongy parenchyma lies towards ...(B)... epidermis. | (A) Adaxial<br>(B) Abaxial |
| **52.** | In grasses, certain ___(A)___ epidermal cells along the veins modify themselves into large, empty, colourless cells. These cells are called ___(B)___ cells. | (A) Adaxial<br>(B) Bulliform |
| **53.** | The ___(A)___ venation in monocot leaves is reflected in the near similar sizes of vascular bundles (except in main veins) as seen in vertical section of the leaves. | (A) Parallel |

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| | | |
|---|---|---|
| **54.** | In a isobilateral leaf, two distinct patches of __(A)__ are present above and below of each large __(B)__ which extend towards upper and lower epidermis, respectively. | (A) Sclerenchyma<br>(B) Vascular bundle |
| **55.** | Name the lateral meristem which is responsible for the formation of secondary xylem, secondary phloem and secondary medullary rays. | Vascular cambium |
| **56.** | The protective layer that surrounds the vascular bundles in a dicot leaf is called as _____. | Bundle sheath |
| **57.** | The vascular bundle in dicot stems is typically arranged in a _____ pattern. | Conjoint |
| **58.** | The innermost layer of the root cortex, associated with the storage of food is called _____. | Endodermis |
| **59.** | On the stem the epidermal hairs are called _____ and these are usually _____ in the shoot system. | Trichomes, multicellular |
| **60.** | Secondary growth in plants is typically found in monocots. | False |
| **61.** | The Pericycle is a single layered tissue that gives rise to the lateral roots & cork cambium. | False |
| **62.** | Which tissues provides flexibility & mechanical support to young growing parts of the plant? | Collenchyma |
| **63.** | The primary function of trichomes in plants is to - | Protect against herbivores & excessive transpiration |
| **64.** | Sieve tube elements lack nuclei at maturity. | True |
| **65.** | Xylem parenchyma cells have thick secondary walls that provide structural support. | False |
| **66.** | What is the primary function of the endodermis in plant roots? | Regulation of solute movement into the stele |
| **67.** | Vessel elements are shorter and wider than tracheids, allowing for more efficient water conduction. | True |
| **68.** | Sieve tubes, which are responsible for translocation in plants, contain nucleus & other organelles. | False |
| **69.** | In monocot roots, pith is large and well developed. | True |

| | | |
|---|---|---|
| **70.** | The oval or round and loosely arranged spongy parenchyma is situated above the palisade cells and extends to the upper epidermis. | False |
| **71.** | The root hairs are usually unicellular whereas trichomes are multicellular. | True |
| **72.** | The parenchymatous cells which lie between the xylem & the phloem are called _____. | Conjunctive tissue |
| **73.** | All tissues on the inner side of the endodermis such as pericycle, vascular bundles and pith constitute the _____. | Stele |
| **74.** | These are usually more than _____ xylem bundles in the monocot root and is known as _____. | Six, Polyarch |
| **75.** | Mesophyll has two types of cells _____ and _____. | Palisade parenchyma, Spongy parenchyma |
| **76.** | In monocot stem, Peripheral vascular bundles are generally _____ than the centrally located ones. | Smaller |
| **77.** | Look at the given figures and identify:<br>  <br>**(a)** **(b)** | a–dicot root,<br>b–monocot root |
| **78.** | Look at the given figures and identify:<br>  <br>**(a)** **(b)** | a–dicot stem,<br>b–monocot stem |



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**79.** Match the column-I with column-II.

| | Column-I | | Column-II |
|---|---|---|---|
| (a) | Monocot stem | (i) | Few xylem bundles |
| (b) | Dicot stem | (ii) | Scattered vascular bundles |
| (c) | Monocot root | (iii) | Many xylem bundles |
| (d) | Dicot root | (iv) | Radical vascular bundles |

a-ii
b-iv
c-iii
d-i

**80.** Match the column-I with column-II.

| | Column-I | | Column-II |
|---|---|---|---|
| (a) | Cuticle | (i) | Guard cells |
| (b) | Bulliform cells | (ii) | Single layer |
| (c) | Stomata | (iii) | Waxy layer |
| (d) | Epidermis | (iv) | Empty colourless cell |

a-iii
b-iv
c-i
d-ii

**81.** Epiblema of roots is equivalent to -

Epidermis

**82.** A piece of wood having no vessels must be belong to:

Gymnosperm

**83.** Identify the plant part whose transverse section shows the following features:-

I. Innermost layer of cortex has casparian strips.

II. Pericycle has parenchymatous cells.

III. Radially arranged two to four xylem & phloem patches.

Dicot stem

**84.** Match the column-I with column-II.

| | Column-I | | Column-II |
|---|---|---|---|
| (a) | Xylem parenchyma | (i) | Absent in most of monocots |
| (b) | Vessel | (ii) | Fruit walls of nut |
| (c) | Phloem parenchyma | (iii) | Food & tannins |
| (d) | Sclereids | (iv) | Long cylindrical tube-like structure |

a-iii
b-iv
c-i
d-ii

**85.** Select incorrect statement with respect to the anatomy of monocot stem :–

(1) Parenchymatous ground tissue is extended from hypodermis to centre.

(2) Hypodermis is made of thick walled living mechanical tissue.

(3) Vascular bundles are conjoint, collateral & closed with endarch xylem.

(4) Pith is absent.

**(2)**

| | | |
|---|---|---|
| **86.** | Fibre (longest plant cell), belongs to which tissue? | Sclerenchyma |
| **87.** | Leaves of some grasses shows rolling & unrolling. It occurs due to - | Bulliform cells |
| **88.** | Gymnosperms are also called soft wood spermatophytes because they lack - | Xylem vessels |
| **89.** | Collenchyma occurs in the stem & petioles of - | Dicots |

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# Cell The Unit of Life

| Q.NO. | Question | Answer |
|---|---|---|
| 1. | Who discovered the first living cell ? | Anton Von leeuwenhoek |
| 2. | Who formulated the cell theory ? | Schleiden and Schwann |
| 3. | Who gave the cell theory a final shape ? | Rudolf Virchow |
| 4. | "Omnis cellula -e- cellula" phrase was given by :- | Rudolf Virchow |
| 5. | Main arena of various cellular activities of the cell is :- | Cytoplasm |
| 6. | Which non-membranous organelle is found in both prokaryotic and eukaryotic cell ? | Ribosome |
| 7. | Who first explained that cells divide and new cells are formed from pre-existing cells ? | Rudolf Virchow |
| 8. | Which non-membranous cell organelle is found in animal cell and helps in cell division ? | Centriole |
| 9. | Who is responsible for extensive compartmentalisation of cytoplasm in eukaryotic cell ? | Membrane bound cell organelles |
| | **Fill in the Blanks** | **Answer** |
| 1. | Some organisms are composed of a single cell and are called "A" organisms while other organisms are composed of many cells are called "B" organisms. | A - unicellular,<br>B - multicellular |
| 2. | _____, a British Zoologist, studied different types of animal cells and reported that cells had a thin outer layer which is today known as the plasma membrane. | Theodore Schwann (1839) |
| 3. | Cell that has membrane bound nucleus is called "A" whereas cell that lack membrane bound nucleus is called "B". | A-eukaryotic,<br>B-prokaryotic |
| 4. | The largest isolated single cell is the _____. | egg of an Ostrich |
| | **True/False** | **Answer** |
| 1. | Anything less than a complete structure of a cell does not ensure independent living | True |
| 2. | Cell is the fundamental structural and functional unit of living organism | True |
| 3. | Various chemical reactions occur in the cytoplasm to keep the cell in the living state | True |
| 4. | All living organisms have a cell wall surrounding the cell membrane | False (absent in Animals) |
| 5. | All eukaryotic cells have similar shape | False<br>(Vary in shape and size) |
| 6. | Shape of the cell may vary with the function they perform | True |

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| CELL WALL | | |
|---|---|---|
| **Q.NO.** | **Question** | **Answer** |
| **1.** | Middle lamella is mainly composed of :- | Ca-Pectate (some Mg pectate also) |
| **2.** | In plants other than algae, cell wall is composed of :- | Cellulose, Hemi cellulose, Pectin, Protein |
| **3.** | In algae cell wall is composed of :- | Cellulose, galactans, mannan, $CaCO_3$ |
| | **Fill in the blanks** | **Answer** |
| **1.** | The cell wall and middle lamella may be traversed by _____ which connect the cytoplasm of neighbouring cells. | plasmodesmata |
| **2.** | The cell wall of a young plant cell, the "A" is capable of growth, which gradually diminish as the cell matures and the "B" is formed on the inner side of the cell. | A- primary wall, B- secondary wall |
| | **True/False** | **Answer** |
| **1.** | Cell wall not only gives shape to the cell, but it protect the cell from mechanical damage & infection | True |
| **2.** | As cell mature the secondary wall is formed on the inner (toward membrane ) side of the cell | True |
| **3.** | Middle lamella holds or glues the different neighbouring animal cells together | False (plant cells) |
| **4.** | Cell wall is a living quasifluid structure forms an outer covering for the plasma membrane of fungi and plants | False (dead and rigid) |
| **5.** | All eukaryotic cells are not identical | True |

| CELL MEMBRANE | | |
|---|---|---|
| **Q.NO.** | **Question** | **Answer** |
| **1.** | Fluid mosaic model was proposed by :- | S.J. Singer & G.L. Nicolson |
| **2.** | $Na^+/K^+$ pump is an example of :- | Active transport |
| **3.** | The ability of protein to move within the membrane is measured as its :- | Fluidity |
| **4.** | Movement of water across the membrane by diffusion is :- | Osmosis |
| **5.** | The detailed structural study of cell membrane was possible after the year:- | 1950 |
| **6.** | The cell membrane is composed of lipids that are arranged in :- | Bilayer |
| **7.** | In the membrane of human RBC the percentage of lipid is approximately. | 40% |
| **8.** | The lipid component of cell membrane mainly consists of :- | Phosphoglycerides |
| **9.** | Cell membrane proteins can classified as integral or peripheral on the basis of :- | Ease of extraction |

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| | Fill in the blanks | Answer |
|---|---|---|
| **1.** | According to **'Fluid Mosaic Model'**, the quasi-fluid nature of ___a___ enables the lateral movement of ___b___ within the bilayer. | a-lipids, b-proteins |
| **2.** | Transport of molecules across the membrane without any requirement of energy is called _____. | passive transport |
| **3.** | Neutral solutes can move across the membrane by the process of _____. | Simple diffusion |
| **4.** | As ___a___ can not pass through the non polar lipid bilayer, they require __b__ to facilitate their transport across the membrane. | a-polar molecules, b-carrier proteins |
| | **TRUE/FALSE** | **Answer** |
| **1.** | The fluid nature of cell membrane is helpful in functions like :-<br>(A) Formation of intercellular junctions<br>(B) Cell growth<br>(C) Cell division<br>(D) DNA replication<br>(E) Secretion<br>(F) Ribosome synthesis | <br>True<br>True<br>True<br>False<br>True<br>False |
| **2.** | The ratio of proteins and lipids is same in all cell membranes | False |
| **3.** | The integral proteins are partially or totally burried in the membrane | True |
| **4.** | The lipids are arranged in the membrane with the polar heads towards the inner side | False |

| **ER AND GOLGI BODY** | | |
|---|---|---|
| **Q.NO.** | **Question** | **Answer** |
| **1.** | The network of tiny tubular structures scattered in the cytoplasm is called :- | ER |
| **2.** | The endoplasmic reticulum bearing ribosomes on its surface is called :- | RER |
| **3.** | In 1898 densely stained reticular structure near the nucleus in the nerve cell of owl was first observed by :- | Camillo Golgi |
| **4.** | Which cell organelle is the major site for synthesis of lipid ? | SER |
| **5.** | Which cell organelle is the important site for synthesis of glycoprotein and glycolipid ? | Golgi apparatus |
| **6.** | In animal cells lipid like steroid hormones are synthesised by which cell organelle ? | SER |
| **7.** | Diameter of cisternae of Golgi apparatus is :- | 0.5 µm - 1 µm |
| **8.** | Which cellular structures are included in endomembrane system ? | ER, Golgi body, lysosome, vacuole |

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| | Fill in the blanks | Answer |
|---|---|---|
| **1.** | Materials to be packaged in the form of vesicles from the ER fuse with "A" face of Golgi apparatus and move towards the "B" face. | A-cis (forming/convex), B-trans (maturing/concave) |
| **2.** | In the absence of ribosomes ER appear smooth and are called _____. | SER |
| | **True/False** | **Answer** |
| **1.** | The Golgi apparatus remain in close association with the endoplasmic reticulum | True |
| **2.** | RER is extensive and continuous with the inner membrane of nucleus | False (Outer nuclear membrane) |
| **3.** | ER divides the intracellular space into luminal and extra luminal part | True |
| **4.** | Varied number of cisternae are present in Golgi apparatus | True |
| **5.** | Cis and trans face of Golgi apparatus are completely same | False (Entirely different) |
| **6.** | Golgi apparatus principally perform the function of packaging of materials to be delivered either to the intracellular targets or secreted outside the cell | True |
| **7.** | Chloroplast is the part of endomembrane system because its function is same as the mitochondria | False |
| **8.** | ER, Golgibody, lysosome and vacuole is included in endomembrane system because their function are coordinated. | True |

| **LYSOSOME/VACUOLE** | | |
|---|---|---|
| **Q.NO.** | **Questions** | **Answer** |
| **1.** | Which cell organelle is formed by ER and Golgi apparatus and has hydrolytic enzymes. | Lysosome |
| **2.** | Which cell structure stores water, sap, excretory product and other materials not useful for the cell ? | Vacuole |
| **3.** | Which membrane is responsible for setting up higher concentration of ions in the vacuole as compare to cytoplasm. | Tonoplast |
| **4.** | Which structure is responsible for excretion in Amoeba ? | Contractile vacuole |
| | **True/False** | **Answer** |
| **1.** | Lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes | True |
| **2.** | Hydrolytic enzymes present in lysosomes are optimally active at basic pH | False (Acidic pH 5) |
| **3.** | Hydrolytic enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids | True |

| | | |
|---|---|---|
| **4.** | In plant cells the vacuoles can occupy upto 90% of the volume of the cell | True |
| **5.** | In many cells, as in protists, food vacuoles are formed by engulfing the food particles | True |

| **MITOCHONDRIA AND CHLOROPLAST** | | |
|---|---|---|
| **Q.NO.** | **Questions** | **Answer** |
| **1.** | The inner compartment of mitochondria is called :- | matrix |
| **2.** | Which organelle perform aerobic respiration in cell ? | Mitochondria |
| **3.** | Which type of leucoplast store oil and fat ? | Elaioplast |
| **4.** | Which membrane of chloroplast is more permeable ? | Outer membrane |
| **5.** | Arrangement of thylakoids like piles of coin is known as :- | Grana |
| **6.** | Which type of ribosomes are found in mitochondria ? | 70S |
| | **Fill in the blanks** | **Answer** |
| **1.** | The number of mitochondria per cell is variable depending on the _____ activity of cell. | physiological |
| **2.** | Mitochondria is typically _____ shaped. | sausage/cylindrical |
| **3.** | Mitochondrial membranes divide its lumen into two _____ compartments. | Aqueous |
| **4.** | Colour giving pigments are present in _____ type of plastid. | Chromoplast |
| **5.** | Chloroplast contain ____ type of ribosomes | 70S |
| **6.** | Folding's of mitochondrial inner membrane are called __a__ present towards __b__. | a-cristae, b-matrix |
| **7.** | Mitochondria divide by _____. | Fission |
| **8.** | Mitochondria's diameter is __a__ and length is __b__. | a - 0.2-1.0 μm (average 0.5 μm), b - 1.0-4.1 μm |
| **9.** | In chromoplast fat soluble _____ pigments are present. | Carotenoids |
| | **True/False** | **Answer** |
| **1.** | Mitochondria store food, waste materials and minerals | False |
| **2.** | Mitochondria does not produce ATP | False |
| **3.** | Mitochondria is a single membrane bound structure | False (double) |
| **4.** | The stroma lamellae connect the thylakoid of different grana | True |
| **5.** | The space limited by the inner membrane of chloroplast is called stroma | True |
| **6.** | Mitochondria are the site of both aerobic and anaerobic respiration | False (only aerobic) |
| **7.** | Plastids are found in plants | True |
| **8.** | Carotene and chlorophyll are fat soluble pigments | True |
| **9.** | Aleuroplast stores protein | True |
| **10.** | Chlamydomonas contains 20–40 chloroplasts in a cell | False (one per cell) |

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| RIBOSOME | | |
|---|---|---|
| **Q.NO.** | **Questions** | **Answer** |
| **1.** | Ribosomes are the granular structures first observed under the electron microscope as dense particles by :- | George Palade (1953) |
| **2.** | Which type of ribosomes are present in prokaryotic cells ? | 70S |
| **3.** | What is the function of ribosomes ? | Protein synthesis |
| **4.** | What is the size of prokaryotic ribosomes ? | 15 nm to 20 nm. |
| **5.** | Several ribosomes may attach to a single mRNA and form a chain, this chain is called :- | polyribosome or polysome |
| **6.** | What are the two sub units of prokaryotic ribosomes? | 50S and 30S |
| **7.** | Ribosomes are composed of which components? | r-RNA and protein |
| | **True/False** | **Answer** |
| **1.** | Sedimentation coefficient is an indirect measurement of density and size | True |
| **2.** | Microbodies are membrane bound vesicles, that contain various enzymes, are present in both plant and animal cells | True |

| CILIA AND FLAGELLA, CENTRIOLE | | |
|---|---|---|
| **Q.NO.** | **Questions** | **Answer** |
| **1.** | A network of filamentous proteinaceous structures present in the cytoplasm is called :- | Cytoskeleton |
| **2.** | Which cell organelle is surrounded by amorphous pericentriolar material ? | Centriole |
| **3.** | Cilia and flagella are emerge from centriole - like structure called:- | basal bodies |
| **4.** | Which small structures of cell work like oars ? | Cilia |
| **5.** | How many radial spokes are present in internal structure of cilia? | 9 |
| **6.** | In axoneme of cilia how many radially arranged peripheral doublet of microtubules are present ? | 9 |
| **7.** | Which cellular structure organisation is like the cartwheel ? | Centriole |
| **8.** | Which cellular structure is responsible for rising of spindle apparatus during cell division in animal cell ? | Centriole |
| **9.** | In centriole radial spoke is made of :- | Protein |
| **10.** | Arrangement of axonemal microtubules in cilia or flagella is :- | 9 + 2 |
| | **Fill in the blanks** | **Answer** |
| **1.** | In cilia the central tubules are connected by __A__ and is also enclosed by a __B__, which is connected to one of tubule of each peripheral doublet by a __C__. | A - central bridge<br>B - central sheath<br>C - radial spoke |
| **2.** | Both the centrioles in a centrosomes lie _____ to each other. | Perpendicular |
| **3.** | In centriole each peripheral fibril is a _____. | Triplet |

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| | | |
|---|---|---|
| **4.** | The central part of the proximal region of the centriole is proteinaceous and called the "A", which is connected with tubules of the peripheral triplet by "B". | A-hub, B-radial spokes |
| | **True/False** | **Answer** |
| **1.** | Eukaryotic flagellum is comparatively smaller than cilia | False |
| **2.** | Bacteria possess flagella but these are structurally different from that of the eukaryotic flagella | True |
| **3.** | The cytoskeleton in a cell is involved in mechanical support and motility | True |
| **4.** | In cilia, peripheral doublets are interconnected by radial spoke | False (Linkers) |
| **5.** | The electron microscopic study of a eukaryotic cilium shows that they are covered with plasma membrane | True |
| **6.** | Centriole is made of nine unevenly spaced peripheral fibrils of tubuline protein | False (evenly spaced) |

| **NUCLEUS** | | |
|---|---|---|
| **Q.NO.** | **Question** | **Answer** |
| **1.** | The interphase has highly extended and elaborate nucleoprotein fibres called :- | Chromatin |
| **2.** | What is the name of basic protein present in chromatin ? | Histone |
| **3.** | What is the approximate length of DNA in a single human cell ? | 2.2 metre |
| **4.** | Which constriction is essentially present in every chromosome ? | Primary constriction |
| **5.** | Other name of primary constriction is | Centromere |
| **6.** | Sometimes a few chromosomes have non staining secondary constriction at a constant location, this gives the appearance of a small fragment called the :- | Satellite |
| **7.** | Nucleus was first described by :- | Robert Brown |
| **8.** | Chromatin name was given by :- | Flemming |
| **9.** | Space between the two parallel membranes of nucleus is called :- | perinuclear space |
| **10.** | The outer membrane of nucleus usually remains continuous with which cell organelle ? | ER |
| **11.** | Nucleolus is a main site for synthesis of :- | ribosomal RNA |
| **12.** | Give the name of two mature eukaryotic cells that lack nucleus :- | (i) RBC (Erythrocytes) of many mammals<br>(ii) Sieve tube cells |
| **13.** | Diameter of space between two parallel membranes of nucleus is:- | 10 to 50 nm |
| **14.** | The spherical structure present in the nucleoplasm is :- | Nucleolus |
| **15.** | Metacentric, submetacentric, telocentric and acrocentric chromosomes are classified on the basis of :- | position of centromere |

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| | | |
|---|---|---|
| **16.** | Name the type of chromosome which has a terminal centromere :- | Telocentric |
| **17.** | In which chromosome the centromere is situated close to its end forming one extremely short and one very long arm ? | Acrocentric |
| **18.** | In which chromosome, centromere is situated middle of chromosome and forming two equal arms ? | Metacentric |
| | **True/False** | **Answer** |
| **1.** | Nucleus controls the activities of organelles and play a major role in heredity | True |
| **2.** | Material of the nucleus is stained by only acidic dye | False |
| **3.** | Nuclear matrix contains chromatin and one or more spherical bodies called nucleoli | True |
| **4.** | Nuclear pores are the passage through which movement of RNA and protein molecules take place in only one direction | False (Both direction) |
| **5.** | The inner membrane of nucleus usually remain continuous with ER and also bears ribosomes on it | False |
| **6.** | Each and every cell of human body contain nucleus | False |
| **7.** | The content of nucleolus is not continuous with the rest of nucleoplasm because it is bound by single membrane | False |
| **8.** | Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis | True |
| **9.** | Some mature eukaryotic cells lack nucleus but yet these cells consider as a living | True |
| **10.** | In submetacentric chromosomes centromere is situated slightly away from the middle of the chromosome | True |
| **11.** | Secondary constriction essentially present in every chromosome | False |
| **12.** | Chromatin contain DNA, some acidic protein called histones some non-histone protein and also RNA. | False (histones = basic proteins) |
| **13.** | Nucleolus is a main site for ribosomal protein synthesis | False |
| | **Fill in the blanks** | **Answer** |
| **1.** | At a number of places the nuclear envelope is interrupted by minute A, which are formed by the B of its two membranes. | A-pores, B-fusion |
| **2.** | During different stages of cell division, cells show structured _____ in place of the nucleus. | Chromosome |
| **3.** | Every chromosome essentially has a "A" on the sides of which disc shaped structures called "B" are present. | A-primary constriction B-kinetochore |

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<table>
<tr><th></th><th>Match the columns</th><th>Answer</th></tr>
<tr><td>1.</td><td>Match the Column-I with Column-II.
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A</td><td>Smallest cell</td><td>i</td><td>Nerve cell</td></tr>
<tr><td>B</td><td>Longest cell</td><td>ii</td><td>Egg of an Ostrich</td></tr>
<tr><td>C</td><td>Largest isolated single cell</td><td>iii</td><td>Mycoplasma</td></tr>
</table></td><td>A-iii, B-i, C-ii</td></tr>
<tr><td>2.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A</td><td>Schleiden & Schwann</td><td>i</td><td>Fluid mosaic model</td></tr>
<tr><td>B</td><td>Anton Von Leeuwenhoek</td><td>ii</td><td>Golgi body</td></tr>
<tr><td>C</td><td>Singer & Nicolson</td><td>iii</td><td>Cell theory</td></tr>
<tr><td>D</td><td>Robert Brown</td><td>iv</td><td>Ribosome</td></tr>
<tr><td>E</td><td>George Palade</td><td>v</td><td>First living cell</td></tr>
<tr><td>F</td><td>Camillo Golgi</td><td>vi</td><td>Nucleus</td></tr>
</table></td><td>A-iii, B-v, C-i,<br>D-vi, E-iv, F-ii</td></tr>
<tr><td>3.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A</td><td>Plasmodesmata</td><td>i</td><td>Holds the different neighbouring cells together</td></tr>
<tr><td>B</td><td>Middle lamella</td><td>ii</td><td>Connect the cytoplasm of neighbouring cells.</td></tr>
<tr><td>C</td><td>Primary cell wall</td><td>iii</td><td>Diminishes as the cell mature</td></tr>
</table></td><td>A-ii, B-i, C-iii</td></tr>
<tr><td>4.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A</td><td>Vacuole</td><td>i</td><td>Protein synthesis and secretion</td></tr>
<tr><td>B</td><td>Lysosome</td><td>ii</td><td>Packaging of materials</td></tr>
<tr><td>C</td><td>Golgi body</td><td>iii</td><td>Digestion</td></tr>
<tr><td>D</td><td>SER</td><td>iv</td><td>Storage of water</td></tr>
<tr><td>E</td><td>RER</td><td>v</td><td>Synthesis of steroidal hormones</td></tr>
</table></td><td>A-iv, B-iii, C-ii, D-v, E-i</td></tr>
<tr><td>5.</td><td>Consider the cell membrane and match the column-I with column-II.
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A.</td><td>Lipid</td><td>i.</td><td>52%</td></tr>
<tr><td>B.</td><td>Protein</td><td>ii.</td><td>40%</td></tr>
<tr><td>C.</td><td>Fluid mosaic model</td><td>iii.</td><td>$Na^+/K^+$ pump</td></tr>
<tr><td>D.</td><td>Active transport</td><td>iv.</td><td>1972</td></tr>
</table></td><td>A - ii, B - i, C - iv, D - iii</td></tr>
<tr><td>6.</td><td>
<table>
<tr><th></th><th>Column-I</th><th></th><th>Column-II</th></tr>
<tr><td>A.</td><td>Chromoplast</td><td>i.</td><td>Svedberg unit</td></tr>
<tr><td>B.</td><td>Ribosome</td><td>ii.</td><td>Carotenoids</td></tr>
<tr><td>C.</td><td>Food vacuole</td><td>iii.</td><td>Important for excretion</td></tr>
<tr><td>D.</td><td>Contractile vacuole</td><td>iv.</td><td>Formed by engulfing the food particles</td></tr>
</table></td><td>A-ii, B-i, C-iv, D-iii</td></tr>
</table>

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**7.**

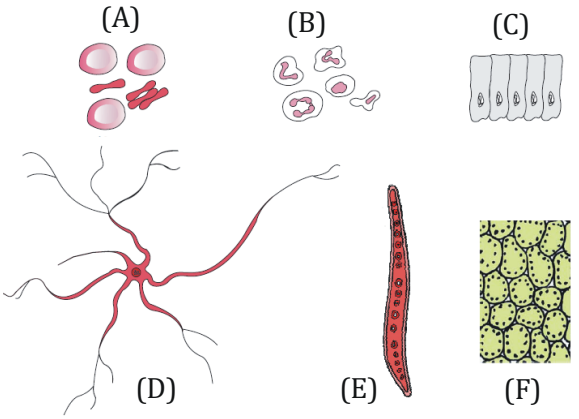
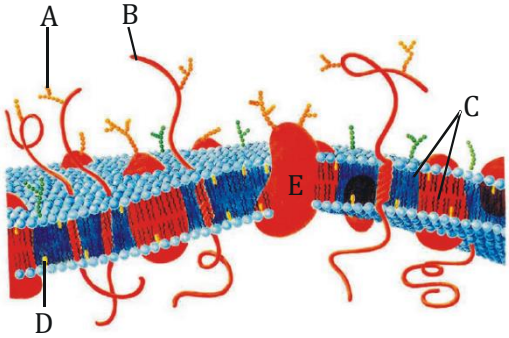
| | Column-I | | Column-II |
|---|---|---|---|
| A. | Power house of cell | i. | Chlorophyll |
| B. | Photosynthesis | ii. | Starch |
| C. | Amyloplast | iii. | Proteins |
| D. | Aleuroplast | iv. | Mitochondria |

A-iv, B-i, C-ii, D-iii

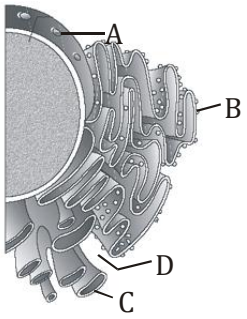
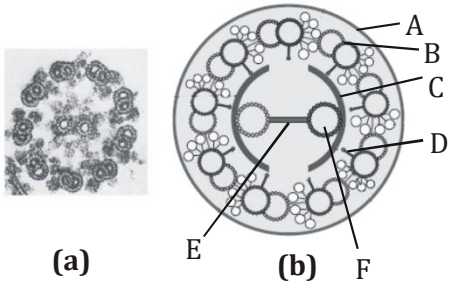
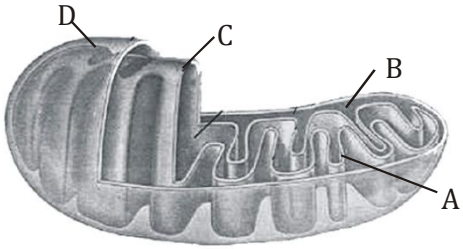
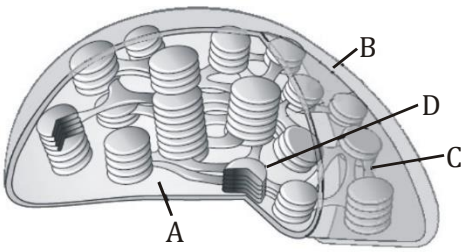
**8.**

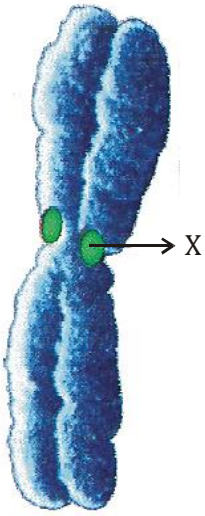
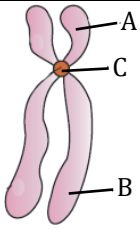
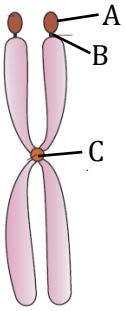
| | Column-I | | Column-II |
|---|---|---|---|
| A. | Chromatin | i | Flat membranous sacs in stroma. |
| B. | Cisternae | ii | Infoldings in mitochondria |
| C. | Thylakoids | iii | Disc shaped sacs in Golgi apparatus |
| D. | Cristae | iv | Highly extended and elaborate nucleoprotein fibres |

A-iv, B-iii, C-i, D-ii

| | **Diagram Based Questions** | **Answer** |
|---|---|---|
| **1.** | Identify the following cells :-<br>  <br> | (A) Red Blood cells<br>(B) White blood cells<br>(C) Columner epithelial cells<br>(D) Nerve cell<br>(E) Tracheid<br>(F) Mesophyll cells |
| **2.** | Given below is the diagram of plasma membrane. Identify the parts labelled A, B, C, D & E.<br>  <br> | A - Sugar<br>B - Protein<br>C - Lipid bilayer<br>D - Cholesterol<br>E - Integral Protein |

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| | | |
|---|---|---|
| **3.** | From the given figure identify the A, B, C and D.<br>  <br> | A - Nuclear pore<br>B - Ribosome (RER)<br>C - SER<br>D – Cytoplasm |
| **4.** | From the given figure identify the A, B, C, D, E and F.<br>  <br> | A - Plasma m embrace<br>B - Peripheral microtubule<br>C - Central sheath<br>D - Radial spoke<br>E - Central bridge<br>F - Central microtubule |
| **5.** | From the given figure identify A, B, C and D.<br>  <br> | A - Cristae<br>B - Intermembrane space<br>C - Inner membrane<br>D - Outer membrane |
| **6.** | From the given figure identify A, B, C and D.<br>  <br> | A - Stroma<br>B - Outer membrane<br>C - Stroma lamella<br>D - Thylakoid |

| | | |
|---|---|---|
| 7. | Identify the parts labelled in given diagrams :-<br>(A)  | X – Kinetochore |
| | (B)  | A → Short arm,<br>B → Long arm,<br>C → Centromere |
| | (C)  | A → Satellite<br>B → Secondary Constriction<br>C → Centromere |

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# Cell Cycle and Cell Division

| Q.No. | Questions | Answer |
|---|---|---|
| 1. | Duration of cell cycle in yeast :- | 90 minutes |
| 2. | The phase when the actual cell division occurs: - | M-phase |
| 3. | The phase between two successive M-phase: - | Interphase |
| 4. | The phase corresponding to the interval between mitosis and initiation of DNA replication: - | $G_1$ phase |
| 5. | The phase which marks the period during which DNA synthesis or replication takes place: - | S-phase |
| 6. | The cells that do not divide further exit $G_1$ phase and enter in an inactive stage called: - | Quiescent / $G_0$ stage |
| 7. | In plant, mitotic cell division is seen in the: - | Diploid cells and haploid cells |
| | **Fill in the blanks** | **Answer** |
| 1. | _____ of cell cycle can vary from organism to organism and also from cell type to cell type. | Duration |
| 2. | The interphase lasts more than _____ of the duration of cell cycle: | 95% |
| 3. | The M-phase starts with __(a)__ corresponding to separation of daughter chromosomes and usually ends with division of cytoplasm called __(b)__ | a- Karyokinesis, b- cytokinesis |
| 4. | The interphase is the time during which the cell is preparing for division by undergoing both _(a)_ and _(b)_ in an orderly manner: - | a- cell growth, b-DNA Replication |
| 5. | If the initial amount of DNA is denoted as 2C then it increases to ____ in S-phase :- | 4C |
| 6. | If the cell had diploid or 2n number of chromosomes at $G_1$ then after S-phase the number of chromosomes will be ____ | Same (2n) |
| 7. | In animal cells, during S-phase _(a)_ begins in the nucleus and _(b)_ duplicates in the cytoplasm | a-DNA Replication, b-Centriole |
| 8. | Cells in $G_0$ stage remain ____ but no longer proliferate unless called on to do so | metabolically active |
| | **True or False** | **Answer** |
| 1. | Average duration of cell cycle of a human is 24-hour cell, cell division proper lasts for only about an hour. | True |
| 2. | Cell growth is a continuous process while DNA replication occurs only during one specific stage in the cell cycle. | True |
| 3. | During prophase, proteins are synthesized in preparation for mitosis. | False |

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| Mitosis | | |
|---|---|---|
| | **Questions** | **Answer** |
| **1.** | Most dramatic period of cell cycle is - | M-phase |
| **2.** | Which stage of mitosis marked by initiation of condensation of chromosomal materials? | Prophase |
| **3.** | In which stage of mitosis morphology of chromosome is most easily studied? | Metaphase |
| **4.** | The stage of mitosis in which splitting of centromeres occurs - | Anaphase |
| **5.** | The stage in which nuclear envelope and nucleolus reforms - | Telophase |
| **6.** | What is the precurser of middle lamella? | Cell plate |
| **7.** | The type of division responsible for the growth of multicellular organisms - | Mitosis |
| **8.** | At which phase of mitosis, chromatids move to opposite poles | Anaphase |
| **9.** | Give a suitable example of multinucleate condition (syncytium) in which karyokinesis is not followed by cytokinesis? | Liquid endosperm in coconut |
| | **Fill in the blanks** | **Answer** |
| **1.** | The number of chromosomes in the parent and progeny cells are same in _____ division. | Mitosis |
| **2.** | Chromosomes are moved to spindle equator in _____ stage. | Metaphase |
| **3.** | Two sister chromatids are held together by the _____ of the chromosome. | Centromere |
| **4.** | Each chromosome move away from the equatorial plate in ____ stage. | Anaphase |
| **5.** | Mitosis is usually restricted to the _____ cells in animals. | Diploid |
| **6.** | Mitosis is also known as _____. | Equational division |
| **7.** | (a) attach to (b) of chromosomes in (c) stage. | (a) Spindle fibres<br>(b) Kinetochores<br>(c) Metaphase |
| **8.** | Nucleolus, Golgi body and endoplasmic reticulum disappear during (a) stage and reform during (b) stage. | (a) Prophase<br>(b) Telophase |
| | **True or False** | **Answer** |
| **1.** | Mitosis is also called as reductional division. | False |
| **2.** | Centriole begins to move towards opposite poles of the cell in anaphase. | False |
| **3.** | Condensation of chromosomes is completed in metaphase. | True |
| **4.** | Small disc shaped structures at the surface of centromeres are called Chiasmata. | False |
| **5.** | Each chromosome start to moves away from equatorial plate in telophase. | False |

| | | |
|---|---|---|
| **6.** | M-phase is the longest phase of cell cycle because karyokinesis occur in this phase. | False |
| **7.** | In telophase stage chromosomes decondense and lose their individuality. | individuality. |

| **Meiosis** | | |
|---|---|---|
| | **Questions** | **Answer** |
| **1.** | In which stage of meiosis, homologous chromosomes separate while sister chromatids remain associated at their centromeres: - | Anaphase I |
| **2.** | Pairing of homologous chromosomes occurs in :- | Zygotene |
| **3.** | Complex structure formed during synapsis is called: - | Synaptonemal complex |
| **4.** | Enzyme required for crossing over: - | Recombinase |
| **5.** | Dissolution of synaptonemal complex occur in: - | Diplotene |
| **6.** | X-shaped structure (Chiasmata) appear in: - | Diplotene |
| **7.** | Terminalisation of chiasmata occur in: - | Diakinesis |
| **8.** | Which stage represent transition to metaphase in meiosis-I? | Diakinesis |
| | **Fill in the blanks** | **Answer** |
| **1.** | Crossing over is the exchange of genetic material between _____ chromatids of homologous chromosomes. | Non-sister |
| **2.** | In oocytes of some vertebrates _____ stage can last for months or years. | Diplotene |
| **3.** | During _____ stage nuclear membrane and nucleolus reappear, cytokinesis follows and two cells are formed called dyad of cells. | Telophase-I |
| **4.** | During _____ stage bivalent chromosomes clearly appear as a tetrad. | Pachytene |
| **5.** | The stage between meiosis-I and meiosis-II is called _____ and is generally short lived. | Interkinesis |
| | **True or False** | **Answer** |
| **1.** | Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms. | True |
| **2.** | During interphase the chromosomes become gradually visible under the light microscope. | False |
| **3.** | Recombination between homologous chromosomes is completed by the end of pachytene. | True |
| **4.** | Increase in the genetic variability in population of organisms from one generation to next occurs by meiosis. | True |
| **5.** | At the end of meiosis-I, four haploid daughter cells are formed. | False |

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<table>
<tr><th></th><th>Match the column</th><th>Answers</th></tr>
<tr><td>1.</td><td>
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>1.</td><td>Centriole duplication</td><td>a.</td><td>$G_2$-phase</td></tr>
<tr><td>2.</td><td>Quiescent stage</td><td>b.</td><td>S-phase</td></tr>
<tr><td>3.</td><td>Interval between M-phase and S-phase</td><td>c.</td><td>$G_0$-phase</td></tr>
<tr><td>4.</td><td>Second gap phase</td><td>d.</td><td>$G_1$-phase</td></tr>
</table>
</td><td>1-b, 2-c, 3-d, 4-a</td></tr>
<tr><td>2.</td><td>
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>1.</td><td>Duration of M-phase</td><td>a.</td><td>S-phase</td></tr>
<tr><td>2.</td><td>DNA replication</td><td>b.</td><td>5%</td></tr>
<tr><td>3.</td><td>Duration of Interphase</td><td>c.</td><td>$G_0$-stage</td></tr>
<tr><td>4.</td><td>Metabolically active cells but no longer proliferation</td><td>d.</td><td>95%</td></tr>
</table>
</td><td>1-b, 2-a, 3-d, 4-c</td></tr>
<tr><td>3.</td><td>
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>A</td><td>Interphase</td><td>i.</td><td>Condensation of chromosomal material</td></tr>
<tr><td>B</td><td>Prophase</td><td>ii.</td><td>Chromatids move to opposite poles</td></tr>
<tr><td>C</td><td>Anaphase</td><td>iii.</td><td>Spindle fibres attach to kinetochores of chromosomes</td></tr>
<tr><td>D</td><td>Metaphase</td><td>iv.</td><td>Resting phase</td></tr>
</table>
</td><td>A-iv, B-i, C-ii, D-iii</td></tr>
<tr><td>4.</td><td>
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>A.</td><td>Telophase</td><td>i.</td><td>Centriole move to opposite poles</td></tr>
<tr><td>B.</td><td>Metaphase</td><td>ii.</td><td>Golgi complex and ER reform</td></tr>
<tr><td>C.</td><td>Anaphase</td><td>iii.</td><td>Chromosome aligned at metaphase plate</td></tr>
<tr><td>D.</td><td>Prophase</td><td>iv.</td><td>Splitting of centromeres</td></tr>
</table>
</td><td>A-ii, B-iii, C-iv, D-i</td></tr>
<tr><td>5.</td><td>
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>1.</td><td>Crossing over</td><td>a.</td><td>Pachytene</td></tr>
<tr><td>2.</td><td>Synapsis</td><td>b.</td><td>Anaphase-II</td></tr>
<tr><td>3.</td><td>Chiasmata appear</td><td>c.</td><td>Diakinesis</td></tr>
<tr><td>4.</td><td>Division of centromere</td><td>d.</td><td>Zygotene</td></tr>
<tr><td>5.</td><td>Terminalisation of chiasmata</td><td>e.</td><td>Diplotene</td></tr>
</table>
</td><td>1-a, 2-d, 3-e, 4-b, 5-c</td></tr>
</table>

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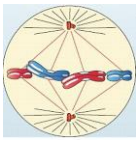
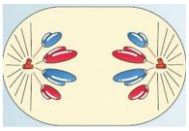
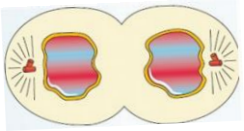
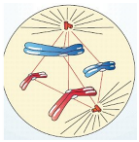
**6.**

| | Column I | | Column II |
|---|---|---|---|
| I. | Mitosis | a. | Evolution |
| II. | Meiosis | b. | Cell repair |
| | | c. | Genetic variability |
| | | d. | Replacement of old cells |
| | | e. | Growth of multicellular organisms. |
| | | f. | Conservation of specific chromosome number of each species |

I – b, d, e
II – a, c, f

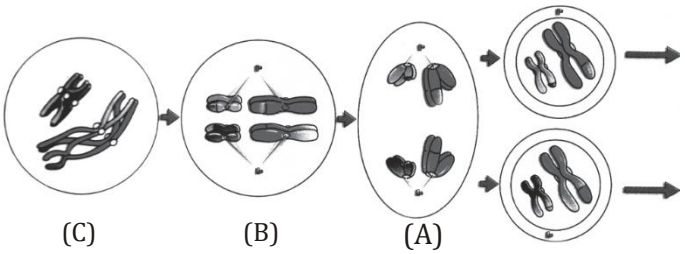
| | **Diagram based Questions** | **Answer** |
|---|---|---|

**1.** Match the following column-I and column II :-

| | Column I | | Column II |
|---|---|---|---|
| I. |  | a. | Metaphase |
| II. |  | b. | Late prophase |
| III. |  | c. | Transition to metaphase |
| IV. |  | d. | Anaphase |
| V. |  | e. | Telophase |

I – (b)
II – (a)
III – (d)
IV – (e)
V – (c)

**2.** Identify the following figures :-



(C) (B) (A)

(A) Anaphase-I
(B) Metaphase-I
(C) Prophase-I

| | Questions | Answers |
|---|---|---|
| **1.** | Electron micrographs of this stage indicate that chromosome synapsis is accompanied by the formation of complex structure called _____. | synaptonemal complex |
| **2.** | _____ is marked by terminalization of chiasmata. | Diakinesis |
| **3.** | The bivalent chromosomes align on the equatorial plate in _____ stage. | Metaphase–I |
| **4.** | Anaphase - I begin with the simultaneous splitting of the centromere of each chromosome. | False |
| **5.** | Chromosomes move toward opposite poles of the cell by shortening of microtubules attached to kinetochores. | True |
| **6.** | Meiosis involves two sequential cycles of nuclear and cell division called __(a)__ and __(b)__ but only a __(c)__ cycle of DNA replication. | a - meiosis I,<br>b - meiosis II,<br>c – single |
| **7.** | Prophase I is divided into five phases based on chromosomal behaviour namely: | Leptotene, Zygotene, Pachytene, Diplotene and Diakinesis. |
| **8.** | There is no replication of DNA during interkinesis. | True |
| **9.** | These structures serve as the sites of attachment of spindle fibres to the chromosomes during cell division. | Kinetochores |
| **10.** | Chromosomes that have reached their respective poles decondense and lose their individuality. | Telophase |

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# Plant Growth and Development

| Q.NO. | QUESTIONS | ANSWERS |
|---|---|---|
| 1. | An irreversible permanent increase in size of an organ or it's parts or even of an individual cell is called. | Growth |
| 2. | Name the cells which have the capacity to divide & self-perpetuate. | Meristematic cell |
| 3. | The form of growth where in new cells are always being added to the plant body by the activity of the meristem is called. | Open form of growth |
| 4. | Name the meristems which appear later in life of dicotyledonous and gymnosperm plants and causes increase in the girth of the organ in which they are active. | Lateral meristems (Vascular cambium and cork cambium) |
| 5. | Write the variety of parameters by which growth can be measured. | Fresh weight, dry weight, length, area, volume, cell number etc. |
| 6. | Growth of a pollen tube is measured in terms of it's | length |
| 7. | The period of growth is generally divided into three phases namely | meristematic, elongation and maturation phase |
| 8. | Which phase of growth is represented by root apex & shoot apex? | Meristematic phase |
| 9. | Write the characteristics of cells which are proximal to the meristematic zone. | Increased vacuolation Cell enlargement New cell wall deposition |
| 10. | Cells of which phase attain maximum size in terms of wall thickening and protoplasmic modification? | Phase of maturation |
| 11. | The increased growth per unit time is termed as | growth rate |
| 12. | Relative growth rate is the measure of the ability of the plant to produce new plant material so it is also referred as | efficiency index |
| 13. | All changes that an organism goes through during it's life cycle from germination of the seed to senescence is called | development |
| 14. | Plants follow different pathways in response to environment or phases of life to form different kinds of structures this ability is called | plasticity |
| 15. | Sum of growth and differentiation is considered as | development |
| 16. | Write the intrinsic and extrinsic growth factors. | Intrinsic = Genetic (intracellular) PGR (intercellular) Extrinsic = light, temperature, water, oxygen and nutrition |

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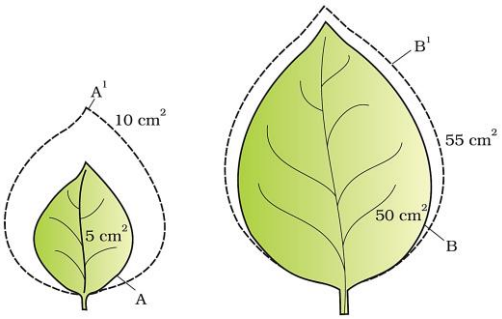
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| | | |
|---|---|---|
| **17.** | Name the plant hormones which are involved in growth promoting activities such as cell division, cell enlargement, pattern formation, tropic growth, flowering, fruiting and seed formation. | Auxin, Gibberellin, Cytokinin |
| **18.** | Plant hormones, which responses to wounds and stresses of biotic & abiotic origin & are also involved in dormancy & abscission, are– | ethylene and abscisic acid |
| **19.** | Name the hormone which influence the tip of coleoptile and bend it towards light. | Auxin |
| **20.** | Inhibitor B, abscission II and dormin are related to which hormone ? | Abscisic acid (ABA) |
| **21.** | Name the volatile substance which was confirmed by H.H. Cousins from ripened oranges. | Ethylene |
| **22.** | Miller, F. Skoog and his co-workers are concered to discovery of | Kinetin (Cytokinin) |
| **23.** | Name the hormone which is used extensively in agricultural and horticultural practices and help to initiate rooting in stem cutting during plant propagation. | Auxin |
| **24.** | In higher plants, the growing apical bud inhibits the growth of the lateral (axillary) buds, this phenomenon is called. | apical dominance |
| **25.** | Write the process that results in the growth of lateral buds and widely applied in tea plantations and hedge making. | Decapitation |
| **26.** | Write the hormone that induce parthenocarpy and also controls xylem differentiation. | Auxin |
| **27.** | Write the hormone which increase the length of grapes, stalks improves the shape of apple and speed up the malting process in brewing industry. | Gibberellin |
| **28.** | Hormone which helps to produce new leaves, chloroplasts in leaves, lateral shoot growth and adventitious shoot formation. | Cytokinin |
| **29.** | A simple gaseous PGR synthesised by senescence tissue and ripening fruits. | Ethylene |
| **30.** | Hormone which enhances the respiration rate during ripening of the fruits (respiratory climactic). | Ethylene |
| **31.** | Name the hormone which promotes root growth and root hair formation. | Ethylene |
| **32.** | The most widely used compound as source of ethylene is | Ethephon |
| **33.** | Hormone that promotes female flowers in cucumbers and increasing the yield. | Ethylene |
| **34.** | Name the hormone which regulate abscission and dormancy. | Abscisic acid |
| **35.** | Hormone that inhibits the plant metabolism and seed germination. | Abscisic acid |
| **36.** | Flowering response of plants to periods of day/night is termed as | Photoperiodism |
| **37.** | If flowering is either quantitatively or qualitatively dependent on exposure to low temperature, this phenomenon is termed as. | Vernalisation |

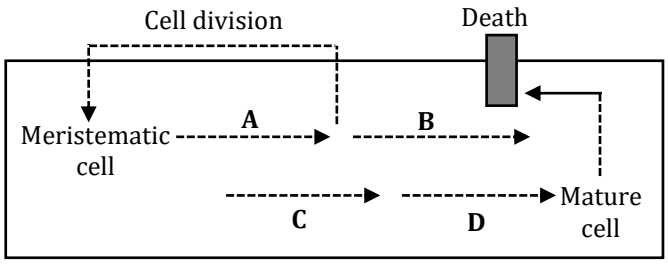
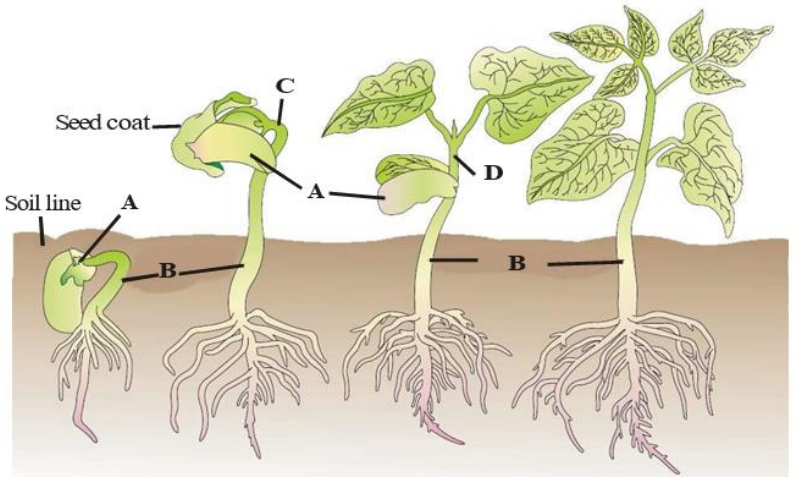
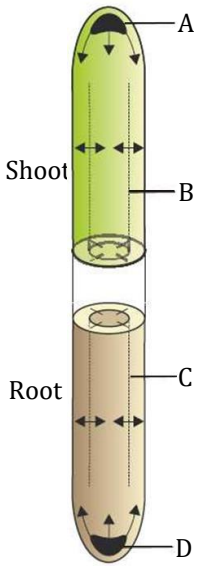
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| Q.NO. | FILL IN THE BLANKS | ANSWERS |
|---|---|---|
| 1. | __(A)__ is regarded as one of the most fundamental and conspicuous characteristics of a living being. | A = Growth |
| 2. | Growth at a cellular level is principally a consequence of increase in the amount of __(A)__. | A = protoplasm |
| 3. | Single maize root apical meristem can give rise to more than __A__ new cells per hour whereas cells in a watermelon may increase in size by upto __B__ times. | A = 17,500<br>B = 3,50,000 |
| 4. | During differentiation cells undergo few to major structural changes both in their ___A___ and ___B___. | A = cell wall<br>B = protoplasm |
| 5. | In a plant, cell/tissues arising out of the same meristem have different structure at maturity it is known as ___A___. | A = open differentiation |
| 6. | Development in plants is under the control of __A__ and ___B___ factors. | A = intrinsic<br>B = extrinsic |
| 7. | Auxin was isolated by ___A___ from tips of coleoptiles of oat seedlings. | A = F.W. Went |
| 8. | The bakane (foolish seedling), a disease of rice seedlings was caused by a fungal pathogen _____. | *Gibberella fujikuroi* |
| 9. | Cytokinin's help overcome the ____A____. They promote ____B___ which helps in the delay of leaf senescence. | A = apical dominance<br>B = nutrient mobilisation |
| 10. | Influence of ethylene on plant include ___A___ of seedlings, ____B____ and ____C___ in dicot seedlings. | A = horizontal growth<br>B = swelling of the axis<br>C = apical hook formation |
| 11. | Ethylene promotes ____A____ and ____B____ of plant organs especially of leaves and flowers. | A = senescence<br>B = abscission |
| 12. | ____A____ promotes rapid internode/petiole elongation in deep water rice plants. | A = Ethylene |
| 13. | ____A____ stimulates the closure of stomata in the epidermis and increase the tolerance of plants to various kinds of stresses. | A = ABA |
| 14. | Abscisic acid acts as an antagonist to _____. | gibberellic acid |
| 15. | The site of perception of light/dark duration are the _____. | leaves |
| 16. | Hormonal substance migrates from leaves to shoot apices for inducing flowering only when the plants are exposed to the necessary _____. | inductive photo period |

| Q.NO. | TRUE AND FALSE | ANSWERS |
|---|---|---|
| 1. | All cells of a plant are descendents of the zygote. | True |
| 2. | The leaves, flowers and fruits of the same tree not only have limited dimensions but also appear and fall periodically and sometime repeatedly. | True |

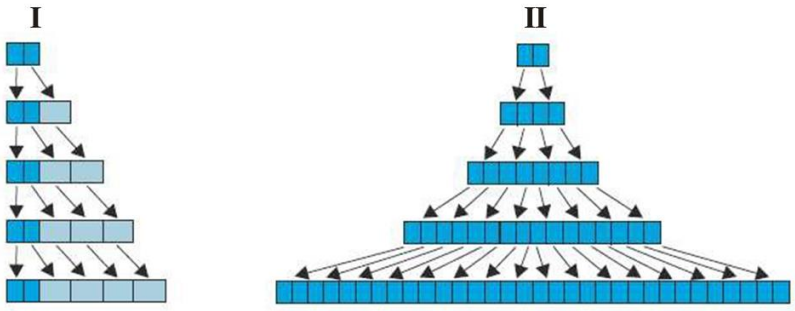
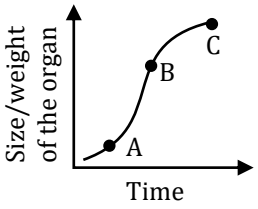
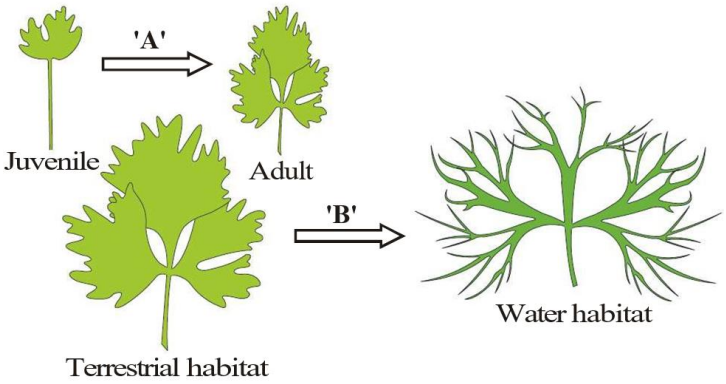
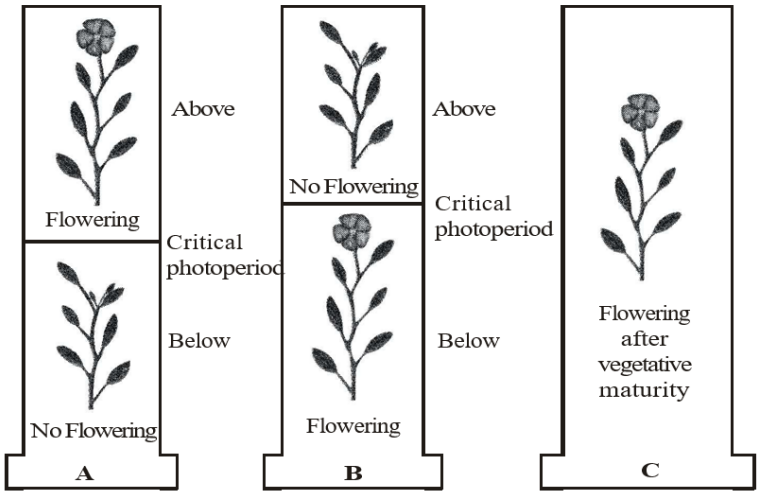
| | | |
|---|---|---|
| **3.** | Development of a mature plant from a zygote (Fertilized egg) follow a precise and highly ordered succession of events. | True |
| **4.** | Plants retain the capacity for unlimited growth throughout their life. | True |
| **5.** | Choose the correct or incorrect statement regarding given figure.<br>  <br><br>(a) Figure (A) shows more relative growth rate as compare to figure (B).<br>(b) Figure (A) and figure (B) shows the similar absolute growth rate. | True<br><br>True |
| **6.** | Water provides the medium for enzymatic activities needed for growth while oxygen helps in releasing metabolic energy essential for growth activities. | True |
| **7.** | Nutrients are required by plants for the synthesis of protoplasm and act as source of energy. | True |
| **8.** | Auxin was first isolated from human urine. | True |
| **9.** | Spraying juvenile conifers with GA hastens the maturity period thus leading to early seed production. | True |
| **10.** | Kinetin occur naturally in plants. | False |
| **11.** | Natural cytokinin's are synthesised in regions where rapid cell division occurs. | True |
| **12.** | Ethylene is used to initiate flowering and for synchronising fruit set in pineapples. | True |
| **13.** | Ethephon hastens fruit ripening in tomatoes and apples and accelerates abscission in flowers and fruits. | True |
| **14.** | ABA plays an important role in seed development, maturation and dormancy. | True |
| **15.** | Events like vernalisation, flowering, dormancy, seed germination and plant movements are controlled only by extrinsic factors. | False |
| **16.** | In certain plants relative duration of light and dark period plays a important role in flowering. | True |
| **17.** | Vernalisation prevents precocious reproductive development late in the growing season. | True |
| **18.** | Spring variety is normally planted in winter and harvested in autumn. | False |
| **19.** | Biennials are monocarpic plants that normally flower and die in the second season. | True |
| **20.** | Effect of vernalisation can not be seen in sugar beet, cabbages and carrots. | False |
| **21.** | Winter varieties of annual plants are planted in autumn, they germinate, and over winter come out as small seedlings, resume growth in spring and are harvested around mid - summer. | True |

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| Q.NO. | DIAGRAM BASED QUESTIONS | ANSWERS |
|---|---|---|
| 1. | Given figure shows the sequence of the development process in a plant cell, identify labelled parts A, B, C and D.<br>  | A = Plasmatic growth<br>B = Differentiation<br>C = Expansion<br>D = Maturation |
| 2. | Find the labelled parts in given figure which shows germination and seedling development in bean.<br>  | A = Cotyledon<br>B = Hypocotyl<br>C = Epicotyl hook<br>D = Epicotyl |
| 3. | Name the labelled meristems in given diagram.<br>  | A = shoot apical meristem<br>B and C = vascular combium of shoot and root respectively<br>D = root apical meristem |
| 4. | Identify the following diagrammatic representation of types of growth and their mathematical expressions. | I = Arithmetic $L_t = L_0 + rt$ |



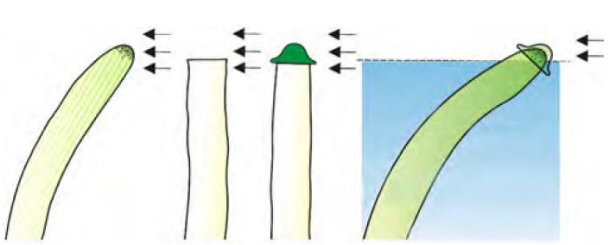
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| | | |
|---|---|---|
| |  | II = Geometric $W_1 = W_0 e^{rt}$ |
| **5.** | Identify the labelled parts A, B and C in given figure.<br>  | A = Lag phase<br>B = Log phase or exponential phase<br>C = Stationary phase |
| **6.** | Identify the phenomena A and B :-<br>  | A = Heterophylly due to phases of life.<br>B = Environmental heterophylly. |
| **7.** | In the given figure identify the group of plant.<br>  | A = Long day plant<br>B = Short day plant<br>C = Day neutral plant |

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<table>
<tr><th>Q.NO.</th><th>MATCH THE COLUMNS</th><th>ANSWERS</th></tr>
<tr><td>1.</td><td>Match these molecules to their chemical composition :-
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>A</td><td>Auxin (IAA)</td><td>i</td><td>indole compounds</td></tr>
<tr><td>B</td><td>Cytokinin's (kinetin)</td><td>ii</td><td>adenine derivatives</td></tr>
<tr><td>C</td><td>Abscisic acid</td><td>iii</td><td>carotenoid derivatives</td></tr>
<tr><td>D</td><td>Gibberellic acid</td><td>iv</td><td>terpene derivatives</td></tr>
<tr><td>E</td><td>Ethylene</td><td>v</td><td>$C_2H_4$</td></tr>
</table></td><td>A = i, B - ii, C = iii,<br>D = iv, E = v</td></tr>
<tr><td>2.</td><td>Match the column-I and Column-II :-
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>1</td><td>Natural auxin</td><td>i</td><td>Indole 3-acetic acid (IAA)</td></tr>
<tr><td>2</td><td>Synthetic auxin</td><td>ii</td><td>Indole butyric acid (IBA)</td></tr>
<tr><td></td><td></td><td>iii</td><td>Naphthalene acetic acid (NAA)</td></tr>
<tr><td></td><td></td><td>iv</td><td>2,4-dichlorophenoxyacetic acid (2,4D)</td></tr>
</table></td><td>1 = a, b<br>2 = c, d</td></tr>
</table>

| Q.No. | Question | Answer |
|---|---|---|
| **1.** | In geometric growth, following mitotic cell division, only one daughter cell continues to divide while other differentiates and matures. | False |
| **2.** | A _____ is a characteristic of living organism growing in a natural environment. | Sigmoid curve |
| **3.** | In $W_1=W_0 e^{rt}$, r is the absolute growth rate and is also the measure of the ability of the plant to produce new plant material, referred to as efficiency index. | False |
| **4.** | _____ hastens fruit ripening in tomatoes and apples and accelerates abscission in flowers and fruits (thinning of cotton, cherry, walnut). | Ethephon |
| **5.** | Following experiment demonstrate about which plant growth regulator :-<br>  <br><br>Charles Darwin & Francis Darwin Experiment | Auxin |

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# Respiration in Plants

| Q.NO. | QUESTIONS | ANSWERS |
|---|---|---|
| **1.** | Name the organism(s) that can prepare their own food by photosynthesis | All green plants, cyanobacteria, some protist and some other eubacteria. |
| **2.** | Name the site(s) where breakdown of complex molecules takes place to yield energy | Cytoplasm & mitochondria |
| **3.** | The breakdown of the C–C bonds of complex compounds through oxidation within the cells, leading to release of considerable amount of energy is called. | Respiration |
| **4.** | Name the structures/organs in plants that are used for gaseous exchange. | Stomata & lenticels |
| **5.** | Write the reaction for complex combustion of glucose molecule. | $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + Energy |
| **6.** | All living organisms retain the enzymatic machinery to partially oxidise glucose without the help of oxygen. This breakdown of glucose to pyruvic acid is called. | Glycolysis |
| **7.** | Name the scientists who gave the scheme of glycolysis | Gustav Embden<br>Otto Meyerhoff & J. Parnas |
| **8.** | Name the site of glycolysis in the cell | Cytoplasm |
| **9.** | How many ATP are directly synthesised in glycolysis from one glucose. | 4ATP |
| **10.** | Write the names of three major ways in which different cells handle pyruvic acid. | Lactic acid fermentation<br>Alcoholic fermentation<br>Aerobic respiration |
| **11.** | Write the enzymes which catalyse the conversion of pyruvic acid to $CO_2$ and ethanol. | Pyruvic acid decarboxylase<br>Alcohol dehydrogenase |
| **12.** | How much amount of energy is released in lactic acid and alcoholic fermentation? | Less than 7% of total energy in glucose. |
| **13.** | Write the enzyme & its coenzymes which catalyse the following reaction: -<br>Pyruvic acid + $NAD^+ \longrightarrow$ Acetyl Co-A + $CO_2$ + NADH | Pyruvate dehydrogenase,<br>$NAD^+$, CoA, $Mg^{2+}$,<br>TPP (Thiamine pyrophosphate),<br>LA (Lipoic acid) |
| **14.** | Name the process by which ATP is formed in electron transport system (ETS) in mitochondria. | Oxidative phosphorylation |
| **15.** | How many ATP are gained during aerobic respiration using one glucose. | 36 or 38 ATP |

| | | |
|---|---|---|
| **16.** | The ratio of the volume of $CO_2$ evolved to the volume of $O_2$ consumed in respiration is called. | Respiratory quotient (RQ) or Respiratory ratio |
| **17.** | Find the value of RQ when fats (e.g. Tripalmitin) are used in respiration. | 0.7 |
| **18.** | Find the value of RQ when respectively carbohydrates and proteins are used in respiration. | 1.0 and 0.9 |

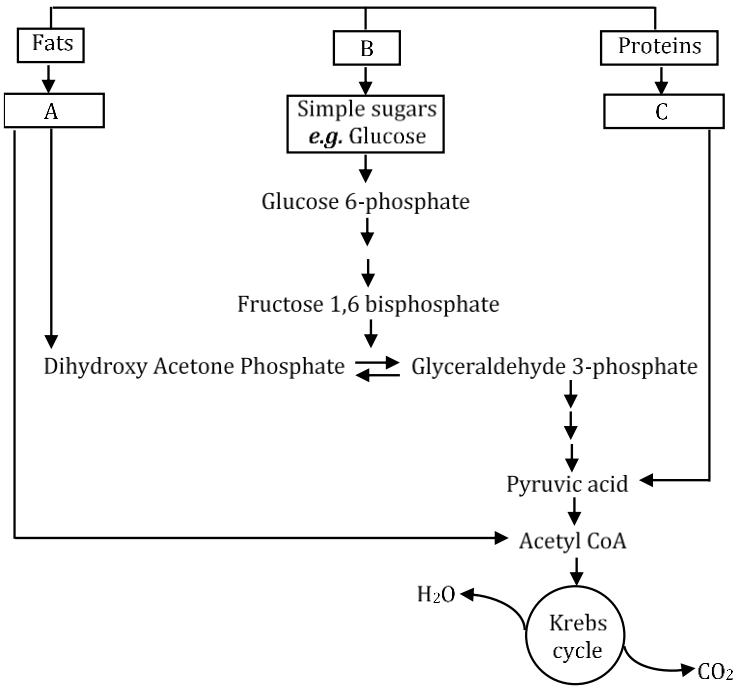
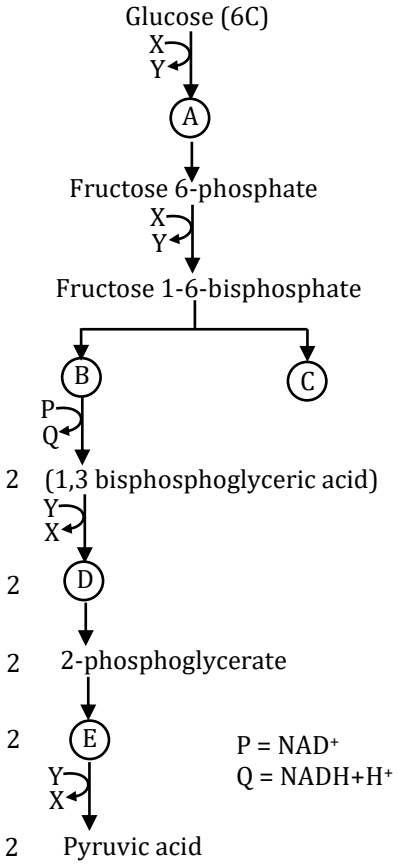
| | **FILL IN THE BLANKS** | **ANSWERS** |
|---|---|---|
| **1.** | Usually (A) are oxidised to release energy but (B) and even (C) can be used as respiratory substances in some plants. | A = carbohydrate<br>B = proteins, fats<br>C = organic acids |
| **2.** | During oxidation energy released in a series of slow stepwise reactions controlled by (A) and is trapped as (B) in the form of (C). | A = enzymes<br>B = chemical energy<br>C = ATP |
| **3.** | The first cells on this planet lived in atmosphere that lacked (A). | A = oxygen |
| **4.** | Fermentation takes place under (A) conditions in many (B) and unicellular (C). | A = anaerobic<br>B = prokaryotes<br>C = eukaryotes |
| **5.** | Yeasts poison themselves to death when the concentration of alcohol reaches about _____. | 13% |
| **6.** | In mitochondrial matrix pyruvate undergoes _____. | Oxidative decarboxylation |
| **7.** | TCA cycle starts with the condensation of acetyl group with (A) and (B) to yield citric acid. | A = oxaloacetic acid<br>B = water |
| **8.** | The metabolic pathway present in the inner mitochondrial membrane through which the electron passes from one carrier to another is called _____. | electron transport system (ETS) |
| **9.** | Oxidation of one molecule of NADH gives rise to ___(A)___ molecules of ATP, while that of one molecule of $FADH_2$ produces ___(B)___ molecules of ATP. | A = three<br>B = two |
| **10.** | If fatty acids were to be respired they would first be degraded to ___(A)___ and enter the pathway. Glycerol would enter the pathway after being converted to ___(B)____. | A = acetyl Co-A<br>B = PGAL |
| **11.** | Respiratory pathway is involved in both anabolism and catabolism it would hence be better to consider the respiratory pathway as an _____. | amphibolic pathway |

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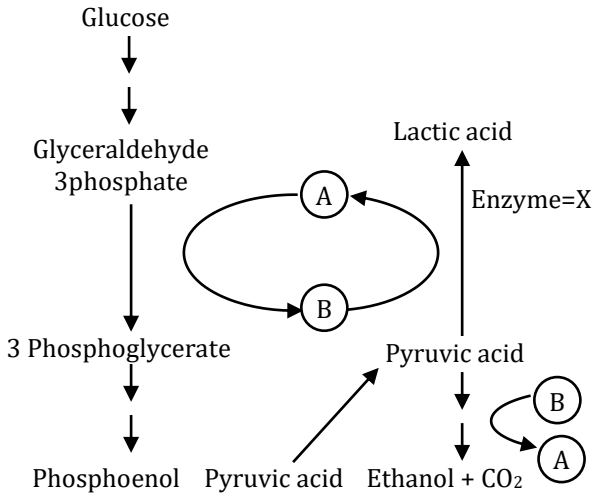
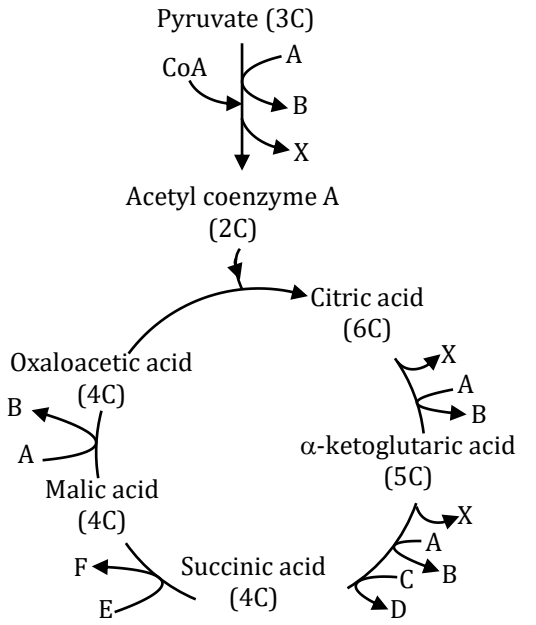
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| Q.NO. | TRUE & FALSE | ANSWERS |
|---|---|---|
| 1. | Energy released by oxidation in respiration is not used directly but is used to synthesise ATP. | True |
| 2. | ATP acts as the energy currency of the cell because it is broken down whenever (and wherever) energy needs to be utilised. | True |
| 3. | Carbon skeleton produced during respiration. (Eg. $\alpha$-ketoglutarate, OAA etc. intermediates) are used as precursor for biosynthesis of other molecules in the cell. | True |
| 4. | Plants do not require $O_2$ for respiration and they never give out $CO_2$. | False |
| 5. | Krebs cycle which is also called aerobic respiration requires $O_2$ supply. | True |
| 6. | In mitochondrial matrix complete oxidation of pyruvate takes place in which all the hydrogen atoms are removed, leaving three molecules of $CO_2$. | True |
| 7. | At inner membrane of mitochondria, passing on of the electrons removed as part of the hydrogen atoms to molecular $O_2$ with simultaneous synthesis of ATP occurs. | True |
| 8. | Oxygen acts as the final hydrogen acceptor in electron transport chain of mitochondria. | True |
| 9. | In respiration, energy of oxidation reduction is utilised for the production of proton gradient required for phosphorylation. | True |
| 10. | Cytochrome c (small protein) is attached to outer surface of the inner mitochondrial membrane. | True |
| 11. | Cytochrome c oxidase complex contains cyto-a & $a_3$ and two copper centres. | True |
| 12. | When the electrons pass from one carrier to another via complex-I to IV in the electron transport chain, they are coupled to ATP synthase (complex-V) for the production of ATP from ADP and inorganic phosphate. | True |
| 13. | Fermentation accounts for only a partial breakdown of glucose whereas in aerobic respiration it is completely degraded to $CO_2$ and $H_2O$. | True |
| 14. | NADH is oxidised to $NAD^+$ rather slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration. | True |

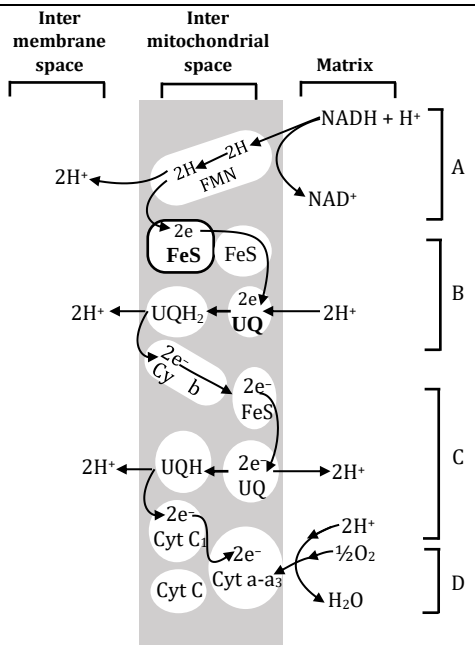
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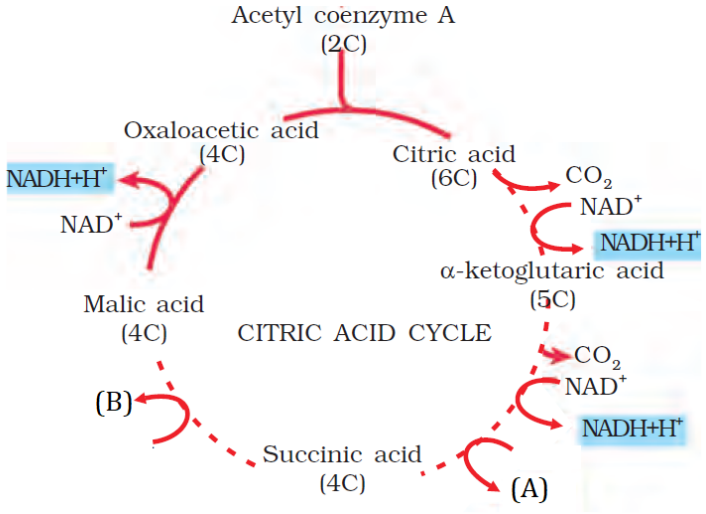
| Q.NO. | DIAGRAM BASED QUESTIONS | ANSWERS |
|---|---|---|
| 1. | Identify A, B and C: -<br>  <br> | A = Fatty acid and glycerol<br>B = Carbohydrate<br>C = Amino acids |
| 2. | In the given figure observe the steps of glycolysis and find out labelled parts :-<br>  <br> | X = ATP Y = ADP<br>A = Glucose-6-phosphate<br>B = Glyceraldehyde-3-phosphate<br>C = Dihydroxy acetone phosphate<br>D = 3-phosphoglyceric acid<br>E = Phosphoenolpyruvate |

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| | | |
|---|---|---|
| **3.** | By observing following anaerobic respiratory pathway find the labelled parts :-<br>  <br> | A = $NAD^+$<br>B = $NADH + H^+$<br>X = Lactate dehydrogenase |
| **4.** | Find the labelled parts in given cycle, that are summarized in following equation.<br>  <br><br>Pyruvic acid + 4A + E + 3H2O + C + Pi $\xrightarrow{\text{Mitochondrial matrix}}$<br>3X + 4B + F + D | X = $CO_2$,<br>A = $NAD^+$,<br>B = $NADH + H^+$,<br>C = GDP,<br>D = GTP,<br>E = FAD,<br>F = $FADH_2$ |
| **5.** | Identify the A, B, C and D :- | A = Oxidation of NADH by NADH dehydrogenase (complex-I)<br>B = Transfer of electrons to ubiquinone via complex-I or complex-II ($FADH_2$)<br>C = Transfer of electrons to |

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| | | |
|---|---|---|
| |  <br>**Electron Transport System (ETS)** | cytoctrome c via cytochrome $bc_1$ complex (complex-III)<br>D = Oxidation of cytochrome c by cytochrome c oxidase (complex-IV) |

| Q.No. | Question | Answer |
|---|---|---|
| **1.** | Energy released by oxidation in respiration is not used directly but is used to synthesize ATP. | True |
| **2.** | Carbon skeleton produced during respiration, (e.g. α-ketoglutarate, OAA etc.) intermediates are used as precursors for biosynthesis of other molecules in the cell. | True |
| **3.** | Krebs cycle which is also called aerobic respiration and requires $O_2$ supply. | True |
| **4.** | In mitochondrial matrix complete oxidation of pyruvate takes place in which all the hydrogen atoms are removed, leaving _____ molecules of $CO_2$. | Three |
| **5.** | Identify A and B in given below diagram :<br>  | A – GTP, B – $FADH_2$ |

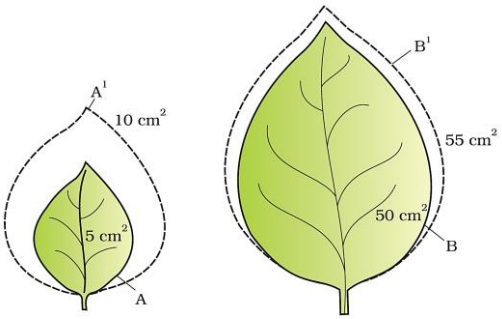
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|---|---|---|
| **6.** | Identify (I) and (II) with reference to given equation :-<br>**(I)**+4 $NAD^+$ + $FAD^+$ + $2H_2O$ + ADP + Pi $\xrightarrow[\text{Matrix}]{\text{Mitochondrial}}$ $3CO_2$+4NADH + $4H^+$ + **(II)** + ATP | (I)–Pyruvic acid;<br>(II)–$FADH_2$ |
| **7.** | Identify (I) and (II) based of RQ value of tripalmitin.<br>$2(C_{51}H_{98}O_6)$ + **(I)** $O_2$ ⟶ **(II)** CO2 + $98H_2O$+energy | (I)–145, (II)–102 |
| **8.** | Cytochrome c (small protein) is attached to outer surface of the inner mitochondrial membrane. | True |
| **9.** | Cytochrome c oxidase complex contains _____ and two _____ centres. | cyt-a & cyt-$a_3$, copper |
| **10.** | Glycolysis occurs in the _____ and produces_____, which in the presence of $O_2$ enters the_____. | cytosol; pyruvate; mitochondrion |

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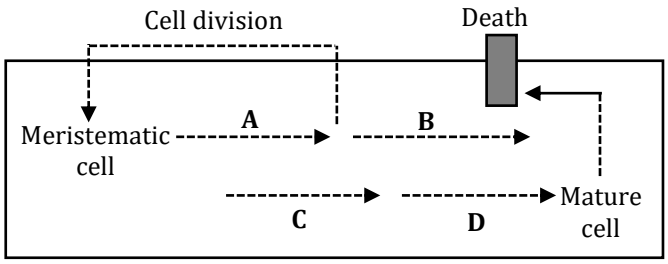
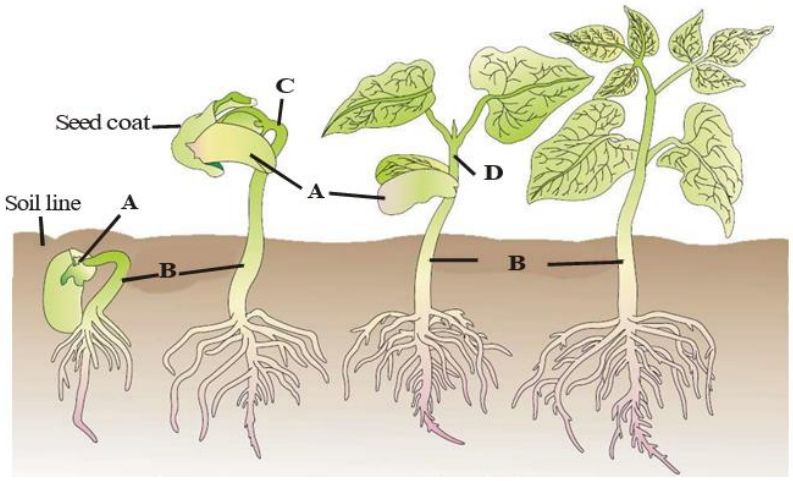
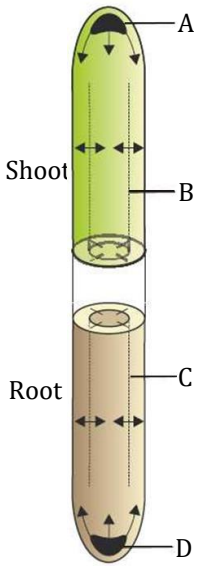
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| Q.NO. | FILL IN THE BLANKS | ANSWERS |
|---|---|---|
| 1. | __(A)__ is regarded as one of the most fundamental and conspicuous characteristics of a living being. | A = Growth |
| 2. | Growth at a cellular level is principally a consequence of increase in the amount of __(A)__. | A = protoplasm |
| 3. | Single maize root apical meristem can give rise to more than __A__ new cells per hour whereas cells in a watermelon may increase in size by upto __B__ times. | A = 17,500<br>B = 3,50,000 |
| 4. | During differentiation cells undergo few to major structural changes both in their ___A___ and ___B___. | A = cell wall<br>B = protoplasm |
| 5. | In a plant, cell/tissues arising out of the same meristem have different structure at maturity it is known as ___A___. | A = open differentiation |
| 6. | Development in plants is under the control of __A__ and ___B___ factors. | A = intrinsic<br>B = extrinsic |
| 7. | Auxin was isolated by ___A___ from tips of coleoptiles of oat seedlings. | A = F.W. Went |
| 8. | The bakane (foolish seedling), a disease of rice seedlings was caused by a fungal pathogen _____. | *Gibberella fujikuroi* |
| 9. | Cytokinin's help overcome the ____A____. They promote ____B___ which helps in the delay of leaf senescence. | A = apical dominance<br>B = nutrient mobilisation |
| 10. | Influence of ethylene on plant include ___A___ of seedlings, ____B____ and ____C___ in dicot seedlings. | A = horizontal growth<br>B = swelling of the axis<br>C = apical hook formation |
| 11. | Ethylene promotes ____A____ and ____B____ of plant organs especially of leaves and flowers. | A = senescence<br>B = abscission |
| 12. | ____A____ promotes rapid internode/petiole elongation in deep water rice plants. | A = Ethylene |
| 13. | ____A____ stimulates the closure of stomata in the epidermis and increase the tolerance of plants to various kinds of stresses. | A = ABA |
| 14. | Abscisic acid acts as an antagonist to _____. | gibberellic acid |
| 15. | The site of perception of light/dark duration are the _____. | leaves |
| 16. | Hormonal substance migrates from leaves to shoot apices for inducing flowering only when the plants are exposed to the necessary _____. | inductive photo period |

| Q.NO. | TRUE AND FALSE | ANSWERS |
|---|---|---|
| 1. | All cells of a plant are descendents of the zygote. | True |
| 2. | The leaves, flowers and fruits of the same tree not only have limited dimensions but also appear and fall periodically and sometime repeatedly. | True |

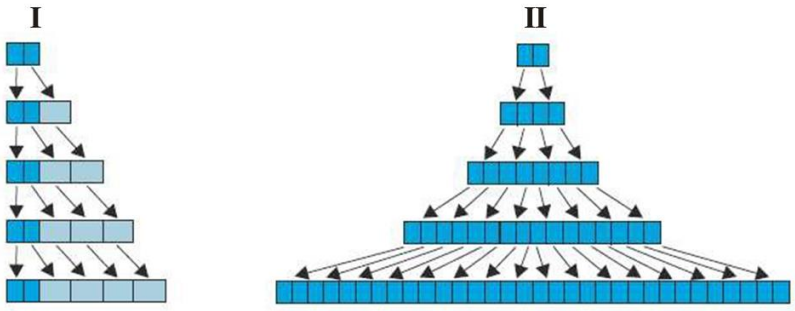
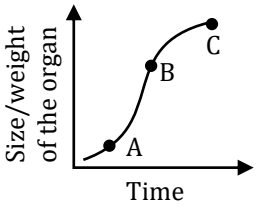
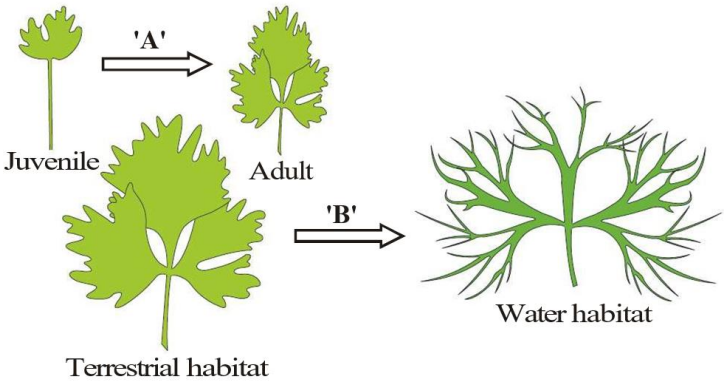
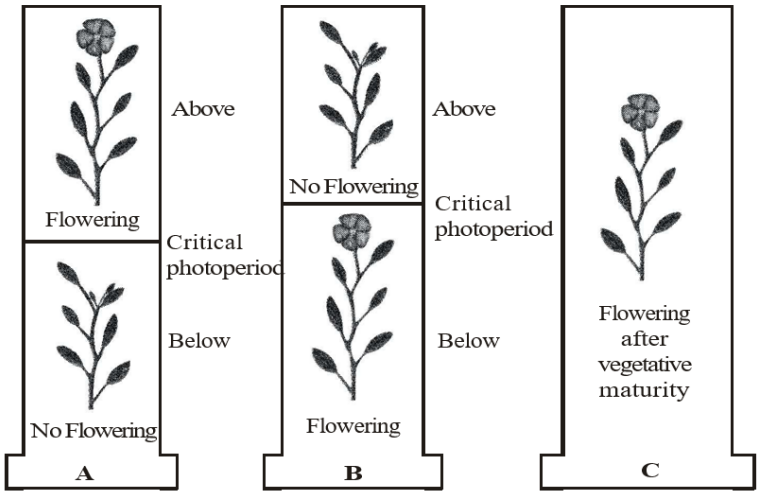
| | | |
|---|---|---|
| **3.** | Development of a mature plant from a zygote (Fertilized egg) follow a precise and highly ordered succession of events. | True |
| **4.** | Plants retain the capacity for unlimited growth throughout their life. | True |
| **5.** | Choose the correct or incorrect statement regarding given figure.<br>  <br><br>(a) Figure (A) shows more relative growth rate as compare to figure (B).<br>(b) Figure (A) and figure (B) shows the similar absolute growth rate. | True<br><br>True |
| **6.** | Water provides the medium for enzymatic activities needed for growth while oxygen helps in releasing metabolic energy essential for growth activities. | True |
| **7.** | Nutrients are required by plants for the synthesis of protoplasm and act as source of energy. | True |
| **8.** | Auxin was first isolated from human urine. | True |
| **9.** | Spraying juvenile conifers with GA hastens the maturity period thus leading to early seed production. | True |
| **10.** | Kinetin occur naturally in plants. | False |
| **11.** | Natural cytokinin's are synthesised in regions where rapid cell division occurs. | True |
| **12.** | Ethylene is used to initiate flowering and for synchronising fruit set in pineapples. | True |
| **13.** | Ethephon hastens fruit ripening in tomatoes and apples and accelerates abscission in flowers and fruits. | True |
| **14.** | ABA plays an important role in seed development, maturation and dormancy. | True |
| **15.** | Events like vernalisation, flowering, dormancy, seed germination and plant movements are controlled only by extrinsic factors. | False |
| **16.** | In certain plants relative duration of light and dark period plays a important role in flowering. | True |
| **17.** | Vernalisation prevents precocious reproductive development late in the growing season. | True |
| **18.** | Spring variety is normally planted in winter and harvested in autumn. | False |
| **19.** | Biennials are monocarpic plants that normally flower and die in the second season. | True |
| **20.** | Effect of vernalisation can not be seen in sugar beet, cabbages and carrots. | False |
| **21.** | Winter varieties of annual plants are planted in autumn, they germinate, and over winter come out as small seedlings, resume growth in spring and are harvested around mid - summer. | True |

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| Q.NO. | DIAGRAM BASED QUESTIONS | ANSWERS |
|---|---|---|
| 1. | Given figure shows the sequence of the development process in a plant cell, identify labelled parts A, B, C and D.<br>  | A = Plasmatic growth<br>B = Differentiation<br>C = Expansion<br>D = Maturation |
| 2. | Find the labelled parts in given figure which shows germination and seedling development in bean.<br>  | A = Cotyledon<br>B = Hypocotyl<br>C = Epicotyl hook<br>D = Epicotyl |
| 3. | Name the labelled meristems in given diagram.<br>  | A = shoot apical meristem<br>B and C = vascular combium of shoot and root respectively<br>D = root apical meristem |
| 4. | Identify the following diagrammatic representation of types of growth and their mathematical expressions. | I = Arithmetic $L_t = L_0 + rt$ |



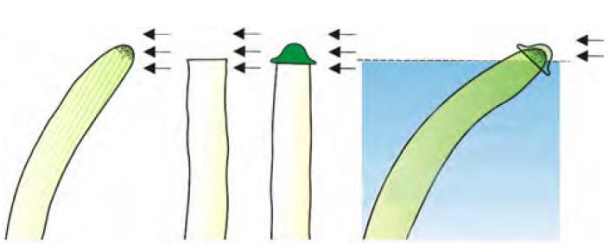
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| | | |
|---|---|---|
| |  | II = Geometric $W_1 = W_0 e^{rt}$ |
| **5.** | Identify the labelled parts A, B and C in given figure.<br>  | A = Lag phase<br>B = Log phase or exponential phase<br>C = Stationary phase |
| **6.** | Identify the phenomena A and B :-<br>  | A = Heterophylly due to phases of life.<br>B = Environmental heterophylly. |
| **7.** | In the given figure identify the group of plant.<br>  | A = Long day plant<br>B = Short day plant<br>C = Day neutral plant |

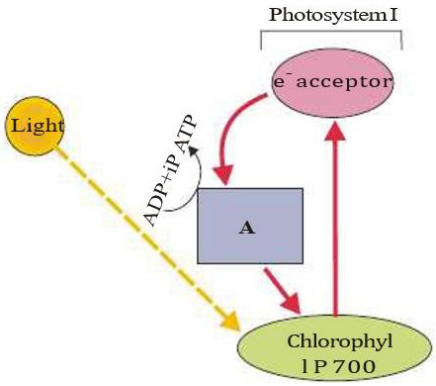
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<table>
<tr><th>Q.NO.</th><th>MATCH THE COLUMNS</th><th>ANSWERS</th></tr>
<tr><td>1.</td><td>Match these molecules to their chemical composition :-
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>A</td><td>Auxin (IAA)</td><td>i</td><td>indole compounds</td></tr>
<tr><td>B</td><td>Cytokinin's (kinetin)</td><td>ii</td><td>adenine derivatives</td></tr>
<tr><td>C</td><td>Abscisic acid</td><td>iii</td><td>carotenoid derivatives</td></tr>
<tr><td>D</td><td>Gibberellic acid</td><td>iv</td><td>terpene derivatives</td></tr>
<tr><td>E</td><td>Ethylene</td><td>v</td><td>$C_2H_4$</td></tr>
</table></td><td>A = i, B - ii, C = iii,<br>D = iv, E = v</td></tr>
<tr><td>2.</td><td>Match the column-I and Column-II :-
<table>
<tr><th></th><th>Column I</th><th></th><th>Column II</th></tr>
<tr><td>1</td><td>Natural auxin</td><td>i</td><td>Indole 3-acetic acid (IAA)</td></tr>
<tr><td>2</td><td>Synthetic auxin</td><td>ii</td><td>Indole butyric acid (IBA)</td></tr>
<tr><td></td><td></td><td>iii</td><td>Naphthalene acetic acid (NAA)</td></tr>
<tr><td></td><td></td><td>iv</td><td>2,4-dichlorophenoxyacetic acid (2,4D)</td></tr>
</table></td><td>1 = a, b<br>2 = c, d</td></tr>
</table>

| Q.No. | Question | Answer |
|---|---|---|
| **1.** | In geometric growth, following mitotic cell division, only one daughter cell continues to divide while other differentiates and matures. | False |
| **2.** | A _____ is a characteristic of living organism growing in a natural environment. | Sigmoid curve |
| **3.** | In $W_1=W_0 e^{rt}$, r is the absolute growth rate and is also the measure of the ability of the plant to produce new plant material, referred to as efficiency index. | False |
| **4.** | _____ hastens fruit ripening in tomatoes and apples and accelerates abscission in flowers and fruits (thinning of cotton, cherry, walnut). | Ethephon |
| **5.** | Following experiment demonstrate about which plant growth regulator :-<br>  <br><br>Charles Darwin & Francis Darwin Experiment | Auxin |

# Photosynthesis in Higher Plants

| Q.NO. | Question | Answer |
|---|---|---|
| 1. | Process in which light energy is converted into chemical energy (organic compounds) is called. | Physico-chemical process e.g. Photosynthesis |
| 2. | An experimental setup to show that $CO_2$ is essential for photosynthesis is | Moll's half leaf experiment |
| 3. | First of all who demonstrate that sunlight is essential for photosynthesis in plants | Jan Ingenhousz |
| 4. | Production of glucose takes place in green part of plant is established by | J.V. Sachs |
| 5. | A first action spectrum of photosynthesis was described by | T.W. Engelmann |
| 6. | Name the algae which was used to describe action spectrum of photosynthesis | *Cladophora* |
| 7. | Who demonstrate that photosynthesis is a light dependent reaction in which hydrogen from a suitable oxidizable compound reduces carbon dioxide to carbohydrate<br>$2H_2O + CO_2 \xrightarrow{\text{Light}} 2A + CH_2O + H_2O$<br>Also conclude $O_2$ evolve during photosynthesis comes from water not from carbon dioxide. | C.V. Niel |
| 8. | Name the pigment(s) which increases the absorption range for photosynthesis. | Chlorophyll b, xanthophylls and carotenoids |
| 9. | Where are the protons and $O_2$, formed during splitting of water, likely to be released. | In lumen of thylakoid |
| 10. | Synthesis of ATP from ADP and inorganic phosphate in presence of light is known as | Photophosphorylation |
| 11. | Identify the A in the given figure of light reaction: -<br>  | A = Electron transport system |
| 12. | Write the requirements for chemiosmotic ATP synthesis | Membrane<br>Proton pump<br>Proton gradient<br>ATP synthase |

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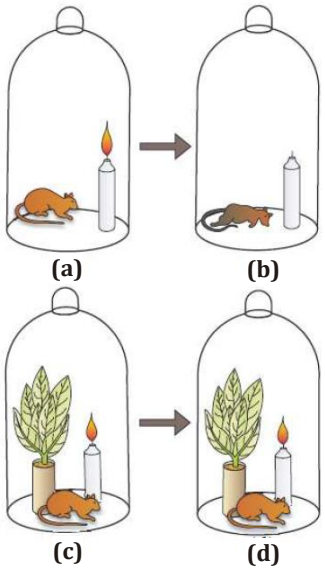
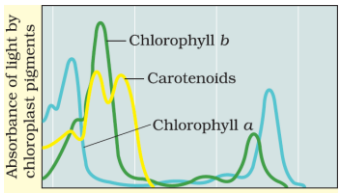
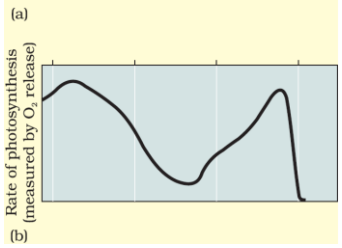
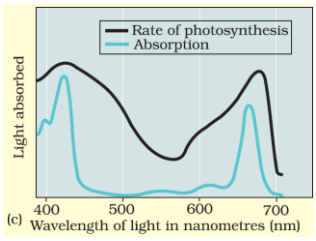
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|---|---|---|
| **13.** | Name of the place where accumulation of protons occur during chemiosmosis in chloroplast and mitochondria respectively. | Thylakoid lumen and intermembrane space of mitochondria |
| **14.** | The phase that leads to synthesis of food (sugar) in photosynthesis is called. | Biosynthetic phase |
| **15.** | The first stable product of Calvin cycle is. | 3-PGA |
| **16.** | The first stable product in $C_4$ pathway is. | OAA |
| **17.** | How many Calvin cycles are required for formation of one glucose? | 6 cycles |
| **18.** | How many ATP and NADPH will be required to make one glucose through the Calvin pathway? | 18 ATP and 12 NADPH |
| **19.** | When in plant leaves, chloroplast containing bundle sheath cells surrounds vascular bundles in several layers that anatomy is known as. | Kranz anatomy ($C_4$ Plants) |
| **20.** | Write the characteristics of bundle sheath cells of $C_4$ plants. | Large number of chloroplast Thick walls impervious to gaseous exchanges no intercellular spaces |
| **21.** | Give the example of $C_4$ plants | Sugarcane, Maize and Sorghum |
| **22.** | Write the enzyme which is present most abundantly in the world | RuBisCO |
| **23.** | For which substrate RuBisCO shows greater affinity? | $CO_2$ |
| **24.** | Name the products of photorespiration which are the result of $O_2$ binding to RuBP | Phosphoglycerate and phosphoglycolate |
| **25.** | Name the internal or plant factors that affect the rate of photosynthesis | Number, size, age and orientation of leaves, mesophyll cells, chloroplasts, internal $CO_2$ concentration and the amount of chlorophyll |
| **26.** | Name the external factors that affect the rate of photosynthesis | Availability of sunlight, temperature, external $CO_2$ concentration and water |
| **27.** | Blackman's law of limiting factor states that if a chemical process is affected by more than one factor, then it's rate will be determined by the factor which is nearest to its – | Minimal value (at suboptimal level) |
| **28.** | For which plants light is a limiting factor | Plants in shade |
| **29.** | Write the concentration of $CO_2$ at which $C_3$ and $C_4$ plants shows saturation respectively. | $C_3$ = 450 $\mu lL^{-1}$<br>$C_4$ = 360 $\mu lL^{-1}$ |

| | | |
|---|---|---|
| **30.** | Among light and dark reaction which reaction is more temperature affected? | Dark reaction (more enzymatic) |
| **31.** | Write the effect of water stress on plants | Reducing the surface area of leaves and metabolic activity and causes stomatal closure. |

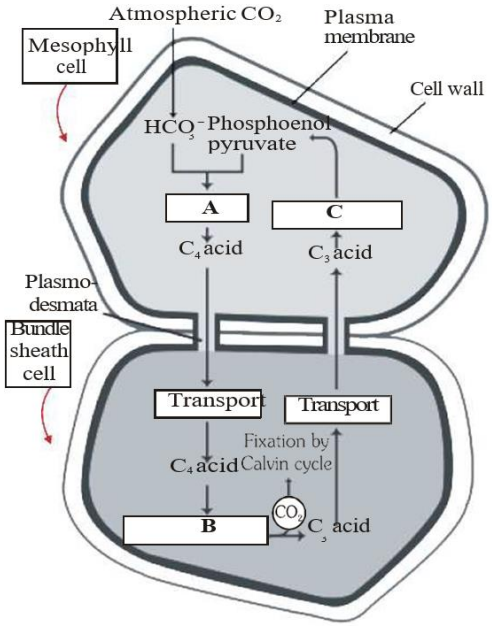
| Q.NO. | FILL IN THE BLANKS | ANSWERS |
|---|---|---|
| **1.** | In photosynthesis directly light driven reactions are called __(A)__ while some reactions are depends on the products of light reaction (ATP and NADPH) conventionally called as __(B)__. | A = light reactions<br>B = dark reactions |
| **2.** | Light reaction or the 'Photochemical' phase include __(A)__, __(B)__, __(D)__ and formation of __(D)__. | A = light absorption<br>B = water splitting<br>C = oxygen release<br>D = ATP and NADPH |
| **3.** | In PS I the reaction centre chlorophyll a has an absorption peak at 700 nm, hence is called __(A)__ while in PS II it has absorption maxima at 680 nm and is called __(B)__. | A = $P_{700}$<br>B = P |
| **4.** | $C_4$ plants are special. They have special type of __(A)__, they tolerate __(B)__, they show a response to __(C)__, they lack a process called __(D)__ and have greater __(E)__. | A = leaf anatomy<br>B = higher temperature<br>C = high light intensities<br>D = photorespiration<br>E = productivity of biomass |
| **5.** | In the photorespiratory pathway, there is neither synthesis of __(A)__ nor of __(B)__. Rather it results in the release of __(C)__ with the utilization of __(D)__. Therefore, photorespiration is a wasteful process. | A = sugar<br>B = ATP<br>C = $CO_2$<br>D = ATP |
| **6.** | The plant or internal factors affecting photosynthesis are dependent on the __(A)__ and the __(B)__ of the plant. | A = genetic predisposition<br>B = growth |
| **7.** | __(A)__ plants have a higher temperature optimum than the plants adapted to __(B)__ climates. | A = tropical<br>B = temperate |
| **8.** | Water stress causes stomata to __(A)__ hence reducing the __(B)__ availability. | A = close<br>B = $CO_2$ |

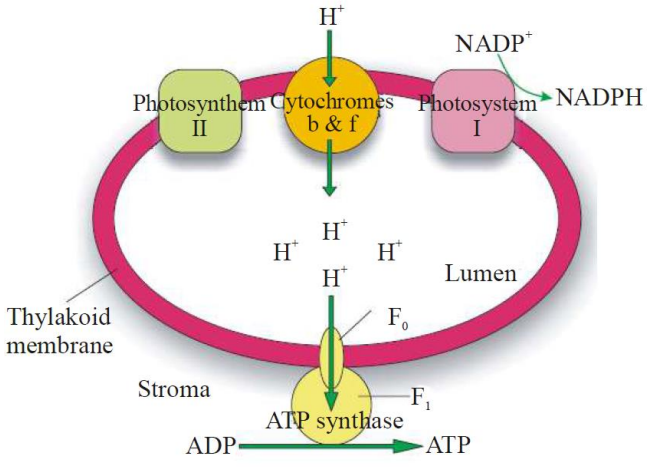
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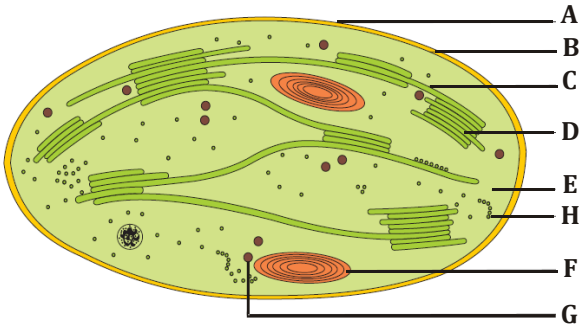
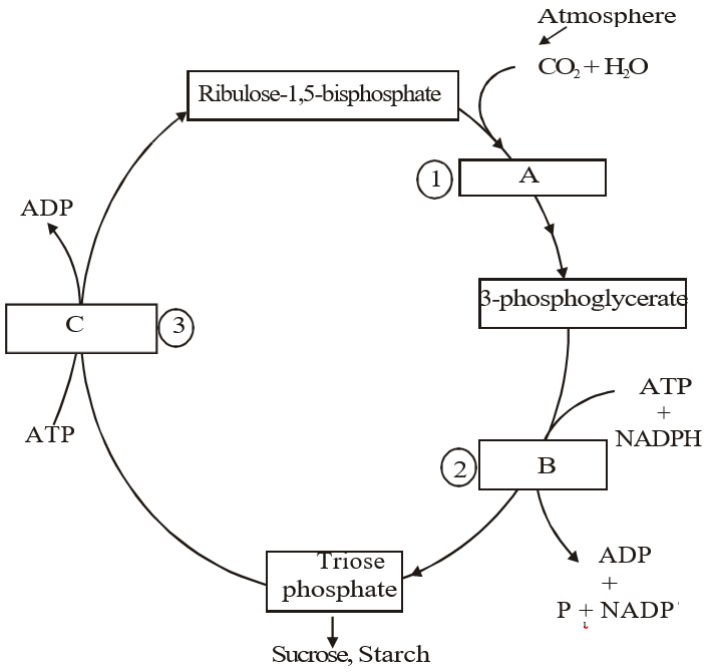
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| Q.NO. | TRUE & FALSE | ANSWERS |
|---|---|---|
| 1. | By observing given figure mark the statement true or false.<br>  <br>(a) (b) (c) (d)<br>By this experiment Priestley hypothesised that plant restore to the air whatever breathing animal and burning candle remove. | True |
| 2. | Observe the following graphs carefully and select the true or false statement regarding it :-<br>  <br><br>  <br><br>  <br><br>(i) Graph a indicates that chlorophyll a shows maximum absorption at wavelengths of blue and red colour<br>(ii) Graph b indicates that at wavelengths of blue and red colour, plants shows maximum rate of photosynthesis.<br>(iii) Graph c indicates complete one to one overlap between the absorption spectrum of chlorophyll a and the action spectrum of photosynthesis. | True<br>True<br>False |
| 3. | Photosystem I (PS I) and photosystem II (PS II) are named in the sequence in which they function during the light reaction. | False |
| 4. | The chlorophyll 'a' molecule forms the 'reaction centre' in photosystem. | True |

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| | | |
|---|---|---|
| **5.** | Calling the biosynthetic phase of photosynthesis as the dark reaction is a misnomer. | True |
| **6.** | Observe the given diagrammatic representation of the Hatch and Slack pathway and find out labelled parts and also choose true & false statements.<br>  <br><br>(i) In the diagram site of primary fixation lacks RuBisCO enzyme.<br>(ii) In mesophyll cell the transporting $C_4$ acid may be malic acid or aspartic acid.<br>(iii) The bundle sheath cells are rich in an enzyme RuBisCO but lack PEP case. | A = Fixation<br>B = Decarboxylation<br>C = Regeneration<br><br>True<br><br>True<br><br>True |

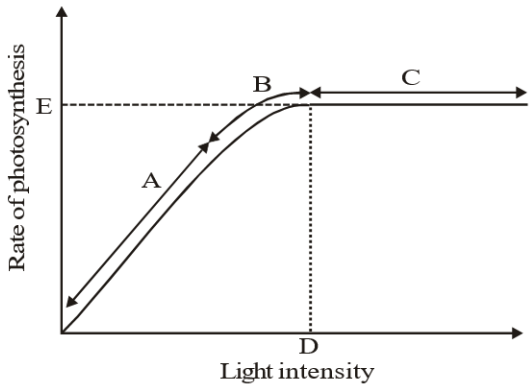
| Q.NO. | DIAGRAM BASED QUESTIONS | ANSWERS |
|---|---|---|
| **1.** | Observe the given figure of ATP synthesis through chemiosmosis and arrange the following events in correct sequence.<br>  <br> | (a)→(b)→(c)→(d)→(e) |

| | | |
|---|---|---|
| | (a) Movement of $e^-$ through H carrier<br>(b) Shifting of protons from stroma to Lumen<br>(c) Facilitated diffusion of protons across the membrane<br>(d) Conformational change in the $F_1$ particle of the ATP synthase<br>(e) Formation of ATP molecule | |
| **2.** | Observe the given figure that represents the section of chloroplast and identify the labelled parts according to their function.<br>  <br><br>(i) More permeable membrane<br>(ii) Less permeable membrane<br>(iii) Site containing only P.S. I<br>(iv) Site containing both P.S. I & P.S. II<br>(v) Site of dark reaction<br>(vi) Stored carbohydrate<br>(vii) Stored fat<br>(viii) Factory of protein | A-i, B-ii, C-iii, D-iv,<br>E-v, F-vi, G-vii, H-viii |
| **3.** | Observe the given figure of Calvin cycle and find the name of steps and number of ATP and NADPH utilised in them for one molecule of $CO_2$.<br>  <br> | **Steps ATP and NADPH**<br>A = Carboxylation Nil<br>B = Reduction 2ATP and 2NADPH<br>C = Regeneration 1ATP |



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| | | |
|---|---|---|
| **4.** | Observe the given graph carefully and choose the correct statements for labelled points (A to E).<br>  <br><br>**Points :-**<br>(A) There is a linear relationship between incident light and $CO_2$ fixation rates.<br>(B) It is a transition point where limiting factor changes.<br>(C) At this region light intensity is not showing detrimental effect on rate of photosynthesis.<br>(D) It shows light saturation point sunlight except for plants in shade or in dense forests.<br>(E) This point show maximum rate of photosynthesis at optimum presence of various parameters | All statements are correct |

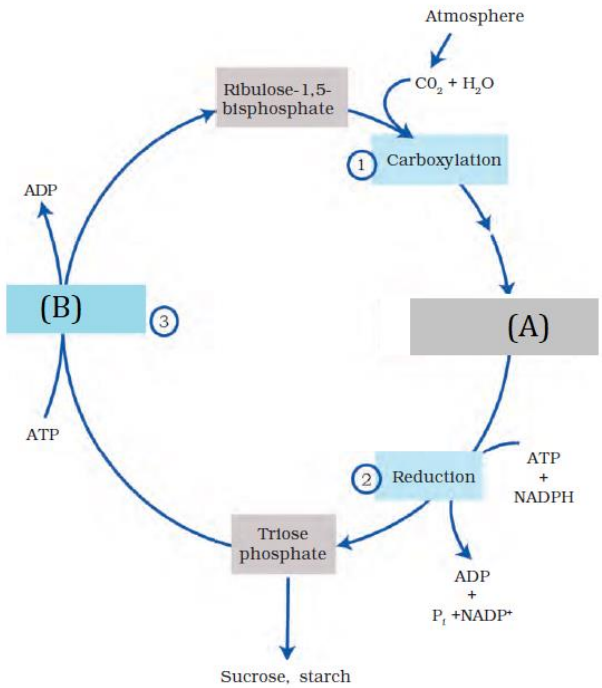
| Q.NO. | MATCH THE COLUMN | ANSWERS |
|---|---|---|
| **1.** | Match the column I with column II :-<br>**Column I** (Intensity of light) / **Column II** (Alignment of chloroplast)<br>(a) High — (i) Along the walls of mesophyll cells<br>(b) Low — (ii) Perpendicular to light<br>(c) Moderate — (iii) Dispersed throughout the cytoplasm | a–i, b–ii, c–iii |
| **2.** | Match the column I with column II :-<br>**Column I** (Pigment) / **Column II** (Colour)<br>(a) Chlorophyll a — (i) Yellow to yellow orange<br>(b) Chlorophyll b — (ii) Bright or blue green<br>(c) Xanthophylls — (iii) Yellow green<br>(d) Carotenoids — (iv) Yellow | a–ii, b–iii, c–iv, d–i |

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**3. Comparison of $C_3$ and $C_4$ plants :-**

| Characteristics | $C_3$ Plants | $C_4$ Plants | Choose from |
|---|---|---|---|
| Cell type in which the Calvin cycle takes place | Mesophyll | Bundle sheath | Mesophyll/Bundle sheath/both |
| Cell type in which the initial carboxylation reaction occurs | Mesophyll | Mesophyll | Mesophyll/Bundle sheath/both |
| How many cell types does the leaf have that fix $CO_2$? | One | Two | **One:** Mesophyll<br>**Two:** Bundle sheath and mesophyll<br>**Three:** Bundle sheath, palisade, spongy mesophyll |
| Which is the primary $CO_2$ acceptor? | RuBP | PEP | RuBP/PEP/PGA |
| Number of carbons in the primary $CO_2$ acceptor | 5 | 3 | 5 / 4 / 3 |
| Which is the primary $CO_2$ fixation product? | PGA | OAA | PGA/OAA/RuBP/PEP |
| No. of carbons in the primary $CO_2$ fixation product | 3 | 4 | 3/4/5 |
| Does the plant have RuBisCO? | Yes | Yes | Yes / No / Not Always |
| Does the plant have PEP Case? | Yes | Yes | Yes / No / Not Always |
| Which cells in the plant have Rubisco? | Mesophyll | Bundle sheath | Mesophyll/Bundle sheath/none |
| $CO_2$ fixation rate under high light conditions. | Medium | High | Low/ high/ medium |
| Whether photorespiration is present at low light intensities? | Negligible | Negligible | High/negligible/sometimes |
| Whether photorespiration is present at high light intensities? | High | Negligible | High/negligible/sometimes |
| Whether photorespiration would be present at low $CO_2$ concentrations? | High | Negligible | High/negligible/sometimes |
| Whether photorespiration would be present at high $CO_2$ concentrations? | High | Negligible | High/negligible/sometimes |
| Temperature optimum | 20-25°C | 30-40°C | 30-40°C/20-25°C/ above 40°C |
| Examples | Wheat / Rice | Maize, Sugarcane *Sorghum* | |

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| Q.No. | Question | Answer |
|---|---|---|
| 1. | In PS II the reaction centre of chlorophyll a has an absorption peak at 700 nm, hence is called $P_{700}$, while in PS I it has absorption maxima at 680 nm, and is called $P_{680}$. | False |
| 2. | The movement of electrons is downhill, in terms of an oxidation-reduction or redox potential scale in ETS. | True |
| 3. | The whole scheme of transfer of electrons in non-cyclic photophosphorylation, starting from the PS II, uphill to the acceptor, down the electron transport chain to PS I, excitation of electrons, transfer to another acceptor, and finally up hill to $NADP^+$ reducing it to NADPH + $H^+$ is called the Z scheme. | False |
| 4. | The excited electron in PS I in cyclic photophosphorylation does not pass on to $NADP^+$ but is cycled back to the PS I complex through the electron transport chain. Hence, results only in the synthesis of ATP, but not of NADPH + $H^+$ | True |
| 5. | _____ occurs when only light of wavelengths beyond 680 nm are available for excitation. | Cyclic photophosphorylation |
| 6. | The use of radioactive $^{14}C$ in algal photosynthesis studies led to the discovery of : | first $CO_2$ fixation product was (3-carbon organic acid) |
| 7. | Carboxylation is the most crucial step of the Calvin cycle where $O_2$ is utilised for the carboxylation of RuBP. | False |
| 8. | The _____ plants respond to higher temperatures while _____ plants have a much lower temperature optimum. | $C_4$, $C_3$ |
| 9. | Identify **A** and **B** in given below diagram :   | A - 3-phosphoglycerate<br>B - Regeneration |

**10.** Fill in the given space in given below table :

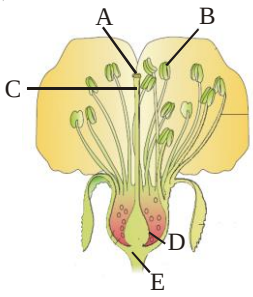
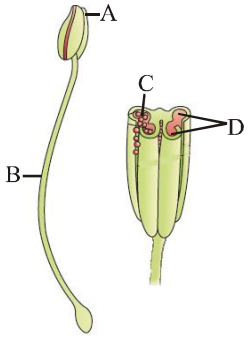
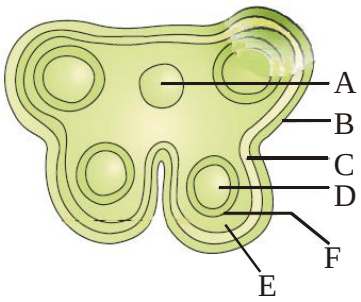
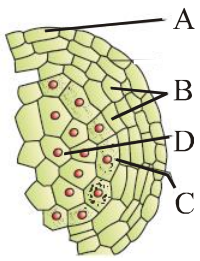
| Characters | $C_3$ Plants | $C_4$ Plants |
|---|---|---|
| Fixation of No. of $CO_2$ | 6 | 18 |
| No. of ATP required | (i) | (ii) |
| No. $NADPH_2$ required | (iii) | (iv) |

(i)-18, (ii)-90,
(iii)-12, (iv)-36

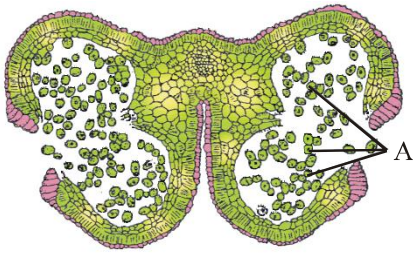
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# Sexual Reproduction in Flowering Plants

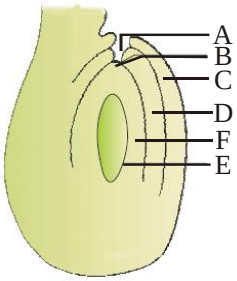
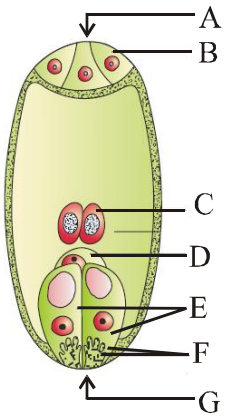
| Q.NO. | QUESTION | ANSWER |
|---|---|---|
| 1. |  Write down the name of A, B, C, D and E in given figure. | A-Stigma,<br>B-Anther,<br>C-Style,<br>D-Ovary,<br>E-Thalamus |
| 2. |  Write down the name of A, B, C and D in given figure. | A-Anther,<br>B-Filament,<br>C-Pollen grain,<br>D-Pollen sacs |
| 3. |  Write down the name of parts A, B, C, D, E and F in given figure. | A-Connective,<br>B-Epidermis,<br>C-Endothecium,<br>D-Sporogenous tissue<br>E-Middle layer,<br>F-Tapetum |
| 4. |  Write down the name of parts A, B, C, and D in given figure. | A-Epidermis,<br>B-Middle layers,<br>C-Tapetum,<br>D-Microspore mother cell |

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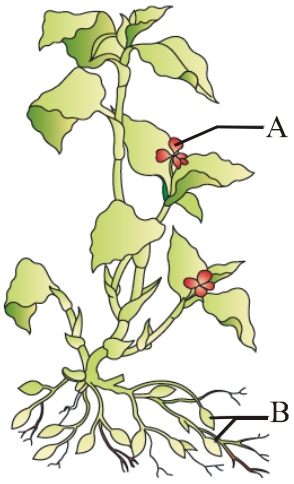
<table>
<tr><td>5.</td><td> 

<br>Write the name of labelled structures 'A'</td><td>A-Pollen grains</td></tr>
<tr><td></td><td>FILL IN THE BLANKS</td><td>ANSWER</td></tr>
<tr><td>6.</td><td>The microsporangium develops further and becomes _____.</td><td>Pollen sac</td></tr>
<tr><td>7.</td><td>The innermost wall layer of microsporangium is the _____.</td><td>Tapetum</td></tr>
<tr><td>8.</td><td>In the centre of each microsporangium a compactly arranged tissue is _____.</td><td>Sporogenous tissue</td></tr>
<tr><td>9.</td><td>There are the two part of a typical ___, the long and slender stalk is called __ and the terminal generally bilobed structure is called ____.</td><td>Stamen, Filament, anther</td></tr>
<tr><td>10.</td><td>A typical angiosperm anther is ____ with each lobe having two theca i.e. they are called ____.</td><td>Bilobed, dithecous</td></tr>
<tr><td>11.</td><td>The hard outer layer of pollen grain, called exine is made up of ____.</td><td>Sporopollenin</td></tr>
<tr><td></td><td>TRUE AND FALSE</td><td>ANSWER</td></tr>
<tr><td>12.</td><td>In pollen grain exine has prominent apertures called germ pores where sporopollenin is absent.</td><td>True</td></tr>
<tr><td>13.</td><td>The inner wall of pollen grain is called intine.</td><td>True</td></tr>
<tr><td>14.</td><td>In microsporogenesis, PMC undergo mitosis and form pollen grains.</td><td>False</td></tr>
<tr><td>15.</td><td>Pollen grains can be stored in liquid nitrogen at –196°C.</td><td>True</td></tr>
<tr><td>16.</td><td>Wheat and rice pollen grains has pollen viability upto months.</td><td>False</td></tr>
<tr><td>17.</td><td>Match the following :-
<table>
<tr><td>1.</td><td>Proximal part of filament</td><td>a.</td><td>Tapetum</td></tr>
<tr><td>2.</td><td>Inner most wall layer of pollen sac</td><td>b.</td><td>attached to the thalamus</td></tr>
<tr><td>3.</td><td>Sporopollenin</td><td>c.</td><td>most resistant material</td></tr>
<tr><td>4.</td><td>Ploidy of the cells of the tetrad</td><td>d.</td><td>Haploid</td></tr>
</table></td><td>1 - b, 2 - a, 3 - c, 4 - d</td></tr>
<tr><td>18.</td><td>In a mature pollen, how many nuclei are present and write their name also.</td><td>two nuclei, 1-vegetative nucleus 1-generative nucleus</td></tr>
<tr><td>19.</td><td>Name any plant that causes pollen allergy.</td><td>Parthenium</td></tr>
<tr><td>20.</td><td>What is role of pollen tablets in athletes and race horses?</td><td>to increase their performance</td></tr>
</table>

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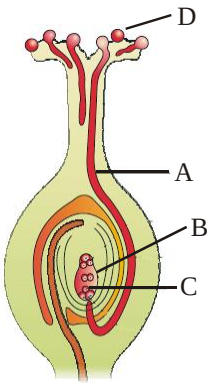
| | | |
|---|---|---|
| **21.** | Where is the use of pollen bank. | Crop breeding programmes |
| **22.** | What is the diameter of pollen grain | 25-50 micrometer |
| **23.** |  <br>In the given figure pistil, identify the labelling's A,B,C & D. | A-Stigma,<br>B-Style<br>C-Ovary,<br>D-thalamus |
| **24.** |  <br>Identify A, B, C, D, E and F in the given figure of ovule. | A-Micropyle,<br>B-Micropylar end,<br>C-Outer integument,<br>D-Inner integument,<br>E-Embryosac,<br>F-Nucellus |
| **25.** |  <br>Identify A, B, C, D, E, F and G | A-Chalazalend<br>B-Antipodal cell<br>C-Polar nuclei<br>D-Egg cell<br>E-Synergid cells<br>F-Filiform apparatus<br>G-Micropylar end |
| | **FILL IN THE BLANKS** | **ANSWER** |
| **26.** | The basal bulged part of the pistil is _____ | Ovary |
| **27.** | _____ serves as a landing platform for pollen grains | Stigma |
| **28.** | The mitotic divisions which occur during embryo sac formation from megaspore are _____. | Strictly free nuclear divisions |

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<table>
<tr><th></th><th>TRUE AND FALSE</th><th>ANSWER</th></tr>
<tr><td>29.</td><td>An ovule has generally one embryo sac.</td><td>True</td></tr>
<tr><td>30.</td><td>Each ovule has one or two protective envelopes, called exine and intine</td><td>False</td></tr>
<tr><td>31.</td><td>Hilum is the region where the body of ovule fuses with funiculus/ funicle.</td><td>True</td></tr>
<tr><td>32.</td><td>Nucellus is a mass of cells enclosed within integuments.</td><td>True</td></tr>
<tr><td>33.</td><td>Only the non-functional megaspore develops into female gametophyte.</td><td>False</td></tr>
<tr><td>34.</td><td>When a embryo sac develops from a single megaspore, it is called monosporic development.</td><td>True</td></tr>
<tr><td>35.</td><td>Match the following :-
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>1</td><td>Central cell</td><td>A</td><td>Egg apparatus</td></tr>
<tr><td>2</td><td>One egg cell and two synergids</td><td>B</td><td>Chalazal end</td></tr>
<tr><td>3</td><td>Antipodal cell</td><td>C</td><td>Meiosis</td></tr>
<tr><td>4</td><td>Megaspore mother cell</td><td>D</td><td>Polar nuclei</td></tr>
<tr><td>5</td><td>Female gametophyte</td><td>E</td><td>Embryosac</td></tr>
</table></td><td>1-D, 2-A, 3-B, 4-C, 5-E</td></tr>
<tr><td>36.</td><td>In flowering plants, female reproductive whorl is called.</td><td>Gynoecium</td></tr>
<tr><td>37.</td><td>What we say if a gynoecium of flower has many carpels.</td><td>Multicarpellary</td></tr>
<tr><td>38.</td><td>If a gynoecium of a flower has many carpels and they are fused, this condition is called.</td><td>Multicarpellary, syncarpous</td></tr>
<tr><td>39.</td><td>The process of formation of megaspores from the megaspore mother cell by meiosis, called.</td><td>Megasporogenesis</td></tr>
<tr><td>40.</td><td>In a typical angiosperm, mature embryosac is _____ celled.</td><td>7/Seven</td></tr>
<tr><td>41.</td><td> 

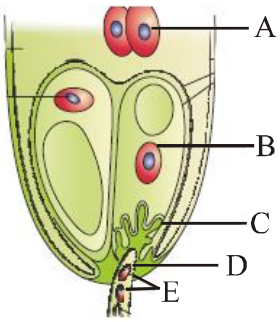
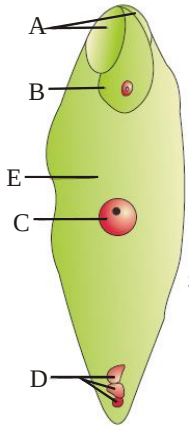
Identify A and B in this figure.</td><td>A-Chasmogamous flower<br>B-Cleistogamous flowers</td></tr>
<tr><td></td><td>FILL IN THE BLANKS</td><td></td></tr>
<tr><td>42.</td><td>The transfer of pollen grains from ___ to ___ is known as ___.</td><td>Anther, stigma, pollination</td></tr>
<tr><td>43.</td><td>Transfer of pollen grains from ___ to stigma of same flower is called _____.</td><td>Anther, Autogamy</td></tr>
</table>

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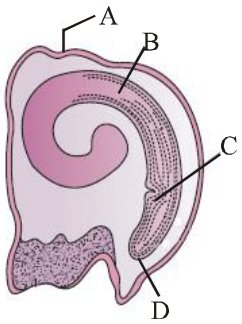
<table>
<tr><td>44.</td><td>For the formation of embryo sac the ____ undergoes one ____ division and three ____ divisions in megaspore nucleus.</td><td>Megaspore mother cell, Meiotic, Mitotic</td></tr>
<tr><td>45.</td><td>Transfer of pollen from anther to stigma of another flower of same plant is called ___.</td><td>Geitonogamy</td></tr>
<tr><td></td><td>TRUE/FALSE</td><td>ANSWER</td></tr>
<tr><td>46.</td><td>Xenogamy is the only type of pollination in which genetically different pollens germinate on stigma.</td><td>True</td></tr>
<tr><td>47.</td><td>Water is the most common pollinating agent in abiotic pollination.</td><td>False</td></tr>
<tr><td>48.</td><td>Wind - pollination is quite common in grasses.</td><td>True</td></tr>
<tr><td>49.</td><td>In Vallisneria the pollinating agent is water.</td><td>True</td></tr>
<tr><td>50.</td><td>Dioecious condition prevents autogamy and geitonogamy.</td><td>True</td></tr>
<tr><td>51.</td><td>Match the following :-
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>1</td><td>Out breeding</td><td>A</td><td>Water Pollination</td></tr>
<tr><td>2</td><td>Chasmogamous and cleistogamous</td><td>B</td><td>Prevents self pollination</td></tr>
<tr><td>3</td><td>Amorphophallus</td><td>C</td><td>Oxalis</td></tr>
<tr><td>4</td><td>Zostera</td><td>D</td><td>Tallest flower</td></tr>
</table></td><td>1-B, 2-C, 3-D, 4-A</td></tr>
<tr><td>52.</td><td>Which genetic mechanism prevents the self-pollination</td><td>Self incompatibility</td></tr>
<tr><td>53.</td><td>Which type of pollination is found in cleistogamous flower</td><td>Self pollination</td></tr>
<tr><td>54.</td><td>How many cells are present in a mature pollen grain</td><td>2-cells</td></tr>
<tr><td>55.</td><td>Which type of pollen grains germinate on the stigma</td><td>Compatible pollen grains.</td></tr>
<tr><td>56.</td><td> 

Identify A, B, C and D in this figure.</td><td>A-Pollen tube,<br>B-Embryosac<br>C-Egg cell,<br>D-Pollen grain</td></tr>
</table>

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| | | |
|---|---|---|
| **57.** |  <br>Identify A, B, C, D and E in this figure. | A-Polar nuclei,<br>B-Synergid cell,<br>C-Filiform apparatus,<br>D-Pollen tube,<br>E-male gametes |
| **58.** |  <br>Identify A, B, C, D and E in this figure of fertilized embryosac. | A-Degenerating synergids<br>B-Zygote<br>C-Primary endosperm nucleus (PEN)<br>D-Degenerating antipodal cells<br>E-Primary endosperm cell (PEC) |
| | **FILL IN THE BLANKS** | **ANSWER** |
| **59.** | Removal of anthers from the flower bud is called ___. | Emasculation |
| **60.** | Two polar nuclei fuse together and form _____. | Secondary nucleus |
| **61.** | First male gamete fuses with egg cell nucleus and form ___. | Zygote |
| **62.** | Ploidy of the primary endosperm nucleus in typical angiosperms is _____. | 3n (Triploid) |
| | **TRUE AND FALSE** | **ANSWER** |
| **63.** | Endosperm is a nutritive tissue. | True |
| **64.** | In coconut the endosperm is free-nuclear endosperm. | True |
| **65.** | Ovary is a ripened ovule. | False |
| **66.** | The study of development of embryo is called embryology. | True |
| **67.** | Two-cotyledons are found in monocot seeds. | False |
| **68.** | In mango, the mesocarp is edible part. | True |
| **69.** | Seeds of pea and ground nut are non endospermic. | True |
| **70.** | Seeds of maize and wheat are endospermic. | True |
| **71.** | What is perisperm? | residual, persistent nucellus in seed |
| **72.** | The term which is used for fruits which are developed without fertilisation is _____. | Parthenocarpic fruits |

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| | | |
|---|---|---|
| **73.** |  <br>Identify A, B, C and D in figure of a seed. | A-Seed coat<br>B-Cotyledon<br>C-Shoot apical meristem<br>D-Root tip |
| **74.** | Occurrence of more than one embryo in a seed is referred to as :- | Polyembryony |
| **75.** | Pollen grains are well preserved due to the presence of _____. | Sporopollenin |
| **76.** | _____ forms the pollen tube. | Intine |
| **77.** | Nucellus is surrounded by protective covering called _____. | Integuments |
| **78.** | The functional megaspore develops into _____. | Female gametophyte |
| **79.** | In groundnut and mustard, pericarp is _____. | Dry |
| | **TRUE AND FALSE** | **ANSWER** |
| **80.** | Aleurone layer is triploid in nature. | True |
| **81.** | In grass family the cotyledon is called scutellum. | True |
| **82.** | Five nuclei and 2 gametes participates in double fertilization. | False |
| **83.** | A typical anther is bilobed and tetrasporangiate. | True |
| **84.** | Ploidy of pollen grain is haploid. | True |
| **85.** | Apomixis results in formation of seeds without fertilization. | True |
| **86.** | Only compatible pollen grain germinates on the stigma of pistil. | True |
| **87.** | Angiosperms exhibit :<br>(1) Syngamy (2) Triple fusion<br>(3) Both (1) and (2) (4) None of the above | **(3)** |
| **88.** | The period for which pollen grains remain viable depends on:<br>(1) Temperature (2) Humidity<br>(3) Species (4) All of the above | **(4)** |
| **89.** | The portion of embryonal axis above the level of cotyledons is:<br>(1) Coloerhiza (2) Coleoptile<br>(3) Epicotyl (4) Hypocotyl | **(2)** |
| **90.** | **Assertion :** Most zygotes divide only after certain amount of endosperm is formed.<br>**Reason:** This is an adaptation to provide assured nutrition to the developing embryo.<br>(1) Both A and R are correct but R is not the correct explanation of A.<br>(2) A is correct but R is incorrect.<br>(3) A is not correct but R is correct.<br>(4) A and R both are correct and R is the correct explanation of A. | **(4)** |

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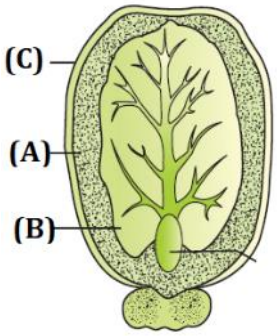
| | | |
|---|---|---|
| **91.** | Some plants such as *Viola* (common pansy), *Oxalis*, and *Commelina* produce two types of flowers _____. | Chasmogamous flowers<br>cleistogamous flowers |
| **92.** | The production of seed without fertilization is called _____. | Apomixis |
| **93.** | The tallest flower is _____. | *Amorphophallus* |
| **94.** | Size of pollen grain is _____. | 25-50 micrometers |
| **95.** | Removal of anthers from the flower bud before the anther dehisces is called _____. | Emasculation |
| **96.** | Identify the diagram given below :-<br>  <br>(1) Multicarpellary, apocarpous gynoecium of *Papaver*<br>(2) Multicarpellary, syncarpous pistil of *Papaver*<br>(3) Multicarpellary, apocarpous gynoecium of *Michelia*<br>(4) Multicarpellary, syncarpous pistil of *Michelia* | **(3)** |
| **97.** | **Assertion :** Pollen tablets increase the performance of athletes and race horses .<br>**Reason :** Exine is made up of sporopollenin.<br>(1) Both A and R are correct but R is not the correct explanation of A.<br>(2) A is correct but R is incorrect.<br>(3) A is not correct but R is correct.<br>(4) A and R both are correct and R is the correct explanation of A. | **(1)** |
| **98.** | Match column A with column B.<br>**Column-A** / **Column-B**<br>(a) *Zostera* — (i) Secondary nucleus and male gamete<br>(b) *Vallisneria* — (ii) Epihydrophily<br>(c) Primary endosperm nucleus — (iii) Synergids and egg cell<br>(d) Egg apparatus — (iv) Hypohydrophily<br>(1) a-ii, b-iii, c-iv, d-I<br>(2) a-iv, b-ii, c-i, d-iii<br>(3) a-ii, b-iii, c-i, d-iv<br>(4) a-iv, b-ii, c-iii, d-i | **(2)** |

**99.** Match column A with column B. **(2)**

| | Column-A | | Column-B |
|---|---|---|---|
| (a) | Apple | (i) | Endosperm |
| (b) | Grapes | (ii) | Pseudocarp |
| (c) | Beet | (iii) | Parthenocarpic |
| (d) | Castor | (iv) | Perisperm |

(1) a-i, b-iv, c-ii, d-iii
(2) a-ii, b-iii, c-iv, d-i
(3) a-i, b-iv, c-iii, d-ii
(4) a-ii, b-iv, c-ii, d-iii

**100.** Identify the following parts A, B, C and D :- **(3)**




(1) A-Seed coat, B- Cotyledon, C-Endosperm
(2) A-Seed coat, B-Hypocotyl, C- Cotyledon
(3) A-Endosperm, B-Cotyledon, C-Seed coat
(4) A-Cotyledon, B-Endosperm, C-Seed coat

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# Principles of Inheritance and Variation

| Q.NO. | QUESTION | ANSWER |
|---|---|---|
| 1. | Dominance is not an autonomous feature of a gene in case more more than one phenotype is influences by the same gene. | TRUE |
| 2. | Mendelian recombinants are due to crossing over. | FALSE |
| 3. | Mendel Conducted Somatic Hybridization experiments on garden pea. | FALSE |
| 4. | Two alleles of a gene are located on homologous sites on non homologous chromosomes. | FALSE |
| 5. | Mutation is a phenomenon which results in alteration of DNA sequences. | TRUE |
| 6. | *Drosophila* complete their life cycle in four weeks. | FALSE |
| 7. | In *Drosophila* single mating could produce large number of progeny flies. | TRUE |
| 8. | *Drosophila* has many types of hereditary variations that can be seen with low power microscopes. | TRUE |
| 9. | A true breeding line is one that, having undergone continuous self pollination, shows the stable trait inheritance and expression for several generations. | TRUE |
| 10. | Mendel selected seven visible characters each with two contrasting traits. | TRUE |
| 11. | The gene which affect more than one phenotype called pleiotropic gene. | TRUE |
| 12. | In a test cross. $F_1$ hybrids are crossed with homozygous dominant plants. | FALSE |
| 13. | Inheritance of one character is always independent to the inheritance of other character. | FALSE |
| 14. | *Pisum sativum* reproduces asexually in nature. | FALSE |
| 15. | Emasculation is the process of removal of pistils from unisexual flowers. | FALSE |
| 16. | Sickle cell anaemia follows the same inheritance pattern as that Haemophilia. | FALSE |
| 17. | Mendel experiments had a large sampling size. | TRUE |
| 18. | In a dissimilar pair of factors one member of the pair dominates the other. | TRUE |
| 19. | Law of segregation is based on the fact that the alleles do not show any blending | TRUE |
| 20. | Multiple alleles can be found only when population studies are made. | TRUE |

| | | |
|---|---|---|
| **21.** | Mendel widely publicised his work. | FALSE |
| **22.** | Mendel's approach of using mathematics to explain biological phenomena was totally new. | TRUE |
| **23.** | Mendel could not provide any physical proof for the existence of factor. | TRUE |
| **24.** | Chromosomes as well as genes occurs in pairs. | TRUE |
| **25.** | Gene for same character segregate at the time of _____. | Gamete formation |
| **26.** | Genes and chromosomes are found in pairs in _____ cells. | Diploid |
| **27.** | Alleles are slightly different forms of the same _____. | Gene |
| **28.** | When modified allele is equivalent to the unmodified allele it will produce _____. | Same phenotype |
| **29.** | *Drosophila* complete their life cycle in about..... weeks | two |
| **30.** | Grasshopper is an example of _____ type of sex detamination | XO |
| **31.** | _____ disease shows its transmission from unaffected carrier female to some of the male progeny | Sex linked recessive |
| **32.** | The possibility of a female becoming a haemophilic is extremely rare because mother of such a female has to be at least carrier and the father should be _____. | haemophilic |
| **33.** | In a monohybrid cross, we study _____ character. | One |
| **34.** | Holandric genes are present on _____. | Y-chromosome |
| **35.** | Behaviour of Chromosomes was parellel to the behaviour of genes explained by _____. | Sutton and Boveri |
| **36.** | During spermatogenesis in human males, _____ type of gametes are produced. | two |
| **37.** | The gene, which affects many phenotypes. | Pleiotropic gene. |
| **38.** | Mendelian recombinants are due to _____. | Independent assortment |
| **39.** | Genes present on same chromosome are inherited together called : | Linked gene |
| **40.** | Characters are controlled by discrete units called :- | Factors |
| **41.** | In which case $F_1$ generation resembled either of the two parents? | Dominance |
| **42.** | In human beings 'ABO' Blood group is controlled by which gene? | Gene I |
| **43.** | How many phenotypes are possible in human 'ABO' Blood group? | Four |
| **44.** | Chromosomal theory of inheritence was proposed by. | Sutton & Boveri |
| **45.** | All genes on single chromosome forms :- | Linkage group |
| **46.** | Holandric genes are always transfer from father to .... | Son |
| **47.** | Sutton united the knowledge of chromosomal segregation with Mendelian principles and called it the _____. | Chromosomal theory of Inheritance |
| **48.** | The unmodified allele, which represents the original phenotype is the dominant allele and the modified allele is generally.... | Recessive |

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**49.** Recessive trait is seen due to _____.

Non-functional enzyme or no enzyme.

**50.** Match the columns :-

| Column-I | | Column-II | |
|---|---|---|---|
| A | Monohybrid cross | 1. | Codominance |
| B | 'AB' Blood group | 2. | Correns |
| C | Dihybrid Cross | 3. | Dominance |
| D | Mirabilis jalapa | 4. | Independent assortment |

A-3, B-1, C-4, D-2

**51.** Match the columns

| Column-I | | Column-II | |
|---|---|---|---|
| A | Hypertrichosis | 1. | Autosomal Recessive |
| B | Colourblindness | 2. | Y-chromosome |
| C | Sickle cell Anaemia | 3. | X-Linked |

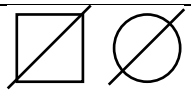
A-2, B-3, C-1

**52.** Match the column

| Column-I | | Column-II | |
|---|---|---|---|
| 1 | Sickle cell anaemia | (A) | Haemophilia |
| 2 | Bleeder's disease | (B) | Hugo de vries |
| 3 | Mutation | (C) | 'O' Blood group |
| 4 | Universal donor | (D) | Mendel |
| 5 | *Pisum sativum* | (E) | Sickle shape RBCs |

1-(E), 2-(A), 3-(B), 4-(C), 5-(D)

**53.**

| Column-I | | Column-II | |
|---|---|---|---|
| (A) |  | (i) | Death |
| (B) |  | (ii) | Five unaffected offsping |
| (C) |  | (iii) | Sex unspecified |
| (D) |   | (iv) | Female |
| (E) |  | (v) | male |

(A)-v, (B)-iv, (C)-iii, (D)-ii, (E)-i

**54.** Sahiwal cow is the result of _____.

Artificial selection and domestication

**55.** Mendel confirmed his inferences from experiments on successive generations of his test plants.

True

**56.** Mendel selected 14 true-breeding pea plant varieties, as pairs that were similar for all characters with contrasting traits

False

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| | | |
|---|---|---|
| **57.** |  <br>This process is not required in :<br>(A) Unisexual flower<br>(B) Bisexual flower | Unisexual flower |
| **58.** | The proportion of plants that were dwarf was _____ of the $F_2$ plants while $3/4^{th}$ of the $F_2$ plants were tall in Mendel's monohybrid cross. | $1/4^{th}$ |
| **59.** | Based on the observations, Mendel proposed that something was being stably passed down, unchanged, from parent to offspring through the gametes, over successive generations. He called these things as _____. | Factors |
| **60.** | Who worked on below mentioned organism.<br>  | T. H. Morgan |
| **61.** | Genes which code for a pair of contrasting traits are known as _____. | Alleles |
| **62.** | By the process of mitosis, the alleles of the parental pair separate or segregate from each other. | False |
| **63.** | The segregation of alleles is a random process. | True |
| **64.** | By simply looking at the phenotype of a dominant trait, it is possible to know the genotypic composition. | False |
| **65.** | In which cross an organism (pea plants here) showing a dominant phenotype (and whose genotype is to be determined), is crossed with the recessive parent? | Test cross |
| **66.** | The law of dominance is used to explain _____. | The expression of only one of the parental characters in a monohybrid cross in the $F_1$ and the expression of both in the $F_2$ generation. |
| **67.** | Which law is based on the fact that the alleles do not show any blending? | Law of segregation |
| **68.** | During _____, the factors or alleles of a pair segregate from each other. | Gamete formation |
| **69.** | Incomplete dominance can be shown in pea plants. | True |

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| | | |
|---|---|---|
| **70.** | When the $F_1$ generation (showing incomplete dominance) was self-pollinated, the $F_2$ generation resulted in the following ratio _____ (in Dog flower). | 1 Red: 2 Pink: 1 White |
| **71.** | In which case, the modified allele is equivalent to the unmodified allele? | When modified allele produces normal/less efficient enzyme |
| **72.** | If the allele produces a non-functional enzyme or no enzyme, the phenotype may be affected. | True |
| **73.** | Which of the three alleles controlling ABO blood grouping, does not produce any sugar on the surface of RBC? | $I^0$ or i |
| **74.** | An example of ABO blood grouping also provides a good example of pleiotropy. | False |
| **75.** | Starch synthesis in garden pea seeds is controlled by……. Gene (how many?) | One |
| **76.** | Dominance is an autonomous feature of a gene or the product that it has information for. | False |
| **77.** | Based upon observations on dihybrid crosses (crosses between plants differing in two traits) Mendel proposed a second set of generalisations that we call _____. | Mendel's Law of Independent Assortment |
| **78.** | When two pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters. This law is called _____. | Mendel's Law of Independent Assortment |
| **79.** | Mendel's approach of using mathematics to explain biological phenomena was widely accepted. | False |
| **80.** | In which year, three Scientists (de Vries, Correns, and von Tschermak) independently rediscovered Mendel's results on the inheritance of characters.? | 1900 |
| **81.** | The structures in the nucleus, that appeared to double and divide just before each cell division, were called _____. | Chromosomes |
| **82.** | Who noted that the behaviour of chromosomes was parallel to the behaviour of genes? | Walter Sutton and Theodore Boveri |
| **83.** | The two alleles of a gene pair are located on homologous sites on homologous chromosomes. | True |
| **84.** | 'Independent pairs segregate independently of each other.' This statement is true for _____. | Genes and chromosomes both |
| **85.** | During Anaphase of meiosis I, the two chromosome pairs can align at the _____ independently of each other. | Metaphase plate |
| **86.** | Who argued that the pairing and separation of a pair of chromosomes would lead to the segregation of a pair of factors they carried. | Sutton and Boveri |
| **87.** | Experimental verification of the chromosomal theory of inheritance was done by _____. | Thomas Hunt Morgan and his colleagues |
| **88.** | Morgan carried out several dihybrid crosses in *Drosophila* to study genes that were sex-linked. The crosses were totally different from the dihybrid crosses carried out by Mendel in peas. | False |

| | | |
|---|---|---|
| **89.** | Morgan coined the term _____ to describe this physical association of genes on a chromosome. | linkage |
| **90.** | Who used the frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes and 'mapped' their position on the chromosome? | Alfred Sturtevant |
| **91.** | The following diagram shows a genetic condition known as ::  | Down's syndrome |
| **92.** | An example of polygenic trait in humans is - | Skin colour |
| **93.** | Who could trace a specific nuclear structure all through spermatogenesis in a few insects, and it was also observed by him that 50 per cent of the sperm received this structure after spermatogenesis. | Henking |
| **94.** | In humans eggs fertilised by sperm having an X-chromosome become _____. | Females |
| **95.** | Example of male heterogamety in mammals. | Humans |
| **96.** | Male honeybees have half the number of chromosomes as female honeybee. | True |
| **97.** | Alteration in chromosomes results in _____. | Abnormalities or aberrations |
| **98.** | _____ are commonly observed in cancer cells. | Chromosomal aberrations |
| **99.** | Mutation also arises due to changes in a single base pair of DNA. This is known as point mutation. A classical example of such a mutation is _____. | Sickle cell anaemia |
| **100.** | Mendelian disorders are mainly determined by alteration or mutation in the single gene. | True |
| **101.** | Most common and prevalent Mendelian disorders are _____. | Haemophilia, Cystic fibrosis, Sickle cell anaemia, Colour blindness, Phenylketonuria, Thalassemia |
| **102.** | Colour blindness is a sex-linked recessive disorder due to defect in _____ resulting in failure to discriminate between red and green colour. | Either red or green cone of eye |
| **103.** | Sickle-cell anaemia is an autosome linked recessive trait. | True |

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# Molecular Basis of Inheritance

| Q.NO. | QUESTION | ANSWER |
|---|---|---|
| 1. | In living world, most abundant genetic material is | DNA |
| 2. | Which end of a polynucleotide chain has a free phosphate moiety ? | 5'–end |
| 3. | 5-methyl uracil is another chemical name for | Thymine |
| 4. | The backbone in a polynucleotide chain is formed due to | Sugars and phosphates |
| 5. | What generates approximately uniform distance between the two strands of DNA double helix | Purine–pyrimidine hydrogen bonding |
| 6. | Flow of information from RNA to DNA is called | Reverse transcription |
| 7. | Histones are rich in which basic amino acids ? | Lysine and Arginine |
| 8. | A segment of DNA coding for a polypeptide is called as | Cistron |
| 9. | The RNA polymerase is only capable of catalysing the process of | Elongation |
| 10. | In eukaryotes, the hnRNA is transcribed by | RNA polymerase II |
| 11. | Which scientist suggested the code should be made up of three nucleotides ? | Geoge gamow |
| 12. | In their experiments, Hershey and Chase used radioative _____ labelled DNA and radioactive _____ labelled protein capsule. | $P^{32}$, $S^{35}$ |
| 13. | The genetic material must have four qualities, These are : | 1. Replicability<br>2. Stability<br>3. Mutability<br>4. Expressibility |
| 14. | Which centrifugation technique was used by Meselson and Stahl in their experiment which proved semi-conservative DNA replication? | CsCl - density gradient Centrifugation |
| 15. | Most Prokaryotic genes are | Polycistronic and Non-split |
| 16. | Most eukaryotic genes are | Monocistronic and Split |
| 17. | Which scientist formed co-polymer of nucleotides | H.G. Khurana |
| 18. | Some amino acids are coded by more than one codon, hence the code is | Degenerate |
| 19. | A classical example of point mutation is | Sickle cell anaemia |
| 20. | Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion. Such mutations are referred to as | Frame-shift mutations |
| 21. | The t-RNA is a compact molecule which in 3-D looks like | Inverted L |
| 22. | An m-RNA also have some additional sequences that are not translated and are referred to as | Untranslated regions (UTR) |
| 23. | In eukaryotes, the regulation of gene expression could be exerted at | Four levels |
| 24. | In most cases the sequence of the operator is bound by | Repressor protein |

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| | | |
|---|---|---|
| **25.** | Each operon has its specific operator and specific | Repressor |
| **26.** | The operon concept was given by | Jacob and Monod |
| **27.** | the i gene of lac operon codes for | Repressor |
| **28.** | The a gene of lac operon codes for | Transacetylase enzyme |
| **29.** | In lac operon, lactose acts as | Inducer |
| **30.** | Regulation of lac operon by repressor is referred to as | Negative regulation |
| **31.** | The repressor is inactivated by interaction with the | Inducer |
| **32.** | The fragments were sequenced using automated DNA sequencers that worked on the principle of a method developed by | F. Sanger |
| **33.** | DNA fingerprinting involves identifying differences in some specific regions in DNA sequence called as | Repetitive DNA |
| **34.** | The bulk DNA forms major peak and the other small peaks are referred to as | Satellite DNA |
| **35.** | If inheritable mutation is observed in a population at high frequency, it is referred to as | DNA polymorphism |
| **36.** | The technique of DNA fingerprinting was initially developed by | Alec Jeffreys |
| **37.** | VNTR belongs to a class of satellite DNA referred to as | Mini-satellite |
| **38.** | The sensitivity of DNA fingerprinting technique has been increased by the use of | Polymerase chain reaction |
| **39.** | HGP was closely associated with the rapid development of new area in biology called | Bioinformatics |

| | **Fill in the blank.** | **ANSWER** |
|---|---|---|
| **40.** | The process of translation requires transfer of genetic information from polymer of _____ to polymer of amino acids. | Nucleotides |
| **41.** | Marshall Nirenberg's _____ for protein synthesis finally helped the code to be deciphered | Cell free system |
| **42.** | AAA $\longrightarrow$ A<br>CCC $\longrightarrow$ B<br>GGG $\longrightarrow$ C<br>UUU $\longrightarrow$ D<br>AUG $\longrightarrow$ E<br>UAA, UAG, UGA $\longrightarrow$ F | A = Lys<br>B = Pro<br>C = Gly<br>D = Phe<br>E = Met and Start codon<br>F = Stop codon |

| | | |
|---|---|---|
| **43.** | Translation refers to process of polymerization of _____ to form a polypeptide. | Amino acids |
| **44.** | The ribosome also act as _____ for the formation of peptide bond | Catalyst |
| **45.** | A translational unit in __A__ is the sequence of RNA that is flanked by the __B__ and the __C__ and codes for a polypeptide | A = m-RNA,<br>B = Start codon,<br>C = Stop codon |
| **46.** | The UTRs are present at both __A__ (Before start codon) and __B__ (After stop codon) | A = 5' end, B = 3' end |
| **47.** | At the end, a _____ binds to stop codon, terminating translation and releasing the complete polypeptide from the ribosome | Release factor |
| **48.** | In prokaryotes, control of the rate of _____ is predominant site for control of gene expression | Transcriptional initiation |
| **49.** | The activity of RNA polymerase at a given promoter is in turn regulated by interaction with _____, which affect its ability to recognise start site | Accessory protein (Regulatory protein) |
| **50.** | The (z) gene codes for __A__, which is primarily responsible for hydrolysis of _____. | A = β-glactosidase<br>B = lactose |
| **51.** | The (y) gene codes for __A__, which increases permeability of cell to __B__. | A = Permease<br>B = β-galactosides (Lactose) |
| **52.** | Human genome project was launched in | 1990 |
| **53.** | The commonly used hosts for cloning in HGP were __A__ and __B__ and the vectors were called as __C__ and __D__. | A = Bacteria, B = Yeast,<br>C = BAC, D = YAC |
| **54.** | The human genome contains _____ million *nucleotide* base pairs. | 3164.7 |
| **55.** | Less than _____ of the human genome codes for proteins | 2 percent |
| **56.** | Chromosome 1 has most genes __A__ and y has the fewest __B__. | A = 2968, B = 231 |
| **57.** | A typical nucleosome contains _____ bp of DNA | 200 |
| **58.** | The nucleosomes in chromatin are seen as _____ structure when viewed under electron microscope. | 'Beads-on-string' |
| **59.** | Transcriptionally active chromatin is called as _____. | Euchromatin |
| **60.** | The biochemical nature of 'transforming principle' in Griffith's experiment was determined by _____. | Avery, MacLeod and McCarty |
| **61.** | The unequivocal proof that DNA is the genetic material came from the experiments of _____. | Alfred Hershey & Martha Chase |
| **62.** | The first genetic material was _____. | RNA |
| **63.** | The DNA in chromosomes also replicates semiconservatively was proved by _____ through his experiments performed on *Vicia faba* using radioactive _____. | Taylor, thymidine |
| **64.** | In *E.coli*, the average rate of polymerisation during DNA replication is approximately _____. | 2000 bp per second |
| **65.** | In a transcription unit, the promoter is said to be located towards _____ of the structural gene. | 5'–end (upstream) |

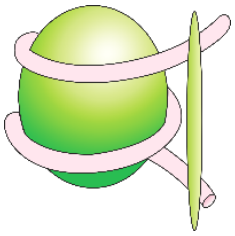
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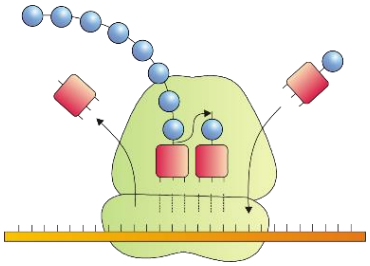
| | | |
|---|---|---|
| **66.** | During DNA replication, leading strand is synthesized ____ while lagging strand is synthesized _____ . | Continuously, Discontinuously |
| **67.** | A piece of DNA if needed to be propagated during recombinant DNA procedures requires a vector because it provides_____. | Origin of replication |
| **68.** | In bacteria, σ-factor helps in transcription _____ whereas ρ-factor helps in transcription _____. | Initiation, termination |

| | **True/False** | **ANSWER** |
|---|---|---|
| **69.** | No complementarity exists between nucleotides and amino acids | True |
| **70.** | Severo Ochoa enzyme (Polynucleotide phosphorylase) was also helpful in polymerising RNA with defined sequence in a template dependent manner. | False |
| **71.** | The code is nearly universal | True |
| **72.** | The codon is read in m-RNA in contiguous fashion | True |
| **73.** | There is one t-RNA for stop codons | False |
| **74.** | Formation of peptide bond requires energy | True |
| **75.** | First phase of translation is charging of t-RNA or aminoacylation of t-RNA | True |
| **76.** | For initiation, the ribosome binds to the m-RNA at the start codon that is recognised only by the initiator t-RNA | True |
| **77.** | The development and differentiation of embryo into adult organism are also a result of the coordinated regulation of expression of several sets of genes. | True |
| **78.** | Regulatory proteins can act only positively | False |
| **79.** | In Lac operon, three structural genes are regulated by a common promoter and regulator gene | True |
| **80.** | The genes present in the operon are needed together to function in same or related metabolic pathway | True |
| **81.** | A very low level of expression of lac operon has to be present in cell all the time | True |
| **82.** | The repressor of the lac operon is synthesized all the time | True |
| **83.** | Lac operon is under control of positive regulation as well | True |
| **84.** | Total number of genes in human genome is approximately 30,000 | True |
| **85.** | The split-gene arrangements represent probably a modern feature of the genome. | False |
| **86.** | The presence of introns is reminiscent of antiquity. | True |
| **87.** | Process of splicing represents the dominance of RNA world. | True |
| **88.** | The process of transription and translation can be coupled in bacteria. | True |
| **89.** | In DNA replication, deoxyribonucleoside triphosphates serve dual purposes as _____ | Substrate and energy source |

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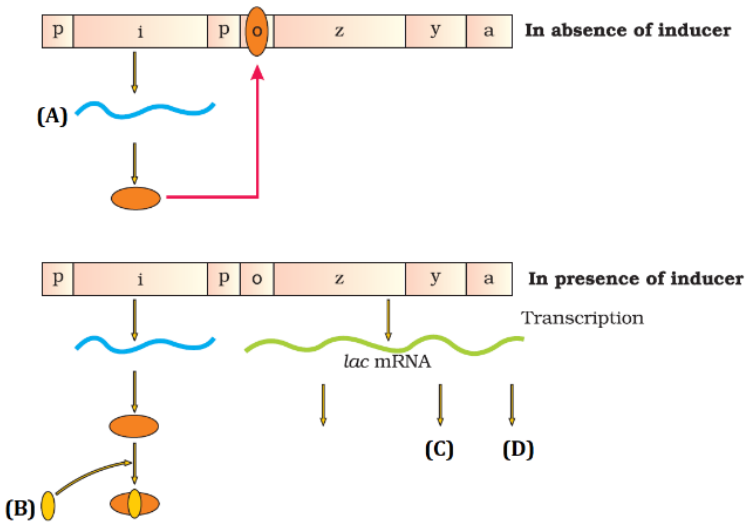
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| | | |
|---|---|---|
| **90.** | At the time of Mendel, the chemical nature of those 'factors' regulating the pattern of inheritance was clear. | False |
| **91.** | _____ is the genetic material, for the majority of organisms. | DNA |
| **92.** | RNA though it also acts as a genetic material in some viruses, mostly functions as a messenger. | True |
| **93.** | The length of DNA cannot be defined as the number of nucleotides (or a pair of nucleotides referred to as base pairs) present in it. | False |
| **94.** | A bacteriophage known as $\phi \times 174$ has 5386 base pairs. | False |
| **95.** | A nitrogenous base is linked to the OH of 1' C of pentose sugar through a _____ linkage to form a nucleoside, | N-glycosidic |
| **96.** | In RNA, every nucleotide residue has an additional –OH group present at 3' -position of the deoxyribose sugar. | False |
| **97.** | DNA is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project outside. | False |
| **98.** | The two chains of DNA are coiled in a right-handed fashion. | True |
| **99.** | Who proposed the Central dogma in molecular biology? | Francis Crick |
| **100.** | Purpose of reverse transcription? | To form DNA from RNA |
| **101.** | The packaging of chromatin at a higher level requires an additional set of proteins that collectively are referred to as _____. | Non-histone Chromosomal (NHC) proteins. |
| **102.** | Euchromatin is said to be transcriptionally inactive chromatin. | False |
| **103.** | Previous discoveries by Gregor Mendel, Walter Sutton, Thomas Hunt Morgan and numerous other scientists had narrowed the search of genetic material to the chromosomes located in the nucleus of most cells. | True |
| **104.** | In Griffith's experiment, R strain bacteria had somehow been transformed by the heat-killed S strain bacteria. Some 'transforming principle', is transferred from the heat-killed S strain. This transforming principle is _____. | DNA |
| **105.** | Name the following structure.  | Nucleosome |
| **106.** | The bacterial cell treats the bacteriophage genetic material as if it was its own and subsequently manufactures more virus particles. | True |

| | | |
|---|---|---|
| **107.** | In some viruses, RNA is the genetic material for example _____. | Tobacco Mosaic viruses, QB bacteriophage, etc. |
| **108.** | Both the nucleic acids (DNA and RNA), have the ability to direct their duplications. | True |
| **109.** | The genetic material should be able enough to change with different stages of life cycle. | False |
| **110.** | The process shown in the following diagram, is known as -  | Translation |
| **111.** | There is now enough evidence to suggest that essential life processes such as _____, evolved around RNA. | metabolism, translation, splicing, etc. |
| **112.** | The semiconservative scheme is related to _____. | DNA replication |
| **113.** | DNA replication time in *E. coli* is _____. | 18 minutes |
| **114.** | The mistakes that take place during DNA replication result into _____. | Mutations |
| **115.** | For long DNA molecules, the two strands of DNA cannot be separated in its entire length during DNA replication process. | True |
| **116.** | The DNA polymerases on their own can initiate the process of replication. | False |
| **117.** | In prokaryotes, the replication of DNA takes place at the S-phase of the cell cycle. | False |
| **118.** | A transcription unit in DNA contains _____. | (i) A Promoter<br>(ii) The Structural gene<br>(iii) A Terminator |
| **119.** | The strand that has the polarity _____ acts as a template during the transcription process. | 3'→5' |
| **120.** | There are additional regulatory sequences that may be present further upstream or downstream to the promoter. | True |
| **121.** | A _____ is defined as the functional unit of inheritance. | gene |
| **122.** | Define a cistron | A segment of DNA coding for a polypeptide |
| **123.** | Sometimes the regulatory sequences are loosely defined as _____. | regulatory genes |
| **124.** | How many types of DNA-dependent RNA polymerases catalyze transcription of RNA in bacteria? | One |

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| | | |
|---|---|---|
| **125.** | RNA polymerase binds to _____ site and initiates transcription (Initiation). | promoter |
| **126.** | In bacteria, transcription and translation can take place in the same compartment. | True |
| **127.** | Primary transcripts in eukaryotes are non-functional. | True |
| **128.** | In tailing, adenylate residues (200-300) are added at _____ end. | 3' |
| **129.** | A direct complementarity exists between nucleotides and amino acids. | False |
| **130.** | If the number of amino acids taking part in protein synthesis was 40, the number of nucleotides in a codon would be | 3 |
| **131.** | Who prepared cell free protein synthesis system to decipher the codons? | Nirenberg |
| **132.** | Some amino acids are coded by more than one code, this is called _____. | Degeneracy |
| **133.** | Mention the change of amino acids that cause the disease Sickle cell anaemia. | Glutamate to valine |
| **134.** | Which loop of tRNA has the bases complementary to the codon? | Anticodon loop |
| **135.** | Amino acids are activated in the presence of ATP and linked to their cognate tRNA – a process commonly called as _____. | charging of tRNA or aminoacylation of tRNA |
| **136.** |  <br>Identify A, B, C, D :- | A = repressor mRNA,<br>B = inducer,<br>C = permease,<br>D = transacetylase, |
| **137.** | Lac operon is under the control of negative and positive regulation. | True |
| **138.** | Two major approaches in methodology of the Human Genome Project are _____. | Expressed sequence tags and sequence annotation. |
| **139.** | Vectors used in Human Genome Project _____. | BAC and YAC |

# Microbes in Human Welfare

<table>
<tr><th>Q.NO</th><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>1.</td><td>Which microorganism grows in milk and convert it to curd ?</td><td>Lactic acid Bacteria (LAB)</td></tr>
<tr><td>2.</td><td>Which microorganism improves the nutritional quality of curd by increasing vitamin $B_{12}$ ?</td><td>LAB</td></tr>
<tr><td>3.</td><td>The puffed-up appearance of dough is due to production of</td><td>$CO_2$ gas</td></tr>
<tr><td>4.</td><td>The dough which is used for making bread is fermented by which microbe ?</td><td>Saccharomyces cerevisiae<br>(Baker's yeast)</td></tr>
<tr><td>5.</td><td>Which traditional drink in some part of southern India is made by fermenting sap of palms ?</td><td>Toddy</td></tr>
<tr><td>6.</td><td>Match the column.
<table>
<tr><th colspan="2">Column-I</th><th colspan="2">Column-II</th></tr>
<tr><td>(1)</td><td>Propionibacterium Sharmanii</td><td>(a)</td><td>Penicillin</td></tr>
<tr><td>(2)</td><td>Saccharomyces</td><td>(b)</td><td>Citric acid</td></tr>
<tr><td>(3)</td><td>Aspergillus niger</td><td>(c)</td><td>Ethanol</td></tr>
<tr><td>(4)</td><td>Penicillium notatum</td><td>(d)</td><td>Swiss cheese</td></tr>
</table></td><td>1-d, 2-c, 3-b, 4-a</td></tr>
<tr><td>7.</td><td>Production on an industrial scale, requires growing microbes in very large vessels called :-</td><td>Fermentors</td></tr>
<tr><td>8.</td><td>From given table choose that alcoholic drinks which are produced without distillation of fermented broth.<br>Whisky, Wine Brandy, Rum, Beer</td><td>Wine, Beer</td></tr>
<tr><td>9.</td><td>Name of the first antibiotic to be discovered :-</td><td>Penicillin</td></tr>
<tr><td>10.</td><td>Match the column.
<table>
<tr><th colspan="2">Column-A</th><th colspan="2">Column-B</th></tr>
<tr><td>(1)</td><td>Acetobacter aceti</td><td>(a)</td><td>Cyclosporin A</td></tr>
<tr><td>(2)</td><td>Clostridium butylicum</td><td>(b)</td><td>Lactic acid</td></tr>
<tr><td>(3)</td><td>Lactobacillus</td><td>(c)</td><td>Butyric acid</td></tr>
<tr><td>(4)</td><td>Trichoderma Polysporum</td><td>(d)</td><td>Acetic acid</td></tr>
</table></td><td>1–d, 2–c, 3–b, 4–a</td></tr>
<tr><td>11.</td><td>Which group of enzymes are used in detergent formulation and are helpful in removing oily stains from the laundary.</td><td>Lipases</td></tr>
<tr><td>12.</td><td>Fill in the Blank<br>(i) .....A..... are the blood-chloesterol lowering agents which is produced by the yeast .....B.....<br>(ii) .....A..... produced by the bacterium Streptococcus and modified by genetic engineering is used as a.....B.....for removing....C... from blood vessel.</td><td>(i) A-Statins<br>B-Monascus purpureus<br>(ii) A-Streptokinase<br>B-Clot buster<br>C-Clots</td></tr>
<tr><td>13.</td><td>In anaerobic sludge digesters during digestion, bacteria produce mixture of gases called biogas. Give the three components of biogas.</td><td>(i) Methane<br>(ii) Hydrogen sulphide<br>(iii) Carbon dioxide</td></tr>
</table>

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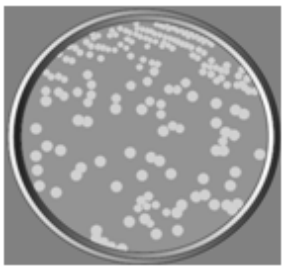
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| | | |
|---|---|---|
| **14.** | Give the name of pathogen which attack on insects and other Arthopods so used as biological control agents. | Baculoviruses |
| **15.** | Majority of baculoviruses used as biological control agents belong to which genus ? | *Nucleopolyhedrovirus* |
| **16.** | The main sources of biofertilisers are.... | Bacteria, fungi and Cyanobacteria |
| **17.** | In Paddy fields, which microorganism serve as important biofertiliser? | Cyanobacteria |
| **18.** | The biogas plant was developed in India by : | Indian Agricultural Research Institute and Khadi & Village Industries Commission. |
| **19.** | Microorganism such as _____ and others commonly called _____ grow in milk and convert it to curd. | *Lactobacillus* ; Lactic Acid Bacteria (LAB) |
| **20.** | In the dough used for making dosa and idli, the puffed-up appearance of it is due to the production of_____ | $CO_2$ gas |
| **21.** | The dough used for making bread is fermented using _____ | Baker's yeast |
| **22.** | The scientific name of Baker's yeast is _____. | *Saccharomyces cerevisiae* |
| **23.** | _____ is a traditional drink in some parts of southern India that is made by fermenting sap from palms. | Toddy |
| **24.** | The large holes in 'Swiss cheese' are due to production of large amount of _____ by a Bacterium named _____. | $CO_2$; *Propino-bacterium sharmanii* |
| **25.** | The 'Rouefort cheese' is ripened by growing specific _____ on them which gives them a particular flavor. | Fungi |
| **26.** | The alcoholic drink _____ and beer are produced without distillation whereas _____,_____ and rum are produced by distillation of fermented broth. | Wine; whisky; brandy |
| **27.** | _____ was the first antibiotic to be discovered by _____. | Penicillin ; Alexander Fleming |
| **28.** | Alexander Fleming while working on _____ observed that the mould named_____ growing in unwashed plate produced penicillin because of which bacteria was not able to grow around mould. | *Staphylococci ; Penicillium notatum* |
| **29.** | Fill the correct acid producer of respective acid: citric acid by _____, acetic acid by _____, butyric acid by _____ , and lactic acid by _____ . | *Aspergillus niger; Acetobacter aceti; Clostridium butylicum; Lactobacillus* |
| **30.** | Streptokinase is produced by bacterium _____. | *Streptococcus* |
| **31.** | Cyclosporin A is used as immunosuppressive agent in organ transplant patients is produced by fungus _____. | *Trichoderma polysporum* |

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| | | |
|---|---|---|
| **32.** | _____ produced by the yeast _____ have been commercialized as blood Cholesterol lowering agents. | Statins ; *Monascus purpureus* |
| **33.** | In primary treatment of waste water, all the solids that settle form the _____, and the supernatant forms effluent. | Primary sludge |
| **34.** | Secondary treatment of waste water is also known as _____. | Biological treatment |
| **35.** | The masses of bacteria associated with fungal filaments to form mesh like structures in Secondary treatment of waste water is called _____. | Flocs |
| **36.** | _____is the measure of organic matter present in water. The more it is, _____is its polluting potential of waste water. | BOD ; more |
| **37.** | In biological treatment of waste water, a small part of _____is pumped back into aeration tank to serve as inoculum and remaining part is pumped into larger tanks called _____. | Activated sludge ; anaerobic sludge digester |
| **38.** | During the digestion in anaerobic sludge digester, bacteria produce a mixture of gases such as_____, hydrogen sulphide and carbon dioxide. These gases forms _____. | Methane ; biogas |
| **39.** | The Ministry of Environment and Forests has initiated _____and _____to save the major rivers of our country from pollution. | Ganga Action Plan ; Yamuna Action Plan |
| **40.** | | |
| **41.** | The microbial biocontrol agents used to control butterfly caterpillars is the bacteria _____. When insect eats spore of this bacterium, the toxin is released in _____of larva and larva is killed. | *Bacillus thuringiensis*; gut |
| **42.** | _____are pathogens that attack insects and other arthropods. The majority of them used as biological control agents are in the genus _____. | *Baculoviruses* ; *Nucleopolyhedrovirus* |
| **43.** | In paddy fields, _____serve as an important biofertilizer. | *Cyanobacteria* |
| **44.** | Which gases are produced by anaerobic bacteria in sewage treatment plant? | Hydrogen sulphide, Carbon dioxide and Methaane |
| **45.** | How deep is the concrete tank used in biogas plant? | 10-15 feet deep |
| **46.** | Toddy a traditional drink of some parts of southern India is made by fermenting sap from | Palm |
| **47.** | The holes in swiss cheese are due to the production of large amount of carbon dioxide by a bacterium named | *Propionibacterium sharmanii* |
| **48.** | Are wine and beer distilled? | No |
| **49.** | Are whisky, brandy and rum distilled? | Yes |
| **50** | Alexander Fleming discovered penicillin while working with which bacteria? | *Staphylococcus* |
| **51.** | What is used as a clot buster for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack? | Streptokinase |

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<table>
<tr><td>52.</td><td>Cyclosporin A is produced by which organism? What is its function?</td><td>Trichoderma polysporum<br>Function-<br>immunosuppressive agent</td></tr>
<tr><td>53.</td><td>Given below diagram represents colonies of<br>  </td><td>Bacteria</td></tr>
<tr><td>54.</td><td>Ladybirds and dragonflies can be used to get rid of which insect pests respectively?</td><td>Aphids and mosquitoes</td></tr>
<tr><td>55.</td><td>Name a free living fungus species very common in root ecosystem and can act as an effective biocontrol agent?</td><td>Trichoderma</td></tr>
<tr><td>56.</td><td>In primary treatment of waste water, all the solids that settle form the _____, and the supernatant forms effluent.</td><td>Primary sludge</td></tr>
<tr><td>57.</td><td>Secondary treatment of waste water is also known as _____.</td><td>Biological treatment</td></tr>
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<tr><td>(a)</td><td>Cyclosporin A</td><td>(i)</td><td>Blood Cholesterol lowering agent</td></tr>
<tr><td>(b)</td><td>Streptokinase</td><td>(ii)</td><td>Immunosuppressive agent</td></tr>
<tr><td>(c)</td><td>Statins</td><td>(iii)</td><td>Clot buster for removing clots</td></tr>
<tr><td>(d)</td><td>Saccharomyces</td><td>(iv)</td><td>Production of ethanol</td></tr>
</table></td><td>a – ii<br>b – iii<br>c – i<br>d - iv</td></tr>
<tr><td>59.</td><td>_____is the measure of organic matter present in water. The more it is, _____is its polluting potential of waste water.</td><td>BOD ; more</td></tr>
<tr><td>60.</td><td>The nodules on the roots of leguminous plants are formed by the symbiotic association of _____.</td><td>Rhizobium</td></tr>
<tr><td>61.</td><td>Microbes can also be used to kill harmful pests, a process called as _____.</td><td>Biocontrol</td></tr>
<tr><td>62.</td><td>The symbiotic association form between some fungi and plants are called _____.</td><td>Mycorrhiza</td></tr>
<tr><td>63.</td><td>Many members of the genus _____form mycorrhiza.</td><td>Glomus</td></tr>
</table>

# Organisms and Populations

<table>
<tr><th>Q.NO.</th><th>QUESTION</th><th>ANSWER</th></tr>
<tr><td>1.</td><td>Match column-I with column-II
<table>
<tr><th colspan="2">Column – I</th><th colspan="2">Column - II</th></tr>
<tr><td>1.</td><td>Commensalism</td><td>a.</td><td>+, –</td></tr>
<tr><td>2.</td><td>Amensalism</td><td>b.</td><td>0, –</td></tr>
<tr><td>3.</td><td>Competition</td><td>c.</td><td>0, +</td></tr>
<tr><td>4.</td><td>Predation</td><td>d.</td><td>–, –</td></tr>
<tr><td>5.</td><td>Mutualism</td><td>e.</td><td>+, +</td></tr>
</table></td><td>1-c, 2-b, 3-d, 4-a, 5-e</td></tr>
<tr><td>2.</td><td>Mutualism is an __x__ relation whereas proto-co-operation is a __y__ relation in which both interacting species are benefitted.</td><td>x = Obligatory<br>y = Non-obligatory</td></tr>
<tr><td>3.</td><td>Interaction in which either one or both participating species are harmed such interactions are called _____.</td><td>Detrimental or negative interactions</td></tr>
<tr><td>4.</td><td>Which organism shows the following properties in an ecosystem.<br>• Act as conduit of energy transfer<br>• Maintain species diversity in community<br>• Regulate the population of prey species<br>• Prudent in nature</td><td>Predators</td></tr>
<tr><td>5.</td><td>More than 10 species of invertebrates became extinct within a year from the American Pacific Coast when Pisaster starfish was removed. This extinction was due to -</td><td>Interspecific competition</td></tr>
<tr><td>6.</td><td>Some insects and frogs are cryptically coloured to avoid being detected by predators. This is called -</td><td>Camouflage</td></tr>
<tr><td>7.</td><td>How much percentage of all insects are phytophagous which feeds on plant sap and other parts of plant?</td><td>≈ 25%</td></tr>
<tr><td>8.</td><td>Match the columns regarding defence mechanisms adopted by prey species to be protected from predators.
<table>
<tr><th colspan="2">Column – A</th><th colspan="2">Column - B</th></tr>
<tr><td>A.</td><td>Cardiac glycosides</td><td>i.</td><td>Acacia, Cactus</td></tr>
<tr><td>B.</td><td>Distasteful due to a special chemical</td><td>ii.</td><td>Insects</td></tr>
<tr><td>C.</td><td>Thorns</td><td>iii.</td><td>Calotropis</td></tr>
<tr><td>D.</td><td>Camouflage</td><td>iv.</td><td>Monarch butterfly</td></tr>
</table></td><td>A-(iii), B-(iv), C-(i), D-(ii)</td></tr>
<tr><td>9.</td><td>When a species is introduced into new geographical area becomes invasive and causes havoc by spreading rapidly, due to _____.</td><td>Absence of its natural predator</td></tr>
<tr><td>10.</td><td>When feeding efficiency of one species is reduced in presence of the other species even if the resources are abundant, it is called -</td><td>Interference competition</td></tr>
</table>

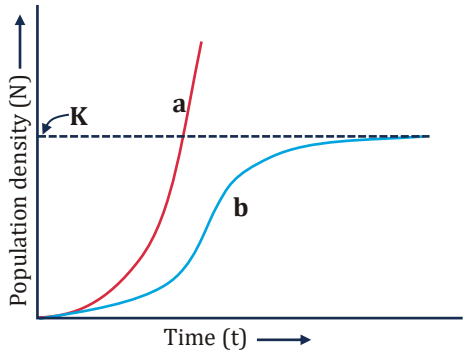
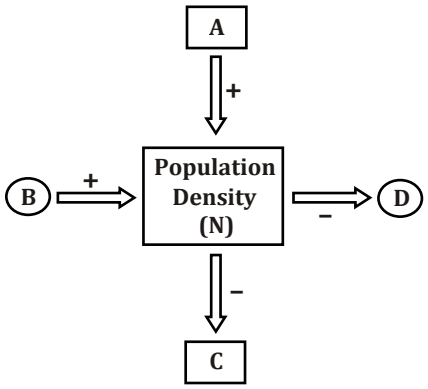
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<table>
<tr><td>11.</td><td>Match the following columns –
<table>
<tr><th colspan="2">Column – A</th><th colspan="2">Column - B</th></tr>
<tr><td>1.</td><td>Resource partitioning</td><td>A.</td><td>Gause</td></tr>
<tr><td>2.</td><td>Competitive exclusion</td><td>B.</td><td>Darwin</td></tr>
<tr><td>3.</td><td>Struggle for existence</td><td>C.</td><td>Mac Arthur</td></tr>
</table></td><td>1 - C, 2 - A, 3 - B</td></tr>
<tr><td>12.</td><td>Which organisms are more adversely affected by competition than carnivores?</td><td>Plants and Herbivores</td></tr>
<tr><td>13.</td><td>To avoid competition species choose different feeding time and different foraging patterns this is called _____.</td><td>Resource partitioning</td></tr>
<tr><td>14.</td><td>How many of the following statements are incorrect for parasites?<br>(A) Presence of un-neccessary sense organs<br>(B) Presence of adhesive organs & suckers<br>(C) Developed digestive system<br>(D) High biotic potential</td><td>Two (A and C are incorrect)</td></tr>
<tr><td>15.</td><td>Human liver fluke is a __A__ which depends on two intermediate hosts __B__.</td><td>A = Trematode parasite<br>B = a snail and a fish</td></tr>
<tr><td>16.</td><td>An organism reduces survival, growth and reproduction of other organism, reduces its population density and makes it physically weak, such organism is called _____.</td><td>Parasite</td></tr>
<tr><td>17.</td><td>How many of the following organisms are considered as Ecto-parasites?<br>Lice, Liver fluke, Ascaris, Ticks, Copepods, Taenia, Cuscuta</td><td>Four<br>(Lice, Ticks, Copepods, Cuscuta)</td></tr>
<tr><td>18.</td><td>How many of the following are specific properties of endo-parasites?<br>(A) More complex life cycle<br>(B) High reproductive potential<br>(C) Simple morphology<br>(D) Simplified anatomy</td><td>Four (A, B, C, D)</td></tr>
<tr><td>19.</td><td>Interaction in which one organism lays its eggs in the nest of another organism is called _x_. For example _y_.</td><td>x = Brood parasitism<br>y = Cuckoo bird and crow</td></tr>
<tr><td>20.</td><td>Match the following organisms according to their commensal relationship.
<table>
<tr><th colspan="2">Column – A</th><th colspan="2">Column - B</th></tr>
<tr><td>1.</td><td>Mango Branch</td><td>A.</td><td>Sea anemone</td></tr>
<tr><td>2.</td><td>Clown fish</td><td>B.</td><td>Barnacles</td></tr>
<tr><td>3.</td><td>Egret bird</td><td>C.</td><td>Cattle</td></tr>
<tr><td>4.</td><td>Whale</td><td>D.</td><td>Orchid</td></tr>
</table></td><td>1 - D, 2 - A, 3 - C, 4 - B</td></tr>
<tr><td>21.</td><td>The mutualistic relationship between fungus and cyanobacteria is called _____.</td><td>Lichen</td></tr>
</table>

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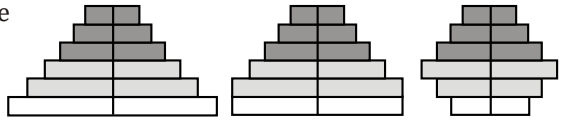
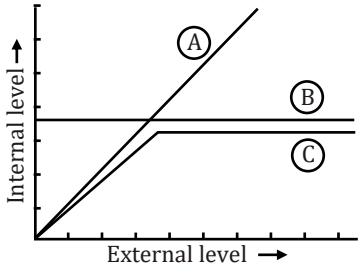
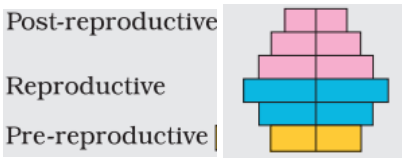
| | | |
|---|---|---|
| **22.** | In an interspecific interaction a filamentous eukaryotic organism absorbs nutrients from the soil for the higher plant. In turn the host provides some energy yielding carbohydrates, such association is called _____. | Mycorrhiza |
| **23.** | Pollinators and seed dispersers are offered reward by plants in form of __A__ and __B__ respectively. | A = Pollen & nectar<br>B = Juicy and nutritious fruits |
| **24.** | In which of the following interaction both interacting species show co-evolution, co-extinction, co-existence?<br>(1) Competition (2) Predation<br>(3) Mutualism (4) Commensalism | Mutualism |
| **25.** | In symbiotic relationship between fig tree and wasp, the wasp uses fig fruit for __x__ and developing seeds for __y__. In turn favours __z__. | x = Oviposition (egg laying)<br>y = Nourishing its larva<br>z = Pollination of fig. |
| **26.** | Select the correct option with respect to the related organisms and interactions.<br>**Interactions** **Related organism**<br>A. Mutualism 1. Sparrow and seed<br>B. Predation 2. Goat and Abingdon tortoise<br>C. Parasitism 3. Flamingoes and fishes<br>D. Competition 4. Monarch butterfly and bird<br>E. Commensalism 5. Hedge plant and *Cuscuta*<br>6. Orchid and mango tree<br>7. Human and *Ascaris*<br>8. Orchid and Bee<br>9. *Balanus* and *Chathamalus* | A – 8<br>B – 1, 4<br>C – 5, 7<br>D – 2, 3, 9<br>E – 6 |
| **27.** | True or False<br>(1) 99% of animals and nearly all plants are conformers.<br>(2) Thermoregulation is energetically expensive for many organisms.<br>(3) Heat loss or heat gain is a function of surface area.<br>(4) All birds and mammals are regulators. | True<br>True<br>True<br>True |
| **28.** | Under unfavourable conditions, a stage of suspended development is called _____. | Diapause |
| **29.** | Many tribes live in the high altitude of Himalaya, have _____ red blood cell count than people living in the plains. | Higher |
| **30.** | Symptoms of altitude sickness are _____. | Nausea, Fatigue,<br>Heart palpitations |

| | | |
|---|---|---|
| **31.** | How does human body compensate to altitude sickness? | Increasing RBC production, Increasing breathing rate, Decreasing binding affinity of Hb |
| **32.** | Mammals from colder climates generally have shorter ears and limbs to minimise heat loss. This is called _____. | Allen's rule |
| **33.** | How does kangaroo rat fulfill its water requirements? | By internal fat oxidation |
| **34.** | How do Antarctic fish manage to keep their body fluids from freezing? | They have antifreeze proteins |
| **35.** | Under normal conditions, _____ and _____ are the most important factors influencing population density. | Births, Deaths |
| **36.** | Exponential growth of population is seen when resources are _____. | Unlimited |
| **37.** | Exponential growth equation is _____. | $N_t = N_0e^{rt}$ or $\frac{dN}{dt} = rN$ |
| **38.** | Logistic growth equation is _____. | $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$ |
| **39.** | Write some examples of organisms that breed only once in their life time. | Pacific salmon fish and Bamboo plant |
| **40.** | Write equations for curves a and b in the graph given below.<br>  <br> | $a = \frac{dN}{dt} = rN$<br>$b = \frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$ |
| **41.** | Identify A, B, C & D<br>  <br> | A – Immigration<br>B – Natality<br>C – Emigration<br>D – Mortality |



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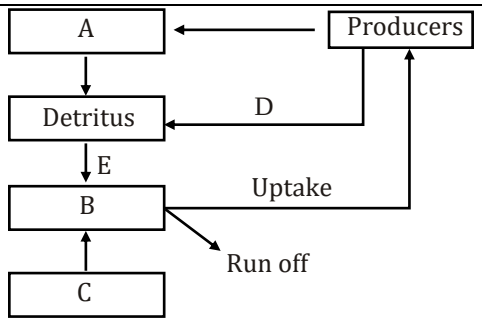
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|---|---|---|
| **42.** | Identify the growth status of the population from given age pyramids :-<br>Post-reproductive<br>Reproductive<br>Pre-reproductive<br>  <br>(A) (B) (C) | A – Expanding<br>B – Stable<br>C – Declining |
| **43.** | Identify A, B & C –<br>  <br> | A – Conformers<br>B – Regulators<br>C – Partial regulators |
| **44.** | A group of individuals resulting from even asexual reproduction is also generally considered a population for the purpose of ecological studies. | True |
| **45.** | Population ecology is an important area because it links ecology to population genetics and evolution. | True |
| **46.** |  <br><br>This shape of pyramid represents which type of population? | Declining population |
| **47.** | _____is the number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration. | Immigration |
| **48.** | If N is the population density at time t, then its density at time t +1 is _____. | $N_{t+1} = N_t + [(B+I) - (D+E)]$ |
| **49.** | Under normal conditions, _____and _____ are the most important factors influencing population density. | births, deaths |
| **50.** | In the equation $\frac{dN}{dt} = rN$ , r represents_____. | intrinsic rate of natural increase |
| **51.** | A population growing in a habitat with limited resources show initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote. | True |
| **52.** | Some organisms breed only once in their lifetime to maximise their reproductive fitness, examples includes _____, _____. | Pacific salmon fish, bamboo |
| **53.** | Intraspecific interactions arise from the interaction of populations of two different species. | False |
| **54.** | _____ act as 'conduits' for energy transfer across trophic levels. | Predator |

| | | |
|---|---|---|
| **55.** | Biological control methods adopted in agricultural pest control are based on the ability of the prey to regulate predator population. | False |
| **56.** | Interaction in which one species benefits and the other is neither harmed nor benefited. | Commensalism |
| **57.** | The mycorrhizae are associations between fungi and the roots of higher plants comes under which type of interaction. | Mutualism |
| **58.** | The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee. | True |
| **59.** | The intrinsic rate of natural increase (r) is a measure of the inherent potential of a population to grow. | True |
| **60.** | What type of interaction occurs when one organism benefits while the other is harmed?<br>(A) Mutualism<br>(B) Amensalism<br>(C) Predation<br>(D) Commensalism | Predation |
| **61.** | Plants have evolved diverse ____A___ and ___B_____ defenses against herbivory. | A- morphological<br>B- chemical |
| **62.** | Which principle states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually. | Gause's 'Competitive Exclusion Principle' |
| **63.** | If two species compete for the same resource, they could avoid competition by choosing, for instance, different times for feeding or different foraging patterns through the mechanism of _____. | resource partitioning |

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# Ecosystem

| Q.NO. | QUESTION | ANSWER |
|---|---|---|
| **1.** | How many of the following can be categorised under man-made ecosystem?<br>Lake, Wetland, Pond, Cropland, River, Aquarium, Estuary | Cropland and Aquarium<br>(Pond may be natural or man made) |
| **2.** | Different species of plants occupy various levels in the forest ecosystem. Such distribution is called _____. | Stratification or vertical stratification |
| **3.** | The basic requirement for any ecosystem to function and sustain is - | Constant solar input |
| **4.** | Production and productivity can be represented by __x__ and __y__ respectively. | x = gm/m$^2$<br>y = gm/m$^2$/yr |
| **5.** | Representation of production and productivity in terms of energy____. | kcal/m$^2$ (Production)<br>kcal/m$^2$/yr (Productivity) |
| **6.** | A rabbit feeds on grass and gains 1 kg weight within two months, this rate of biomass gain is termed as _____. | Secondary productivity. |
| **7.** | The NPP of whole biosphere is ___A___ out of which NPP of oceans is ___B___. | A = 170 billion tons<br>B = 55 billion tons |
| **8.** | Despite covering 70% of surface, oceans show very low productivity because _____. | Producers are micro-organisms (Phytoplankton),<br>$N_2$ deficiency, light limiting factor in aquatic ecosystem |
| **9.** |  <br>Identify A, B, C, D and E in above cycle. | A = Consumer<br>B = Soil solution<br>C = Rock minerals<br>D = Litterfall<br>E = Decomposition |
| **10.** | How much amount of carbon is fixed in biosphere through photosynthesis annually? | $4 \times 10^{13}$ kg |
| **11.** | The price tag given to fundamental ecosystem services is approximately _____. | US $33 trillion |
| **12.** | If 1,000,000 J of sunlight is provided to a food chain what is the amount of energy present at top consumer level if food chain have four trophic level. | 10 J<br>(Because only 1% of total sunlight is available to plants) |
| **13.** | _____is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. | Primary production |

| | | |
|---|---|---|
| **14.** | Primary Productivity is expressed in terms of _____or _____. | weight ($gm^{-2}$), energy ($kcal\ m^{-2}$). |
| **15.** | _____ of an ecosystem is the rate of production of organic matter during photosynthesis. | Gross primary productivity |
| **16.** | The annual net primary productivity of the whole biosphere is approximately _____ (dry weight) of organic matter. | 170 billion tons |
| **17.** | Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute:<br>(1) Detritus<br>(2) Duff<br>(3) Humus<br>(4) Productivity | detritus |
| **18.** | Fragmentation, Leaching, Catabolism, Humification And Mineralization<br>These steps are included in the process of_____ | decomposition |
| **19.** | Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation. | True |
| **20.** | Catabolism leads to accumulation of a dark coloured amorphous substance called humus. | False |
| **21.** | Decomposition is largely an anaerobic process. | False |
| **22.** | In a particular climatic condition, decomposition rate is slower if detritus is rich in ___A___, and quicker, if detritus is rich in ____B____. | A - lignin and chitin<br>B - nitrogen and sugars |
| **23.** | **Assertion:** Temperature and soil moisture are the most important climatic factors that regulate decomposition.<br>**Reason:** Warm and moist environment favour decomposition. | Both assertion and reason are true but reason is not a correct explanation of assertion. |
| **24.** | Of the incident solar radiation less than 50 per cent of it is _____. | photosynthetically active radiation |
| **25.** | Plants capture only 2-10 per cent of the ISR and this small amount of energy sustains the entire living world. | False |
| **26.** | Unidirectional flow of energy takes place from the sun to consumers and then to producers. | False |
| **27.** | Ecosystems do not follow the Second Law of thermodynamics. | False |
| **28.** | Producers in an aquatic ecosystem are:<br>(1) Woody trees<br>(2) Phytoplankton's<br>(3) Algae<br>(4) Both (2) and (3) | Both (2) and (3) |
| **29.** | Death of organism is the beginning of the detritus food chain/web. | True |
| **30.** | Primary consumers will be _____ | herbivore |

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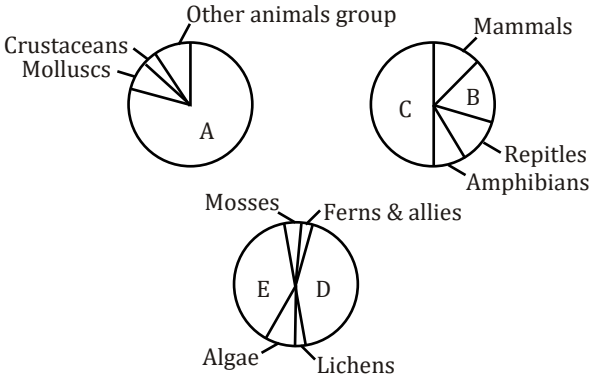
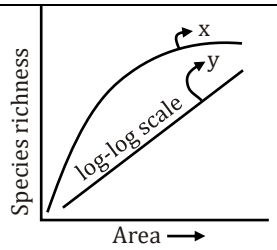
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<table>
<tr><td>31.</td><td>Primary carnivores are also known as</td><td>Secondary consumers</td></tr>
<tr><td>32.</td><td>Decomposers are heterotrophic organisms.</td><td>True</td></tr>
<tr><td>33.</td><td>_____secrete digestive enzymes that breakdown dead and waste materials into simple, inorganic materials, which are subsequently absorbed by them.</td><td>Saprotrophs</td></tr>
<tr><td>34.</td><td>In an _____ecosystem, GFC is the major conduit for energy flow.</td><td>aquatic</td></tr>
<tr><td>35.</td><td>Some of the organisms of GFC are prey to the DFC animals</td><td>False</td></tr>
<tr><td>36.</td><td>Based on the source of their nutrition or food, organisms occupy a specific place in the food chain that is known as their _____.</td><td>trophic level</td></tr>
<tr><td>37.</td><td>_____A____belong to the first trophic level, ____B_____ to the second and ____C_____ to the third.</td><td>A- Producers<br>B- Herbivores<br>C- carnivores</td></tr>
<tr><td>38.</td><td>The standing state is measured as the mass of living organisms (biomass) or the number in a unit area.</td><td>False</td></tr>
<tr><td>39.</td><td>The base of each pyramid represents the _____while the apex represents _____.</td><td>Producers, top level consumer</td></tr>
<tr><td>40.</td><td>Pyramid of numbers in a grassland ecosystem is</td><td>Upright</td></tr>
<tr><td>41.</td><td>Inverted pyramid of biomass represents small standing crop of _____A____ supports large standing crop of ____B_____.</td><td>A- phytoplankton<br>B- zooplankton</td></tr>
<tr><td>42.</td><td>The trophic level represents a functional level, not a species as such.</td><td>True</td></tr>
<tr><td>43.</td><td>Energy at a higher trophic level is always more than at a lower level.</td><td>False</td></tr>
<tr><td>44.</td><td>The pyramid of biomass in sea is generally _____ because the biomass of fishes far exceeds that of phytoplankton.</td><td>inverted</td></tr>
<tr><td>45.</td><td>Ecological pyramid does not consider the same species belonging to two or more trophic levels.</td><td>True</td></tr>
<tr><td>46.</td><td>Match the following biotic components with their examples:
<table>
<tr><td>A</td><td>Producers</td><td>1</td><td>Herbivores, carnivores, omnivores</td></tr>
<tr><td>B</td><td>Consumers</td><td>2</td><td>Green plants, algae.</td></tr>
<tr><td>C</td><td>Decomposers</td><td>3</td><td>Fungi, bacteria</td></tr>
</table></td><td>A-2, B-1, C-3</td></tr>
</table>

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<table>
<tr><td>47.</td><td>Match the following ecological productivity types with their descriptions:<table><tr><td>A.</td><td>Gross primary productivity (GPP)</td><td>1.</td><td>The total amount of energy produced by photosynthesis in an ecosystem.</td></tr><tr><td>B.</td><td>Net primary productivity (NPP)</td><td>2.</td><td>The energy remaining after respiration and used for the growth and reproduction of primary producers.</td></tr></table></td><td>A-1, B-2</td></tr>
<tr><td>48.</td><td>Nutrient cycling is of two types _____ and _____.</td><td>Gaseous, sedimentary</td></tr>
<tr><td>49.</td><td>Ecosystem services includes purification of air and water by forests.</td><td>True</td></tr>
<tr><td>50.</td><td>The number of trophic levels in the grazing food chain is restricted as the transfer of energy follows 10 per cent law.</td><td>True</td></tr>
<tr><td>51.</td><td>The _____ of an ecosystem represents the sum total of all the living organisms present in an area.</td><td>Biotic component</td></tr>
<tr><td>52.</td><td>In an ecosystem, the _____ represents the transfer of energy from one trophic level to another.</td><td>energy flow</td></tr>
</table>

# Biodiversity and Conservation

| Q.NO. | QUESTION | ANSWER |
|---|---|---|
| 1. | Term biodiversity was popularised by - | Edward Wilson |
| 2. | Biodiversity is the combined diversity at all levels of biological organisation what are those levels? | (i) Genetic diversity<br>(ii) Species diversity<br>(iii) Ecological diversity |
| 3. | How many species of plants and animals are described according to IUCN (2004)? | > 1.5 million |
| 4. | How much percentage of recorded species are animals and plants respectively? | Animals - > 70%<br>Plants - 22% |
| 5. |  <br>Study the above pie charts and match the following<br>(i) Angiosperms (ii) Fishes (iii) Birds<br>(iv) Fungi (v) Insects | A-Insects (v)<br>B-Birds (iii)<br>C-Fishes (ii)<br>D-Angiosperm (i)<br>E-Fungi (iv) |
| 6. | India has only _x_ % of world's land area but have _y_ % of global species diversity. | x = 2.4%<br>y = 8.1% |
| 7. | The species diveristy _____ when we move towards the equator whereas it _____ when we move towards the poles. | Increases<br>Decreases |
| 8. |  <br>Study the above graph and answer the following questions –<br>(i) Identify x and y<br>(ii) The relation between species richness and area can be denoted as generally<br>(iii) The value of Z lies in range of ___, regardless of the taxonomic group or the region (whether it is plants in Britain, birds in California or molluscs in New York state.)<br>(iv) On a log scale the relation between species richness and area is | (i) $x = (S = CA^Z)$,<br>$y = (\log S = \log C + Z \log A)$<br>(ii) Rectangular hyperbola<br>(iii) Z = 0.1 to 0.2<br>(iv) Straight line |

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<table>
<tr><td>9.</td><td>According to rivet popper hypothesis, if ecosystem is considered as airplane then what does the rivets on wings represent.</td><td>Key stone species</td></tr>
<tr><td>10.</td><td>Match the following organism according to their place of extinctions :-
<table>
<tr><td>A.</td><td>Dodo</td><td>i.</td><td>Australia</td></tr>
<tr><td>B.</td><td>Steller's sea cow</td><td>ii.</td><td>Mauritius</td></tr>
<tr><td>C.</td><td>Quagga</td><td>iii.</td><td>Africa</td></tr>
<tr><td>D.</td><td>Thylacine</td><td>iv.</td><td>Russia</td></tr>
</table></td><td>A(ii), B(iv), C(iii), D(i)</td></tr>
<tr><td>11.</td><td>Loss of biodiversity in a region may result in<br>(A) Increased variability in plant productivity, water use, pests and disease cycle.<br>(B) Decline in plant production<br>(C) Lowered resistance to environmental perturbations<br>How many of the above statements are correct?</td><td>A, B, C (All)</td></tr>
<tr><td>12.</td><td>Select the terms which are related to loss of biodiversity and considered under The Evil Quartet –<br>Co-extinction, Proto-cooperation, Over exploitation, Predation, Endemism, Alien - species invasion, Habitat loss and fragmentation.</td><td>Four<br>(Co-extinction, Over exploitation, Alien - species invasion, Habitat loss and fragmentation)</td></tr>
<tr><td>13.</td><td>Which area on earth is also called lungs of planet? Why?</td><td>Amazon rains forest<br>(It produces 20% of total $O_2$ in earth atmosphere)</td></tr>
<tr><td>14.</td><td>Select the species which are extincted due to over exploitation :-<br>Nile Perch, Water hyacinth, African catfish, Passenger pigeon, Lantana, Cichlid fish, Steller's sea cow, Quagga, Dodo.</td><td>Two<br>(Passenger pigeon, Steller's sea cow)</td></tr>
<tr><td>15.</td><td>Exploring molecular, genetic, species - level diversity for products of economic importance is termed as</td><td>Bioprospecting</td></tr>
<tr><td>16.</td><td>Reasons of biodiversity conservation can be grouped into how many categories?</td><td>(1) Narrowly utilitatian<br>(2) Broadly utilitatian<br>(3) Ethical</td></tr>
<tr><td>17.</td><td>India has more than ___A___ genetically different strains of rice, and ___B____ varieties of mango.</td><td>A- 50,000<br>B- 1,000</td></tr>
<tr><td>18.</td><td>Robert May places the global species diversity at about 7 million.</td><td>True</td></tr>
<tr><td>19.</td><td>India has only 8.1 per cent of the world’s land area, its share of the global species diversity is an impressive 2.4 per cent.</td><td>False</td></tr>
</table>

| | | |
|---|---|---|
| **20.** | Species area relationship, on log scale the relationship becomes _____. | linear |
| **21.** | Which scientist found that plots with more species showed less year-to-year variation in total biomass and showed that in his experiments, increased diversity contributed to higher productivity. | David Tilman |
| **22.** | Tropical rain forests once covering more than _____ of the earth's land surface, these rain forests now cover no more than _____. | 14 per cent, 6 per cent. |
| **23.** | Many species extinctions in the last 500 years for example _____ and _____were due to overexploitation by humans. | Steller's sea cow, passenger pigeon |
| **24.** | In which of the four evil quartet following situation is included- the recent illegal introduction of the African catfish Clarias gariepinus for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers. | Alien species invasions |
| **25.** | _____ are the regions with very high levels of species richness and high degree of endemism. | Biodiversity hotspots |
| **26.** | Zoological parks, botanical gardens and wildlife safari parks are example of _____. | Ex-situ conservation |

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